



Research Report

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China's Techno-Industrial Strategy in the Xi Era

Producing Under Pressure

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About This Report

In this report, we examine how China’s techno-industrial policy has evolved under Xi Jinping into a more centralized, security-linked, finance-driven system. We analyze how the Party-state sets industrial priorities and mobilizes institutions, financial resources, firms, and real economy inputs to advance industrial and technological goals.

The report’s purpose is to clarify how China’s contemporary techno-industrial system operates in practice. Our analysis focuses on institutional design, policy instruments, implementation mechanisms, and emerging constraints, particularly tighter fiscal conditions and changing center-local dynamics. We also examine how these domestic arrangements are projected abroad through overseas investment, industrial parks, and standards-setting efforts.

The report is intended to provide policymakers and analysts with a broad understanding of China’s techno-industrial policy system in the Xi era to support more informed analysis of China’s economic trajectory and its implications for global economic competition and governance.

This report draws in part on a June 2025 expert workshop held jointly with the Center for Strategic and International Studies Trustee Chair in Chinese Business and Economics. Research for the main analysis concluded in fall 2025; the report reflects developments through March 2026 and incorporates selected proof-stage updates available through June 2026.

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Summary

In this report, we examine how China’s techno-industrial policy has evolved under Xi Jinping into a more centralized, security-linked, finance-driven system. We explain how this system operates in practice by organizing Xi-era policy into five channels (fiscal, financial, real economy, Party-firm, and overseas) and 18 instruments through which the Party-state defines priorities, mobilizes institutions and financial resources, disciplines firms and local governments, and projects these arrangements abroad. Our analysis draws on Chinese-language policy documents and Party directives, official statistics, financial disclosures, sectoral data, a June 2025 expert workshop, and original analysis of China National Knowledge Infrastructure academic literature. Research was conducted through fall 2025 and reflects developments through March 2026.

Under Xi, China’s techno-industrial policy has shifted from prioritizing growth and catch-up to prioritizing national security, technological self-reliance, and frontier leadership. The Party-state has overhauled institutions, financing, and corporate governance to direct resources toward priority industries. China’s techno-industrial ecosystem has produced impressive gains in technological capabilities, manufacturing scale, and supply chain resilience in priority sectors. But the system also generates significant tensions: Centralization risks suppressing the local experimentation that fueled earlier gains, mandates and designations spread the costs of industrial policy to firms and institutions in ways that are difficult to measure, and the politicization of investment and corporate governance may degrade the quality of economic decisionmaking even as it tightens alignment with state priorities. Techno-industrial policy does not resolve China’s underlying demand and productivity problems that increasingly drag on aggregate growth and prosperity.

Key Findings

- **Fiscal constraints are reshaping the policy toolkit.** After roughly 2021–2022, tighter local budgets reduced room for discretionary subsidies and other outlays. Several measurable channels, such as direct subsidies, tax breaks, and government procurement, have plateaued or declined since their peaks around 2022. The 2026 Government Work Report and the 15th Five-Year Plan (FYP) (2026–2030) signal that Beijing intends to keep fiscal deficits roughly steady as a share of gross domestic product (GDP) while sustaining industrial policy support by curbing wasteful local incentives.
- **The Party-state is shifting from direct spending toward mandates.** The Party-state increasingly acts as an investor and coordinator through guidance funds, targeted credit, tax incentives, procurement mandates, and capital-market reforms. These tools operate through market channels but under tighter political direction. The effect is to spread the costs of industrial policy across firms and institutions rather than bear them on the public budget. Whether this amounts to greater efficiency or to a proliferation of unfunded obligations is an open question. For firms, the incentive is to signal alignment with strategic sectors and Party guidance to gain access to resources, a logic that rewards political positioning alongside—and sometimes over—commercial performance.
- **The “investor state” faces structural limits.** Government guidance fund commitments have grown dramatically since 2014, but corruption, short-termism, and geographic restrictions have exposed serious weaknesses in the local fund model. Beijing is centralizing oversight, but earlier reform rounds also set ambitious goals that were unevenly implemented. Structural incentives—such as short fund horizons, politicized investment decisions, and jurisdictional restrictions—are features of the system’s design and are unlikely to be resolved by tightening oversight alone.
- **Party-state corporate control has deepened.** State-owned enterprises (SOEs) remain preferred instruments in upstream and security-sensitive sectors, while private firms are steered through regulation,

Party embedding, minority state stakes (“golden shares”), and champion-designation programs, such as the “Little Giants” (小巨人) and “Single Champions” (单项冠军). Beijing’s intent is to reserve upstream, capital-intensive, and security-sensitive sectors for SOEs while delegating the fast-moving, innovation-driven segments (artificial intelligence, electric vehicles, biotechnology, drones) to private firms, steering them through incentives, designations, and political oversight.

- **The external dimension is explicitly techno-industrial.** The Belt and Road Initiative (BRI)—especially its Digital Silk Road, Green BRI, standards, and industrial park components—creates external demand for Chinese technology, locks in Chinese equipment and standards in partner markets, and aligns out-bound investment with domestic industrial priorities.
- **The 15th FYP codifies greater discipline and deeper intervention.** The published FYP outline calls for “extraordinary measures” (超常规措施) on key core technologies, including integrated circuits, and identifies ten “new industry tracks” (新产业新赛道) for priority cultivation. It also elevates the “unified national market” (全国统一大市场) to a core objective, aiming to standardize rules, procurement, and competition enforcement across provinces, curbing the duplicative local subsidy races that have fueled overcapacity while pooling national demand.
- **China’s techno-industrial strategy is generating structural global friction.** China’s manufacturing trade surplus (roughly 10 percent of GDP) is comparable as a share of the domestic economy to those of Germany, Japan, South Korea, and Taiwan. But China’s economy is far larger, making the scale of its surplus unprecedented. Depreciation of China’s real effective exchange rate since 2021, driven by comparatively weak domestic inflation, has compounded the competitive advantages generated by industrial policy and scale. If manufacturing output continues to outpace domestic demand, the surplus must be absorbed abroad, inviting growing resistance through tariffs, industrial policy responses, and trade remedies.

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Introduction

China's rapid industrial and technological ascent has spurred great interest in the mechanics of its development model. Today, the country is the world's dominant industrial producer and is rising in measures of innovation and technological capacity, catching or even surpassing global leaders across some industries and scientific domains. Yet, on measures of economic and financial efficiency, China's trajectory is more worrisome. The institutions of the Chinese Communist Party (CCP) and the Chinese state play changing roles in this process. Under Xi Jinping, even as China reaches closer to the technological frontier, Party-state guidance of the economy is intensifying.

China's techno-industrial ambitions are reshaping the global economic landscape. No economy so large and so deeply integrated into global markets has ever attempted an industrial transformation as broad, state-guided, and security-driven as the one Beijing is now pursuing. These policies influence supply chains, tilt standards, and increasingly set the pace of technological change. China's push to dominate advanced manufacturing and frontier technologies poses both a competitive challenge and a test of coordination for the United States and its allies, forcing them to rethink how innovation, security, and industrial resilience intersect. Understanding how China's techno-industrial system works is essential for assessing both China's growth trajectory and the ways its industrial strategy will reshape the global economy, particularly as Western nations pursue their own industrial policies.

This report provides a structured overview of China's techno-industrial development strategy and its evolution under Xi Jinping. As defined by the International Monetary Fund (IMF), *industrial policy* refers to "targeted government interventions aimed at supporting specific domestic firms, industries, or narrowly defined economic activities to achieve certain national (economic or non-economic) objectives."¹ In China's case, those interventions now extend along the "technology chain," from research and development (R&D) and standards to commercialization and diffusion. We use *techno-industrial policy* to describe Beijing's fusion of traditional industrial policy with science and technology (S&T) policy to promote not just specific industries but also the underlying technologies and production systems that enable them.² We therefore treat industrial policy and aspects of S&T policy as an integrated policy. Given the breadth of this system, our analysis focuses on the industrial side of the fusion. We touch on S&T where it intersects with industry, but we do not provide a full treatment of "pure S&T" topics, such as the basic research funding system, national lab governance, academic evaluation, or higher education administration.³

¹ IMF, *Industrial Policy Coverage in IMF Surveillance—Broad Considerations*, Policy Paper No. 2024/008, March 11, 2024, p. 1.

² The term *techno-industrial policy* is used in the China context in Chen Ling and Barry Naughton, "The Emergence of Chinese Techno-Industrial Policy: From Megaprojects to Strategic Emerging Industries, 2003–2011," Instituto Nacional de Ciência e Tecnologia em Políticas Públicas, Estratégias e Desenvolvimento, January 2013, one of the earliest works to explicitly characterize China's post-2003 integration of industrial and S&T policy under a unified techno-industrial strategy.

³ For a more detailed study on S&T policy, see Tai Ming Cheung, *China's Competitive Model for Strategic Science, Technology, and Innovation Development*, University of California Institute on Global Conflict and Cooperation, November 2025.

This report adds to a growing body of research on China's industrial and technology policy by providing a systematic assessment of techno-industrial policy and by situating it within China's historical trajectory and current political economy.⁴ Prior studies have examined such mechanisms as subsidies and credit distortions,⁵ decentralization and local competition,⁶ or Party-state control of firms.⁷ But none have mapped how these fiscal, financial, and organizational tools operate together as an integrated system under Xi Jinping, especially in recent years.⁸

The rest of this report proceeds as follows:

- In the next section, we provide a brief history of China's industrial policies.
- In Chapter 2, we discuss China's current techno-industrial objectives and show how those priorities have shifted since the mid-2010s.
- In Chapter 3, we examine the politics and institutions of implementation, including the turn toward Party-led “top-level design” (顶层设计) verticalization in key bureaucracies and the evolving balance between central steering and decentralized execution.⁹
- In Chapter 4, we catalog the policy toolkit and how it has been reshaped and has become more structured, more financialized, and more explicitly tied to security.
- In Chapter 5, we explain how fiscal constraints are changing the mix of policy instruments.
- In Chapter 6, we analyze financial system channels that align capital with industrial strategy.
- In Chapter 7, we review “real economy” (实体经济) levers—such as land, energy, zones, and inputs—and how they have been disciplined and redirected.
- In Chapter 8, we turn to Party-firm coordination and evolving state-private linkages.

⁴ This report also draws on a June 2025 expert workshop convened jointly with the Center for Strategic and International Studies Trustee Chair in Chinese Business and Economics, which helped inform our assessment of how China's techno-industrial policy instruments operate in practice.

⁵ Daniel Garcia-Macia, Siddharth Kothari, and Yifan Tao, *Industrial Policy in China: Quantification and Impact on Misallocation*, International Monetary Fund, WP/25/155, August 2025; Camille Boullenois, Agatha Kratz, and Daniel H. Rosen, *Far from Normal: An Augmented Assessment of China's State Support*, Rhodium Group, March 17, 2025.

⁶ Kaicheng Luo, Shaoda Wang, and David Y. Yang, *Laboratories of Autocracy: Landscape of Central-Local Dynamics in China's Policy Universe*, National Bureau of Economic Research, Working Paper 34219, September 2025; Chen Ling and Barry Naughton, “An Institutionalized Policy-Making Mechanism: China's Return to Techno-Industrial Policy,” *Research Policy*, Vol. 45, No. 10, December 2016.

⁷ Philipp K€oncke, Lucas Erlbacher, and Stefan Schmalz, “The Anatomy of Chinese Capitalism: Sectoral Variation in Party-State Permeation and Corporate Governance,” *Competition & Change*, Vol. 30, No. 2, April 2026; Yingyao Wang, “Policy Articulation and Paradigm Transformation: The Bureaucratic Origin of China's Industrial Policy,” *Review of International Political Economy*, Vol. 28, No. 1, 2021.

⁸ Among recent studies, a 2024 Bruegel report comes closest to this report's broad conception of China's industrial policy, explicitly emphasizing that it extends well beyond direct subsidies to encompass financial and administrative instruments of industrial policy (Alicia Garc a-Herrero and Robin Schindowski, *Unpacking China's Industrial Policy and Its Implications for Europe*, Bruegel, Working Paper 11/2024, May 13, 2024). However, the authors' analysis is oriented toward cross-sector assessment and implications for Europe rather than a systems-level mapping of how fiscal, financial, and organizational levers are integrated and coordinated under Xi.

A May 2026 U.S. Chamber of Commerce report prepared with Rhodium Group independently characterizes China's next-generation industrial policy as a systemwide strategy spanning upstream inputs, industrial equipment, downstream applications, services, and frontier technologies. See Camille Boullenois, Malcolm Black, and Alessia Caruso, *China's Next-Generation Industrial Policy*, prepared for the U.S. Chamber of Commerce, Rhodium Group, May 11, 2026.

⁹ For concise definitions of select Chinese policy terms, slogans, and institutional phrases used throughout this report, see the appendix.

- In Chapter 9, we show how these policies are projected abroad via the Belt and Road Initiative (BRI), standards diplomacy, and outbound investment discipline.
- In Chapter 10, we conclude by assessing priorities for the 15th Five-Year Plan (FYP) (2026–2030) and what they imply for the coming policy phase and geopolitical tensions.

A Very Brief History of China's Industrial Policies

Since the late Qing dynasty, every generation of Chinese leadership has pursued the same core goal: restoring national strength by matching and eventually surpassing the industrial and technological capabilities of leading global powers. Methods have shifted from early state-led modernization to Mao Zedong's mobilized self-reliance, reform-era pragmatism to Xi's "grand steerage," but the underlying aim remains building a strong, independent, and technologically advanced China. Today's industrial policy language and concepts reflect this enduring project.¹⁰

China's industrial policy has transformed in response to changes in political priorities, economic strategy, and the global environment (Table 1.1). During the planned economy (1949–1976), industrialization was central to socialist development, guided by Soviet planning. The First FYP (1953–1957) prioritized heavy industry, aiming for an "independent and relatively complete industrial system" (独立的比较完整的工业体系). The Great Leap Forward (1958–1962) disrupted this focus, mobilizing communes and small workshops for rapid catch-up through "self-reliance" (自力更生). After a period of retrenchment (1961–1963), the Third Front (1964–1976) relocated factories and research institutes inland to build a defense-industrial base, embedding strategic self-sufficiency, a concept that would later reappear as "supply-chain security."¹¹

After 1978, "reform and opening" (改革开放) dismantled rigid central planning but retained industrial policy as a tool of state guidance. Early reforms prioritized light industry, exports, and foreign investment. In the 1990s, the focus shifted to "pillar industries" (支柱产业), such as automobiles, petrochemicals, and telecommunications. This was supported by enterprise reform and World Trade Organization (WTO) accession. Marketization and selective state support began to blend, setting the stage for more-targeted development.

For much of the reform era, China's transformation was driven less by top-down industrial policy and more by the coevolution of the market and the government. Township and village enterprises, agricultural liberalization, foreign direct investment (FDI), housing privatization, and state enterprise reform spurred growth, enabling China to become the world's top exporter without a unified national industrial strategy.¹²

Unlike Japan and Korea, which built sectoral targeting industrial plans onto functional markets, China started as a command economy. Early planning efforts, such as the 1978 Ten-Year Plan, were top-down and largely ineffective. FYPs in the 1980s and 1990s struggled to keep pace with market-driven growth. Deng Xiaoping's 1992 Southern Tour marked a shift toward experimental liberalization. Policymakers studied Japan's model but lacked the institutional capacity to replicate it. Such initiatives as the 863 and 973 Plans

¹⁰ Barry Naughton, "Grand Steerage as the New Paradigm for State-Economy Relations," in Frank N. Pieke and Bert Hofman, eds., *CPC Futures: The New Era of Socialism with Chinese Characteristics*, East Asian Institute, National University of Singapore, 2022.

¹¹ Mao's ambition for total societal transformation often clashed with China's industrialization drive. He frequently elevated ideological mobilization over material incentives and prioritized "reds" over "experts." His simultaneous pursuit of communist egalitarianism and nationalist industrialization frequently meant that neither goal was fully advanced. For a fuller elaboration of this argument, see Lucien Bianco, *Stalin and Mao: A Comparison of the Russian and Chinese Revolutions*, trans. by Krystyna Horko, Chinese University of Hong Kong Press, 2018.

¹² Yingyi Qian, *How Reform Worked in China: The Transformation from Plan to Market*, MIT Press, 2016; Yuen Yuen Ang, *How China Escaped the Poverty Trap*, Cornell University Press, 2016.

TABLE 1.1
Industrial Policy Phases in the People's Republic of China

Era	Objectives and Policy Concepts	Favored Sectors
Soviet-Model Foundation (1949–1957)	Build a socialist-heavy industrial base under central planning; “independent and relatively complete industrial system”	Steel, coal, machinery, Soviet-aided plants
Great Leap and Retrenchment Cycle (1958–1963)	Rapid catch-up via mass mobilization, then retrenchment; self-reliance, “walking on two legs”	Communes, backyard furnaces, small-scale steel
Defense and Regional Rebalancing (1964–1976)	Develop inland industrial base for national defense; strategic self-reliance, Third Front	Armaments, inland heavy industry, infrastructure in interior regions
Reform and Opening Beginnings (1978–1992)	Restore growth and efficiency via reform and opening; “socialist market economy”	Township and village enterprises, special economic zones, light industry exports
Marketization and Pillar Industries (1993–2005)	Deepen marketization and integrate into the global economy; “grasp the large, release the small,” pillar industries strategy	State-owned enterprise (SOE) restructuring, automotive, petrochemicals, WTO accession
Strategic Emerging Industries and State Guidance (2006–2013)	Upgrade industrial structure and innovation capacity; “indigenous innovation,” state-guided development of strategic sectors	ICT, high-end equipment, post-global financial crisis stimulus sectors
Early Xi Era: Financialized Techno-Industrial State (2014–2019)	Strengthen technological capabilities through state coordination and finance; Made in China 2025, supply-side structural reform	Government guidance funds, semiconductors, robotics, advanced manufacturing
Current Xi Era: Innovation, Security, and Productivity (2020–Present)	Integrate innovation with economic security and resilience; “dual circulation,” “new quality productive forces,” industrial and supply chain security	Artificial intelligence (AI), semiconductors, new energy vehicles, supply chain resilience programs

NOTE: Quotation marks identify official or widely used Chinese policy terms and slogans, not author-created labels or quotations from a single source. Translations are by the authors unless otherwise indicated.

introduced techno-nationalist ideas but were fragmented; vertical industrial policy faded in the 1990s as ministries were dismantled.¹³

A turning point came in 2003 with the creation of the National Development and Reform Commission (NDRC) and the release of the Medium- and Long-Term Plan for Science and Technology Development (MLP) in 2006.¹⁴ The MLP promoted “indigenous innovation” (自主创新) and launched 16 megaprojects in strategic sectors, including semiconductors, nuclear power, and broadband.¹⁵

Beijing's response to the 2008 global financial crisis triggered unprecedented state-led investment via local government financing vehicles (LGFVs), stimulating local initiatives. At the central level, the new Ministry of Industry and Information Technology (MIIT) complemented the NDRC, coordinating top-down

¹³ The 863 Program (launched in 1986) and the 973 Program (launched in 1997) were China's flagship national S&T initiatives in the reform era. The 863 Program funded strategic high-technology fields, such as aerospace, information and communications technology (ICT), advanced materials, and biotechnology to reduce reliance on foreign technology. The 973 Program focused on basic research underpinning long-term scientific capacity. Both reflected early techno-nationalist ambitions but operated through fragmented bureaucratic channels, with limited coordination across ministries or between research and industry. See Barry Naughton, *The Rise of China's Industrial Policy: 1978 to 2020*, Universidad Nacional Autónoma de México, 2021.

¹⁴ State Council of the People's Republic of China, “Outline of the National Medium- and Long-Term Science and Technology Development Plan (2006–2020),” *State Council Gazette*, No. 9, 2006a.

¹⁵ Naughton, 2021.

industrial policy. Stimulus funds supported the MLP and helped launch the Strategic Emerging Industries (SEIs) (战略性新兴产业) initiative in 2010, targeting such sectors as biotechnology, new energy, and next-generation information technology (IT) through planning, subsidies, and direct lending.¹⁶

Between 2009 and 2013, techno-industrial policy became more structured. Sector-specific FYPs proliferated, and SEIs and the MLP reinforced each other. Beijing regained authority over industrial development, supported by new financial tools and cross-ministerial coordination.¹⁷

By 2013, the architecture for modern techno-industrial policy was largely established. Bureaucratic capacity was rebuilt, planning cycles were aligned with the technology strategy, and the policy toolkit was expanded. Under Xi Jinping, this approach intensified.

¹⁶ State Council of the People's Republic of China, "State Council Decision on Accelerating the Cultivation and Development of Strategic Emerging Industries," State Council Document No. 32 of 2010, October 18, 2010. The 2010 decision identifies seven priority sectors, including biotechnology, new energy, and next-generation IT, as "pillar" industries and sets targets for their contribution to gross domestic product (GDP), emphasizing such policy tools as fiscal support, credit, and planning under NDRC coordination.

¹⁷ For examples of sector-specific FYPs issued during this period, see NDRC, *12th Five-Year Plan for the Bioindustry*, 2011; MIIT, *12th Five-Year Plan for the New Materials Industry*, 2012; MIIT, *12th Five-Year Plan for the Next-Generation IT Industry*, 2011.

Goals of China's New Techno-Industrial Policy

China's long-term techno-industrial ambitions have consistently centered on economic development and national strength, but under Xi Jinping the near-term focus has shifted from catch-up and export competitiveness toward self-reliance, resilience, and technological leadership. Earlier leaders emphasized expanding manufacturing capacity, integrating into global supply chains, and acquiring foreign technologies to fuel modernization and absorb labor rather than pursuing global technological primacy.¹

Xi's approach is an intensification and securitization of these long-standing aims. Techno-industrial policy is explicitly linked to national security, with "independent and controllable" (自主可控) technologies seen as essential buffers against external pressure.² Indigenous innovation, rooted in mid-2000s initiatives and later embodied in such efforts as Made in China 2025 (中国制造2025), has become more central, and the ambition has shifted from narrowing gaps with advanced economies to targeting global leadership in such frontier domains as AI, biotechnology, quantum technologies, and clean energy.³ Reflecting this shift, the share of government documents dedicated to industrial policy has surged at both the central and local levels since 2018, even as the broader objective of industrial upgrading and moving up the global value chain has remained constant across leadership transitions.⁴

These goals are visible across successive FYPs. The 12th FYP (2011–2015) promoted SEIs (next-generation IT, new energy vehicles, biotechnology, new materials, high-end equipment, energy saving/environmental protection, new energy) with targets and action plans.⁵ The 13th FYP (2016–2020) reframed the hierarchy around innovation-driven development and "Develop China into a Manufacturing Powerhouse," tying execution to Made in China 2025.⁶ The 14th FYP (2021–2025) put S&T self-reliance forward as support for devel-

¹ Naughton, 2021.

² Chinese Communist Party Central Committee, *Proposal of the Central Committee of the Chinese Communist Party on Drawing Up the 14th Five-Year Plan for National Economic and Social Development and Long-Range Objectives for 2030*, trans. by Etcetera Language Group, Inc., ed. by Ben Murphy, Center for Security and Emerging Technology, December 1, 2020.

³ Wendy Chang, Rebecca Arcesati, and Antonia Hmaid, *China's Drive Toward Self-Reliance in Artificial Intelligence: From Chips to Large Language Models*, Mercator Institute for China Studies, July 22, 2025.

⁴ Hanming Fang, Ming Li, and Guangli Lu, *Decoding China's Industrial Policies*, National Bureau of Economic Research, Working Paper 33814, May 2025. The authors built a new large language model-derived dataset from about 3 million Chinese government documents (2000–2022) and identified around 0.77 million industrial policy documents. Inside these documents, objectives concentrate on promoting strategic industries (22 percent), fostering innovation (17 percent), and technology adoption (8 percent), and there is a sizable social welfare component (26 percent); cities tilt more toward emerging industries, employment, and urbanization, while the center leans into strategic industries and innovation.

⁵ Joseph Casey and Katherine Koleski, "Background: China's 12th Five-Year Plan," U.S.-China Economic and Security Review Commission, June 24, 2011.

⁶ The People's Republic of China, *The 13th Five-Year Plan for Economic and Social Development of the People's Republic of China (2016–2020)*, trans. by Compilation and Translation Bureau, Central Committee of the Communist Party of China, Central Compilation & Translation Press, 2016.

opment and paired it with “dual circulation” (双循环) and expanded “new infrastructure” (新基建), moving the innovation system to the center of economic strategy.⁷

The 15th FYP (2026–2030), published in March 2026, confirms the priorities signaled in the Fourth Plenum proposals: building a “modernized industrial system” (现代化产业体系) and consolidating and strengthening the “foundation of the real economy” (实体经济根基), followed by accelerating “high-level S&T self-reliance and self-strengthening” (高水平科技自立自强) and developing “new quality productive forces” (新质生产力).⁸ The sequencing implies that S&T is positioned as the enabler of industrial outcomes, consistent with a push to convert years of S&T buildout into industrial and productivity gains amid slower growth. This reflects Beijing’s long-standing belief that national power ultimately rests on the strength of the “real economy” (实体经济), which is understood as broad-based manufacturing prowess, underpinned by a “complete industrial system” (完整工业体系) and resilient “independent and controllable” industrial chains in strategic sectors. The outline also identifies ten “new industry tracks” (新产业新赛道) for priority cultivation.⁹

To identify and prioritize China’s techno-industrial policy goals in the Xi era, we conducted a targeted review of Chinese-language academic literature using China National Knowledge Infrastructure (CNKI), China’s largest scholarly database (Table 2.1). To avoid noise and low-quality material in the complete CNKI corpus, we restricted the search to peer-reviewed academic journals published between 2015 and 2025 and excluded low-impact papers, retaining only articles that met a minimum download threshold. By comparing keyword frequency growth from 2015 to 2025, we approximated which goals the system itself has elevated over time.¹⁰ We selected 2015 as the baseline because it marked the formal elevation of the Innovation-Driven Development Strategy, signaling a new phase in which China’s techno-industrial policy became more coordinated and explicitly centered around technological self-reliance.¹¹ The results reveal 13 core goals, which provide a structured indicator of how key objectives have been elevated in mainstream academic discourse, shaped by and often responding to official priorities.

Several structural and external factors explain why China’s techno-industrial policy goals have shifted under Xi. Three stand out: changes in China’s stage of development and the need for economic rebalancing, heightened national security concerns and rising geopolitical tensions, and organizational and ideological imperatives.

⁷ People’s Republic of China, *Outline of the People’s Republic of China 14th Five-Year Plan for National Economic and Social Development and Long-Range Objectives for 2035*, trans. by Etcetera Language Group, Inc., ed. by Ben Murphy, Center for Security and Emerging Technology, May 12, 2021. *Dual circulation* is the strategy of prioritizing domestic demand while maintaining engagement with global markets.

⁸ “Outline of the Fifteenth Five-Year Plan for National Economic and Social Development of the People’s Republic of China,” Xinhua, March 13, 2026.

⁹ These ten tracks are integrated circuits, embodied intelligence (具身智能) (i.e., physical AI and humanoid robotics), bio-manufacturing, new-generation batteries, commercial aerospace, domestically built large aircraft, low-altitude equipment (drones and urban air mobility), green hydrogen, brain-computer interfaces, and high-end medical devices.

¹⁰ For each of the goals, we selected two authoritative phrases from Party documents and speeches to capture how the Chinese system defines each objective. These phrases are included in Table 2.1. These categories should be understood as overlapping goal families in China’s techno-industrial discourse rather than mutually exclusive priorities. Rankings are based on the percentage increase in frequency of these terms since 2015.

The results reveal a clear hierarchy. Such goals as self-reliance and technology leadership have surged to the forefront, while traditional objectives, such as employment and global competitiveness, have either stagnated or declined. This exercise does not capture internal funding flows or implementation outcomes. But it does provide a proxy for salience within policy-related academic discourse and signals which goals are most likely to drive policy design and resource allocation going forward.

¹¹ Naughton, 2021.

TABLE 2.1
Changes in Prioritization of Techno-Industrial Policy Goals (2015–2025)

Rank	Goal	Chinese Phrasing
1	Self-reliance	自立自强, 科技自立自强
2	Technology leadership	关键核心技术, 科技强国
3	Structural transformation ^a	产业结构优化升级, 高质量发展转型
4	Decarbonization	绿色科技, 绿色低碳发展
5	Supply-side quality upgrade	供给侧结构性改革, 提高供给质量
6	Regional development	区域协调发展, 产业布局优化
7	R&D and commercialization	产学研融合, 科技成果转化
8	Industrial upgrading	产业升级, 制造业高端化
9	Productivity growth	劳动生产率, 全要素生产率
10	Economic diversification	多元化发展, 构建多元化产业体系
11	National prestige	实现中华民族伟大复兴, 大国重器
12	Global competitiveness	国际竞争力, 打造世界一流
13	Employment and wages	扩大就业, 就业结构调整

^a *Structural transformation* refers to economywide restructuring associated with the shift to high-quality development, including changes in the sectoral mix and the optimization and upgrading of China's industrial structure. This is broader than *industrial upgrading*, which we use to describe moving manufacturing toward higher-end products, processes, and technologies.

Why These Goals? Structural, Security, and Ideological Drivers

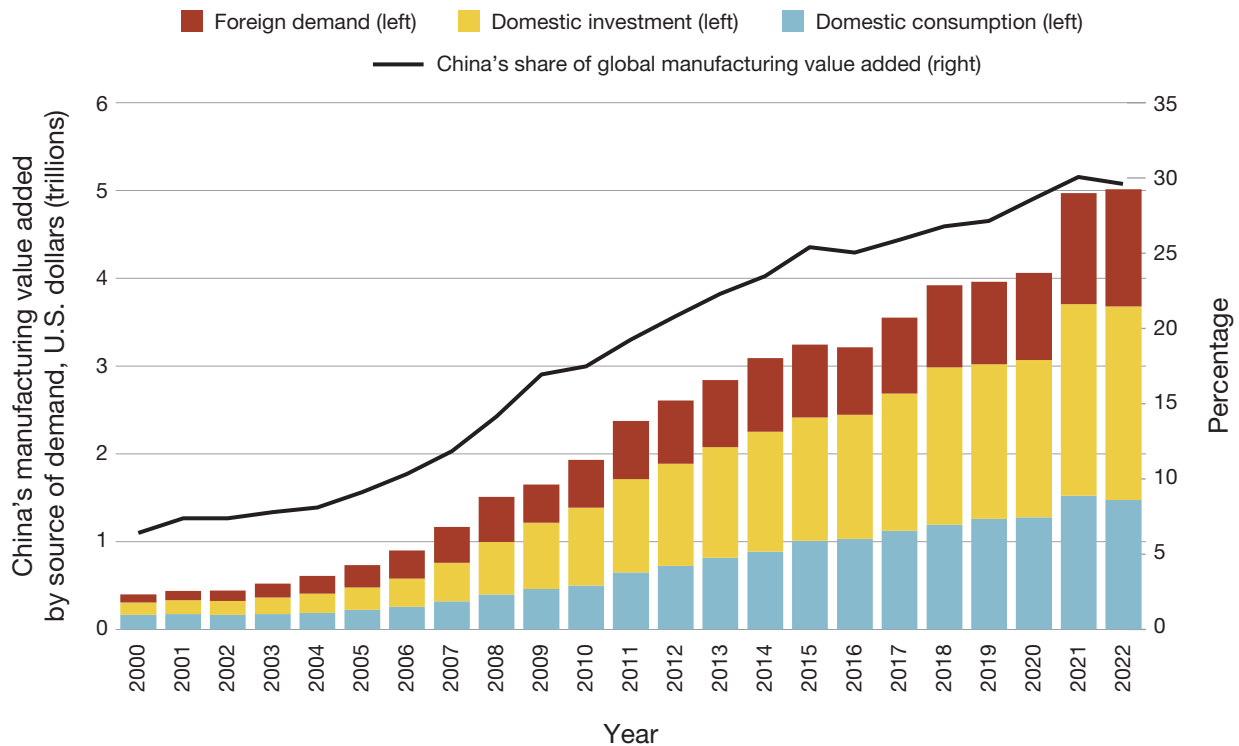
Stage of Development and Economic Rebalancing

China's strategy to move up the value chain had been worked out long before the 2010s, combining “market-for-technology” arrangements with investments in human capital. During the 2000s, local officials courted foreign investors, fostering production ecosystems that became “learning-by-doing” platforms for Chinese firms. At the same time, university enrollment expanded more than fivefold after 1999, laying the groundwork for a more skilled industrial workforce.¹²

By the early 2010s, China had established itself as the world's largest manufacturer and a global exporter. It was approaching the technological frontier in such areas as telecommunications, high-speed rail, and renewable energy, while lagging in “chokepoint” technologies, such as semiconductors, jet engines, and advanced manufacturing equipment. The catch-up model of absorbing foreign technologies and competing on low-cost exports was delivering diminishing returns. Rising wages eroded China's competitiveness in labor-intensive industries, while productivity growth slowed. At the same time, global trade demand weakened following the 2008 financial crisis. These pressures created the need for a shift toward higher-value, innovation-driven growth, including to sustain China's export competitiveness, which remains important for its manufacturing sector (Figure 2.1).

¹² Shulin Gu and Bengt-Åke Lundvall, “China's Innovation System and the Move Toward Harmonious Growth and Endogenous Innovation,” Copenhagen Business School and Aalborg University, DRUID Working Papers 06-07, 2006; Loren Brandt and Eric Thun, “The Fight for the Middle: Upgrading, Competition, and Industrial Development in China,” *World Development*, Vol. 38, No. 11, November 2010; Bin Huang, Massimiliano Tani, Yi Wei, and Yu Zhu, “Returns to Education in China: Evidence from the Great Higher Education Expansion,” *China Economic Review*, Vol. 74, August 2022.

FIGURE 2.1

China Is the World's Top Manufacturer and Needs Foreign Markets

SOURCE: Features data from Organisation for Economic Co-operation and Development (OECD), Trade in Value-Added, database, 2025c.

National Security and Geopolitical Tensions

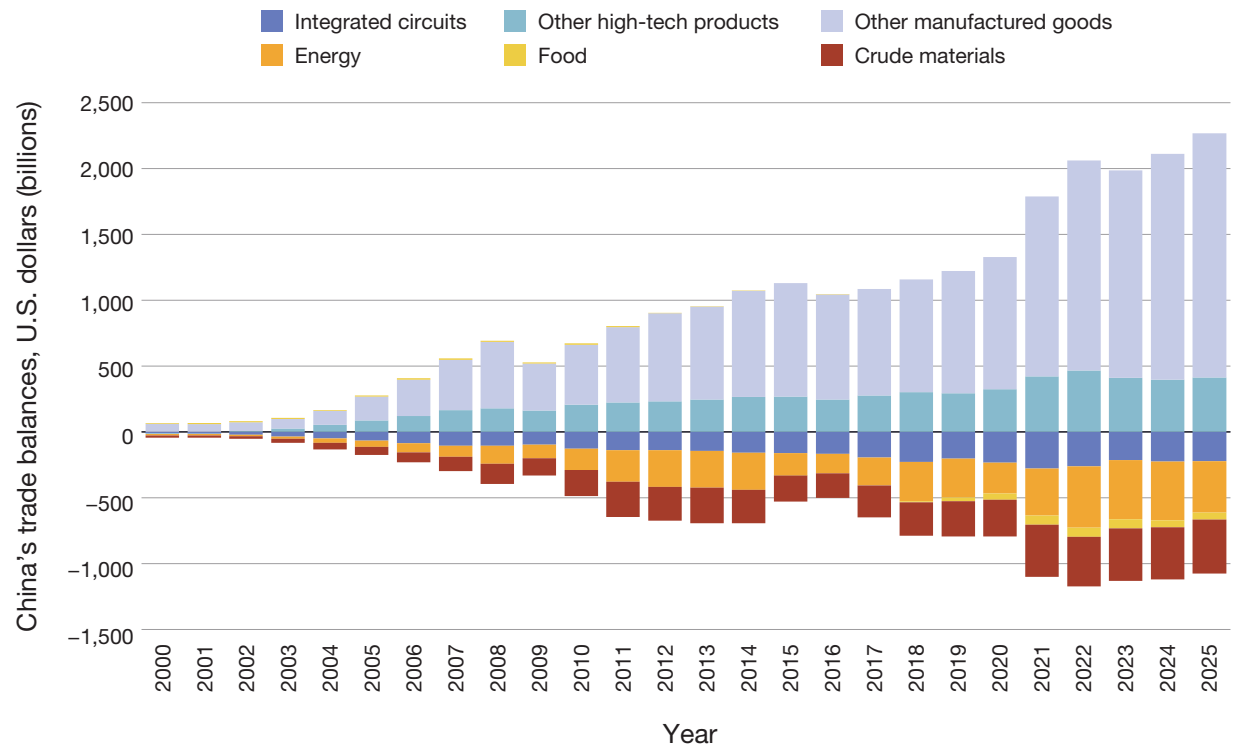
Xi elevated the concept of “comprehensive national security” (总体国家安全观), which explicitly framed economic and technological resilience as core objectives.¹³ Techno-industrial policy was redefined as much as a security tool as an economic one.¹⁴ Civil-military fusion exemplifies this approach, directly linking industrial upgrading to military modernization. The result is that national security has moved from a secondary benefit of industrial upgrading to a central goal.

External shocks after 2016 reinforced Beijing's concerns and accelerated the reorientation of policy goals. The 2018 U.S. ban on component sales to Zhongxing Telecommunications Equipment Corporation revealed how vulnerable Chinese firms were to foreign chokepoints. Restrictions on Huawei, particularly access to semiconductors, hobbled its global expansion. The U.S.-China trade war expanded to include tariffs, investment restrictions, and export controls, while subsequent coordination with U.S. allies curtailed China's access to advanced semiconductors and chipmaking equipment. These developments underscored that foreign dependence was a strategic liability (Figure 2.2). Although China had long aspired to reduce reliance

¹³ Xi introduced the concept on April 15, 2014, during the first meeting of the Central National Security Commission. The concept was then added to the amended CCP Constitution in 2017. See Xi Jinping, “Uphold the Holistic National Security Concept and Follow the Path of National Security with Chinese Characteristics,” Xinhua, April 15, 2014.

¹⁴ In April 2026, China formalized this frame through State Council Order No. 834 on industrial and supply-chain security, which created mechanisms for security investigations and countermeasures, and Order No. 835 on countering foreign states' unlawful extraterritorial jurisdiction. See “China Issues Regulations on Industrial, Supply Chain Security,” Xinhua, April 7, 2026; and “China Issues Rules on Countermeasures Against Foreign States' Unlawful Extraterritorial Jurisdiction,” Xinhua, April 13, 2026.

FIGURE 2.2

China Is a Net Importer of Integrated Circuits and Commodities

SOURCE: Features data from the General Administration of Customs via CEIC, China Premium Database, undated.

NOTE: Positive values indicate a trade surplus; negative values indicate a trade deficit.

on imported technology, these geopolitical shocks transformed self-reliance and supply chain security into urgent goals.

Ideological, Organizational, and Political Goals

China's techno-industrial policy has political and ideological objectives that go beyond economic upgrading. First, it is a tool to assert Party leadership and centralization. Techno-industrial policy, which had already emerged under Hu Jintao as a coordinating device, is now cast as a Party-directed project aimed at securing the "commanding heights" of the economy, thereby justifying the creation of new mechanisms for central design, supervision, and cadre discipline.¹⁵

Second, techno-industrial upgrading is framed as proof of regime superiority and a pillar of national rejuvenation.¹⁶ Industrial success is used to demonstrate the effectiveness of China's governance model and to bolster the Party's claim that only it can deliver modernization, security, and broad-based improvements

¹⁵ Chen and Naughton, 2016.

¹⁶ Wen Yan, "Accelerate the Realization of High-Level Scientific and Technological Self-Reliance and Self-Strengthening, and Victoriously Advance the Historic Cause of Building a Strong Country and Realizing the Great Rejuvenation of the Chinese Nation—Studying Xi Jinping's 'On Self-Reliance and Self-Strengthening in Science and Technology,'" *People's Daily*, September 8, 2023.

in living standards often framed as “common prosperity” (共同富裕).¹⁷ Xi presents global rivalry as “systems competition” (制度竞争), in which advances in frontier industries validate “socialism with Chinese characteristics” (中国特色社会主义) and the Party’s claim to long-term rule.¹⁸

Third, techno-industrial policy underpins China’s posture in geopolitical competition. Technological breakthroughs and supply-chain control are portrayed as essential for withstanding “foreign containment” and reducing vulnerability to external chokepoints.¹⁹

Finally, Xi links techno-industrial goals to the domestic agenda of “common prosperity” (共同富裕).²⁰ In official usage, *common prosperity* refers to achieving broadly shared affluence by narrowing regional and income gaps and improving public services, without abandoning growth or pursuing strict equality of outcomes. By promising that high-tech upgrading will create better jobs, strengthen the “real economy” (实体经济), and support redistribution, Xi casts techno-industrial policy as a vehicle for social equity.²¹ Whether techno-industrial policy actually advances “common prosperity” (共同富裕) in practice is less clear. The distributional effects of high-tech upgrading are not automatic and can be uneven across regions and skill groups, and external assessments emphasize that broadly shared gains depend heavily on complementary reforms (education, social mobility, and stronger household support) beyond industrial upgrading alone.²²

¹⁷ There is, for example, a whole genre of official writing that takes high-speed rail, big S&T projects, and industrial achievements as proof of the superiority and effectiveness of China’s system and governance. See Lu Dongfu, “Creating a Shining Brand for China’s High-Speed Rail,” *Qiushi*, No. 15, 2021.

¹⁸ On Xi’s phrasing of *systems competition* (制度竞争): A “well-founded system is the biggest strength a country has, and competition in terms of systems is the most essential rivalry between countries” (Xi Jinping, “Uphold and Improve the Chinese Socialist System and Modernize State Governance,” in *The Governance of China*, Vol 3, October 31, 2019b). On the development of a steering system and its pre-dating of Xi, see Sebastian Heilmann and Oliver Melton, “The Reinvention of Development Planning in China, 1993–2012,” *Modern China*, Vol. 39, No. 6, November 2013; Barry Naughton, “The Return of Planning in China: Comment on Heilmann–Melton and Hu Angang,” *Modern China*, Vol. 39, No. 6, November 2013.

¹⁹ Communist Party Member Network, “Why Should We Accelerate the Achievement of High-Level Scientific and Technological Self-Reliance?” *100 Questions and Answers on Studying the Report of the 20th CPC National Congress*, January 20, 2023.

²⁰ Research Center for Party Innovation Theory, National University of Defense Technology, “Building New Drivers and Advantages for Development Through Technological Self-Reliance and Self-Improvement,” *Red Flag Manuscript*, December 10, 2022.

²¹ Zheng Shanjie, “Accelerate the Construction of a Modern Industrial System Supported by the Real Economy,” *Qiushi*, July 1, 2023.

²² Dirk Muir, Natalija Novta, and Anne Oeking, *China’s Path to Sustainable and Balanced Growth*, International Monetary Fund, WP/2024/238, November 2024; Yunzhe Nie, *Do Industrial Policies Promote Equitable Economic Growth in China? Policy Proposal to Reduce Inequality and Enhance Effectiveness*, Mossavar-Rahmani Center for Business and Government, Harvard Kennedy School, M-RCBG Associate Working Paper Series, No. 263, July 2025.

The Politics of Techno-Industrial Policy: Centralization and Institutional Design

Nothing has been more central to Xi's tenure as General Secretary than his effort to reinvigorate, strengthen, and direct the CCP. Xi's project has been to coordinate China's often fragmented political system by recentralizing authority in the Party center. At the heart of the project is the concept of "top-level design" (顶层设计), a method of systemwide planning and coordination in which the Party leadership defines the architecture, objectives, and authority relationships that guide the entire bureaucracy.¹ At the same time, much of China's developmental success has been rooted in decentralized local implementation.² Xi's organizational ambition is to harness that local dynamism while tightening central steering through "top-level design" (顶层设计). To date, the results are mixed. Beijing has expanded tools for standardization and compliance, yet local incentives and protectionist practices persist, and implementation gaps remain evident in such campaigns as the "unified national market" (UNM).

Organizational restructuring represents concrete reallocations of personnel and resources (Table 3.1). These institutional changes can offer credible signals of intent and, in broad strokes, are visible enough to allow analysts to trace how the Party-state retools itself to exercise industrial policy.

Institutional Reorganization and the State Apparatus

Since the early post-Mao era, China has periodically overhauled its economic governance structure. The broadest transition has been the shift from a system of direct administrative planning toward one designed to guide market activity through policy. Reform created stronger tools and institutions for market-compatible macroeconomic guidance and for the execution of industrial policy. By the 1990s, an industrial policy coalition was ascendant, and organizational reforms sequentially provided the levers it would soon wield.³

¹ Gunter Schubert and Björn Alpermann, "Studying the Chinese Policy Process in the Era of 'Top-Level Design': The Contribution of 'Political Steering' Theory," *Journal of Chinese Political Science*, Vol. 24, No. 2, 2019; Alex He, "Statement for the Record Before the U.S.-China Economic and Security Review Commission Hearing on 'CCP Decision-Making and the 20th Party Congress,'" U.S.-China Economic and Security Review Commission, January 27, 2022.

² Xi has mandated more experimentation and formalized it as a governing requirement, yet evidence shows that pilot programs often lose effectiveness when scaled, casting doubt on their utility and highlighting instead the role of broader incentive-alignment mechanisms in driving catch-up growth. For recent research findings, see Shaoda Wang and David Y. Yang, "Policy Experimentation in China: The Political Economy of Policy Learning," *Journal of Political Economy*, Vol. 133, No. 7, July 2025; and Abbey S. Heffer and Gunter Schubert, "Policy Experimentation Under Pressure in Contemporary China," *China Quarterly*, Vol. 253, March 2023. On past research that has perhaps overemphasized the role of policy experimentation and its efficacy, see Sebastian Heilmann, *Red Swan: How Unorthodox Policy-Making Facilitated China's Rise*, Chinese University of Hong Kong Press, 2018.

³ Although it is now common to date the rise of China's industrial policy to the mid-2000s—or to Xi's tenure—by the 1990s there were already many indicators. One could reinterpret Yasheng Huang's argument, for example, in his 2008 book, *Capi-*

TABLE 3.1

Organizational Changes That Later Enabled Party-State Led Techno-Industrial Policy

	Restructuring for Industrial Policy (1993–2003)	Industrial Policy Breakthrough (2003–2012)	Party-Led Grand Steerage (2012–Present)
Core reorganization	- Planning ministries merged or abolished to streamline economic governance - Industrial policy consolidated under a new State Economic and Trade Commission, alongside macro-planning authorities	- Creation of the National Development and Reform Commission (NDRC) to unify planning and industrial strategy - Establishment of the State-owned Assets Supervision and Administration Commission (SASAC) to manage state capital - Creation of the Ministry of Industry and Information Technology (MIIT) to coordinate manufacturing and technology upgrading	- Party leadership elevated above economic agencies - Leading small groups upgraded into high-level Party commissions - Strategic oversight of science, technology, and finance centralized - Applied industrial and technology functions consolidated under MIIT, while the science ministry focuses on basic research and system design
Key initiatives	- First modern industrial policy catalogues - Policy bank credit and SOE restructuring (“grasp the large, release the small”)	- Medium- and Long-Term Science and Technology Plan (2006–2020) - Strategic Emerging Industries (SEI) Initiative (2008)	- Made in China 2025 - National Integrated Circuit Fund (“Big Fund”) - Little Giants program and mission-oriented “challenge” mechanisms
Overall effect	Lays groundwork for more coherent industrial policy by combining planning and restructuring.	A more coordinated state apparatus emerges, linking planning, ownership, and sectoral guidance.	Concentrates authority in the Party, enabling tighter alignment of policy instruments with national priorities.

SOURCE: Features information from State Council of the People's Republic of China, “China Releases Plan on Reforming Party and State Institutions,” March 16, 2023; and Naughton, 2021.

NOTE: Reforms summarized here were not necessarily designed primarily to implement industrial policy. The table highlights how such reforms cumulatively expanded the Party-state's later capacity to steer industrial outcomes.

Reforms in 1998 broke with the fragmented ministerial system. Folding the old industrial ministries into a unified macro-guidance body paved the way for “grasp the large, release the small” (抓大放小), China's first systematic effort to cultivate national champions and rationalize state ownership. In 2003, Beijing merged the old planning commission and the State Economic and Trade Commission into the NDRC. China's economic policy apparatus became sufficiently centralized to steer the economy through industrial policy.⁴ It enabled such integrated initiatives as the MLP (2006–2020), China's first comprehensive roadmap for major technological breakthroughs.⁵

talism with Chinese Characteristics, that rather than a statist shift away from market reform (favoring SOEs over township and village enterprises), in the 1990s, it was the Party-state bureaucracy reconsolidating (Yasheng Huang, *Capitalism with Chinese Characteristics: Entrepreneurship and the State*, Cambridge University Press, 2008). See also Sebastian Heilmann and Lea Shih, *The Rise of Industrial Policy in China, 1978–2012*, Harvard-Yenching Institute, 2013.

⁴ John Minnich, *Scaling the Commanding Heights: The Logic of Technology Transfer Policy in Rising China*, MIT Political Science Department, Research Paper No. 2023-2, June 29, 2023.

⁵ On the drafting of the plan, see Tai Ming Cheung, Thomas Mahnken, Deborah Seligsohn, Kevin Pollpeter, Eric Anderson, and Fan Yang, *Planning for Innovation: Understanding China's Plans for Technological, Energy, Industrial, and Defense Development*, report prepared for the U.S.-China Economic and Security Review Commission, University of California Institute on Global Conflict and Cooperation, July 28, 2016.

The 2008 establishment of the MIIT further professionalized industrial policy. MIIT became the hub for roadmaps for SEIs. Its expanding role, culminating in its leadership of Made in China 2025, reflected Beijing's focus on coordinated upgrading and strategic self-reliance.⁶

Meanwhile, these central reforms cascaded downward. Provinces and cities built sprawling ecosystems of development zones, financing vehicles, and local bureaus that combined subsidies, investment initiatives, and industrial upgrading. This fueled rapid expansion but also created duplication, protectionism, and weak coordination.⁷

Two major Party and state institutional reform packages under Xi, released in March 2018 and March 2023, aimed to both rein in this sprawl and recenter authority within the Party.⁸ In 2018, the leadership elevated several top-level central leading groups to standing Party commissions with permanent offices, including the commissions for comprehensively deepening reform and for finance and economics, a structural change intended to strengthen centralized direction over economic strategy. In 2023, the leadership deepened this approach in domains central to techno-industrial policy by establishing a Central Financial Commission and a Central Science and Technology Commission and by aligning key regulators and ministries more directly with Party-led “top-level design” (顶层设计) and supervision.

The shift moved China from administrative consolidation toward what might be called “grand steerage”: unified Party direction of techno-industrial priorities across the state and market.⁹ Together, these reforms created a system capable of intervening at both macro and micro levels, from guiding national investment patterns to launching firm-level initiatives.

China's recent reforms reflect a deeper rewiring of organizational power. A system once defined by fragmented authority between territorial (kuai, 块) and functional (tiao, 条) lines shifted gradually under Hu and decisively under Xi toward vertical control in key domains.¹⁰ These domains include finance and financial regulation, S&T and innovation policy, and data and cybersecurity governance, in which upgraded Party commissions and their offices increasingly sit atop ministerial and regulatory systems. Xi's upgraded Party commissions have dedicated offices and more convening capacity, sitting atop increasingly verticalized bureaucratic channels.¹¹ Yet centralization is far from complete. Top-level guidance has strengthened significantly under Xi Jinping, but implementation remains regionally decentralized: Local development and reform commissions and industry departments still primarily answer to local governments rather than to a unified vertical hierarchy. With the UNM initiative elevated as a core goal of the 15th FYP, Beijing is clearly trying to push implementation toward more nationally standardized rules and enforcement.

⁶ Yukyung Yeo, “Remaking the Chinese State and the Nature of Economic Governance? The Early Appraisal of the 2008 ‘Super-Ministry’ Reform,” *Journal of Contemporary China*, Vol. 18, No. 62, 2009; Heilmann and Shih, 2013.

⁷ Jonathon P. Sine, “The Rise and Fall of LGFVs,” *Cogitations*, Substack, April 2, 2024; Sebastian Heilmann, Lea Shih, and Andreas Hofem, “National Planning and Local Technology Zones: Experimental Governance in China's Torch Programme,” *China Quarterly*, Vol. 216, December 2013.

⁸ “CPC Releases Plan on Deepening Reform of Party and State Institutions,” Xinhua, March 21, 2018; “China Releases Plan on Reforming Party and State Institutions,” Xinhua, March 16, 2023.

⁹ Barry Naughton, Tai Ming Cheung, Siwen Xiao, Yaosheng Xu, and Yujing Yang, *Reorganization of China's Science and Technology System*, University of California Institute on Global Conflict and Cooperation, Working Paper No. 10, July 2023.

¹⁰ Pei Zhang and Jiacheng Zhao, “Policy Experimentation Trapped in Tiao-Driven Authoritarian Environmentalism: A Case Study of National Park Policy in China,” *China Quarterly*, Vol. 263, September 2025; Andrew C. Mertha, “China's ‘Soft Centralization’: Shifting Tiao/Kuai Authority Relations,” *China Quarterly*, Vol. 184, December 2005; Kenneth Lieberthal and Michel Oksenberg, *Policy Making in China: Leaders, Structures, and Processes*, Princeton University Press, 1988.

¹¹ Wendy Leutert and Sarah Eaton, “Deepening Not Departure: Xi Jinping's Governance of China's State-Owned Economy,” *China Quarterly*, Vol. 248, Suppl. S1, November 2021; Naughton et al., 2023.

This top-down reliance on decentralized execution creates a familiar dilemma. Xi's centralized techno-industrial architecture depends on local officials whose incentives often pull in other directions. Although authority over strategy has been recentered at the top, responsibility for execution still rests with local governments whose promotion incentives remain closely tied to visible economic activity. China's decentralized production ecosystem still generates duplication and waste, fueling what Chinese policy discourse now calls "involution" (内卷), a cycle of intensifying competition that erodes margins and quality without producing genuine innovation, while adding to local debt burdens.¹²

Beijing has tried to adjust the cadre evaluation system to curb these distortions. Economic growth remains the dominant driver of promotions, but Xi has added broader performance indicators.¹³ So far, their effect on career outcomes is limited, and certain industrial policy-linked targets, such as rapid growth in "new quality productive forces," may actually spur more duplicative investment. Xi obliquely complained about such waste at the 2025 Central Urban Work Conference: "When launching projects, people often mention a few things: artificial intelligence, computing power, new-energy vehicles—are all provinces now developing these? Should every province in the country be doing so?"¹⁴

A parallel strategy has been expanding auditing, inspection, and disciplinary oversight. The 2018 creation of the National Supervisory Commission and the verticalization of the Central Commission for Discipline Inspection produced a unified Party-state disciplinary system. The new Central Auditing Commission strengthened top-level control over state auditing while Xi also broadened the deployment of central inspection teams and pushed digitalized supervision to tighten oversight. In market governance, the establishment of the State Administration for Market Regulation centralized competition and regulatory enforcement, including the Fair Competition Review System (公平竞争审查制度) and the broader National Unified Market (全国统一大市场) framework overseen by the NDRC.¹⁵

Taken together, three decades of reforms have produced a highly centralized national techno-industrial policy apparatus layered atop a still-decentralized economic system. Xi's organizational project aims to impose greater discipline and strategic coherence on this landscape by strengthening the Party center's steering capacity and constraining debt-driven, uncoordinated local initiatives—while relying on local adaptability to carry industrial policy into practice.

¹² *Involution* denotes a pattern in which intensifying internal effort and competition produce diminishing or no marginal returns. In current Chinese industrial-policy discourse, it is used to describe subsidy races, local protectionism, copycat production, and price wars. See Jin Sheping, "Achieving High-Quality Development by Breaking the 'Involution' Competition," *People's Daily*, June 29, 2025.

¹³ Lizzi C. Lee and Shengyu Wang, "Xi Just Can't Shake GDP Worship," *Foreign Policy*, March 18, 2026.

¹⁴ This assessment is based on recent evidence on cadre-promotion incentives and policy experimentation. For a meta-analysis of cadre evaluation and performance, see Lei Liu, Qianhui Li, May Chu, Wenbo Guo, and Lixu Tang, "Beyond Economic Performance: A Meta-Analytic Study of the Relationship Between Performance Indicators and Cadre Promotion in China," *Policy Studies*, May 13, 2025. For a discussion on GDP targeting, see Andrew Batson, "Why GDP Growth Targets Are Underrated," *Tangled Woof* blog, April 18, 2022. For Xi's speech, see "Xi Addresses Central Urban Work Conference, Listing Priorities for Urban Development," *Xinhua*, July 17, 2025.

¹⁵ The Fair Competition Review System (公平竞争审查制度) remains horizontally organized. An intensification of Beijing's pursuit of its UNM objective would require verticalizing the State Administration for Market Regulation's chain of authority to bring enforcement directly under central control rather than remaining the prerogative of local officials.

Sharpening the Tools of Techno-Industrial Policy

Under Xi, China's industrial policy toolkit has grown larger, more structured, more financialized, and more explicitly tied to security and self-reliance. Earlier approaches leaned on broad subsidies, cheap credit, and underpriced inputs. Those instruments remain, but they have been reshaped by two forces: tighter fiscal constraints and Xi's political preference for mechanisms that bind firms more closely to state priorities. Nominally consistent with the 2013 Third Plenum pledge to give markets a "decisive role," many tools now operate through market-like channels, such as equity investment funds, tax deductions, procurement mandates, and capital market reforms that harness private actors. But these channels are increasingly governed by centralized regulatory frameworks and Party guidance rather than market signals or local discretion, raising the question of whether market mechanisms are being empowered or instrumentalized.¹

The result is a toolkit organized around five channels: fiscal levers (subsidies, tax incentives, R&D support, procurement, talent programs), financial levers (state equity, targeted credit, capital markets), real economy levers (market inputs, special zones, FDI measures), Party-state corporate relations (SOE privileges, Party-state regulation, private-sector mobilization), and overseas initiatives (BRI, financing, industrial parks, standards, outbound investment) (see Table 4.1). These channels are analytically distinct but operationally intertwined. Across these domains, broad, sometimes ad hoc support has given way to instruments that are more selective and centrally influenced. For firms, the incentives are to signal alignment with strategic sectors and Party guidance to gain access to resources.²

Coordination of these five channels is driven less by a single industrial-policy agency than by Party-led priority setting and planning, which are then translated into policy packages that combine fiscal measures, financial guidance, regulation, and implementation mandates for ministries and localities. Recent institutional reforms have explicitly strengthened centralized leadership over S&T policy and Party oversight of the financial system, reinforcing Beijing's ability to align innovation, finance, and industrial objectives.³

In practice, coordination is strongest at the level of agenda-setting and mobilization and weaker at the level of execution. Instruments often stack, with the same sector supported through multiple channels, which can accelerate scaling but lead to inconsistent incentives or center-local slippage. Leadership increasingly acknowledges this coordination problem and calls for tighter cross-policy alignment, including stronger interdepartmental communication and mechanisms that use limited fiscal funds to mobilize more financial resources.⁴

¹ "Communiqué of the Third Plenary Session of the 18th Central Committee of the Communist Party of China," Xinhua, November 12, 2013.

² Franklin Allen, Junhui Cai, Xian Gu, Jun "QJ" Qian, Linda Zhao, and Wu Zhu, "Centralization or Decentralization? The Evolution of State-Ownership in China," SSRN, October 20, 2024.

³ "China Releases Plan on Reforming Party and State Institutions," Xinhua, March 16, 2023.

⁴ See, for example, NDRC, "Report on the Implementation of the 2024 National Economic and Social Development Plan and the Draft 2025 National Economic and Social Development Plan," submitted to the Third Session of the Fourteenth National

TABLE 4.1

Evolution of China's Techno-Industrial Policy Instruments in the Xi Era

Instrument	Pre-Xi (Before 2012)	Xi Era (2012–Present)
Fiscal		
Direct subsidies	Broad and often ad hoc; discretionary local grants and SOE loss support.	More targeted and program-linked; increasingly constrained by fiscal pressure and concentrated in priority sectors.
Corporate tax breaks	Narrow; limited to selective preferential rates.	Broadened and systematized; increasingly substituted for direct spending as primary fiscal channel.
S&T and R&D support	Targeted but limited in scale; R&D deduction modest; small government S&T budget.	Composition shift: direct government funding flat while tax-based incentives (R&D super deductions) now dominant.
Government procurement	Fragmented and local; indigenous innovation catalogues rolled back under WTO pressure.	Formalized and demand-shaping; “secure and controllable” mandates and “first-set/first-batch” rules, but falling as share of GDP.
Talent programs	Overseas recruitment focus; fragmented across ministries and localities.	Consolidated and security-linked; selection steered to strategic fields; career incentives standardized.
Financial		
Government equity	Limited and experimental; early-stage state investment tools.	Massive expansion of guidance funds, then centralization; corruption and short-termism exposed limits of local fund model.
Credit support	Broad, state-led lending; weak targeting and bias toward SOEs.	Targeted structural tools (PBOC relending); industrial loans surged post-COVID-19 but are now tightening as overcapacity concerns grow.
Capital markets	Secondary financing channel; limited role in industrial policy.	Elevated as policy tool: registration-based IPOs for tech (STAR Market, BSE); S&T bonds; golden shares embed state oversight.
Real economy		
Below-market inputs	Widespread input subsidies; cost-based industrial support.	Continues but more disciplined; discounts channeled through provincial catalogues to priority sectors.
Industrial parks and special zones	Export-oriented and regionally driven; loosely coordinated.	Reoriented toward strategic industries and innovation; more integrated with national priorities.
Trade and foreign investment measures	Broad liberalization; emphasis on openness and export growth.	Selective opening: manufacturing caps lifted, services still restricted; “high-standard opening” serves domestic capacity-building.

Table 4.1—Continued

Instrument	Pre-Xi (Before 2012)	Xi Era (2012–Present)
Party-state corporate relations		
State-owned enterprise advantages	Privileged but weakly disciplined; soft budget constraints.	Consolidated into national champions; stronger discipline and mission orientation but still privileged in credit and procurement.
Party embedding and regulatory steering	Present but uneven; limited operational influence.	Institutionalized: Party cells in corporate governance, golden shares in platforms, antitrust and data regulation as discipline tools.
Private-sector harnessing and reassurance	Fragmented support; bias toward state firms in key sectors.	Systematic mobilization via Little Giant and Single Champion designations; conditional support paired with tightened Party oversight.
Overseas		
Belt and Road Initiative	Fragmented outward investment; limited coordination.	State-coordinated external strategy integrating finance, tech exports, and standards; post-2019 shift to smaller, targeted projects.
Financial and institutional infrastructure	Expanding but ad hoc; limited policy coordination.	Lending down after 2021; composition pivoting to green, high-tech, and manufacturing sectors.
Overseas industrial parks	Limited scale and coordination; trade facilitation focus.	Expanded to 100+ zones in 49 countries; aligned with supply chain security and industrial transfer.
Standards and rule-making power	Reactive adoption of international standards; limited leadership.	Proactive standard-setting tied to BRI projects; expanded leadership in ISO/IEC and ITU.

NOTE: Categories are used as a descriptive framework. Many instruments operate across channels and are coordinated through planning and joint policy packages rather than within a single unified program. Quoted policy terms are defined in the appendix and cited in the corresponding chapter discussions. BSE = Beijing Stock Exchange; COVID-19 = coronavirus disease 2019; IPO = initial public offering; ISO/IEC = International Organization for Standardization/International Electrotechnical Commission; ITU = International Telecommunication Union; PBOC = People's Bank of China; STAR Market = Shanghai Science and Technology Innovation Board.

Measuring China's spending on industrial policy is notoriously difficult. Unlike most OECD members, China does not publish detailed budgetary breakdowns by economic function or disclose the value of subsidies in its WTO notifications. Available series from the Ministry of Finance (MOF) (budgetary S&T outlays), the National Bureau of Statistics (R&D expenditures by source of funds), and firm-level reports (subsidies disclosed in listed company filings) each capture a different slice of support and are not harmonized. Many important channels, such as below-market credit from state banks, government procurement preferences, and regulatory advantages for SOEs, are largely unquantifiable in aggregate, at least with any precision. The World Bank, IMF, and OECD have similarly stressed that China's subsidy and fiscal data lack transparency and comparability with international standards.⁵

People's Congress, March 5, 2025b, which emphasizes strengthening macro-policy coordination, improving interdepartmental communication, ensuring that policies "form synergy," and combining fiscal and financial tools to amplify policy impact.

⁵ Official assessments by major international institutions have emphasized that China's subsidy and fiscal data are difficult to measure and fall short of international transparency standards. For example, the OECD has urged more detailed government notifications and corporate disclosures in order to improve monitoring of China's industrial subsidies; the IMF has noted that "official information on the size of [industrial policies] support or the effects of such policies is for the most part unavailable"; and the World Bank's China Economic Update series frequently highlights fiscal support measures while

According to a Center for Strategic and International Studies (CSIS) report, China's industrial policy spending likely exceeded 1.7 percent of GDP in 2019 (even on a conservative basis), more than that of the United States (0.39 percent), Japan (0.50 percent), South Korea (0.67 percent), and other economies in the study.⁶ A recent IMF study using firm-level and land-registry data estimated that China's industrial policy spending reached roughly 4 percent of GDP in 2023. The IMF's higher estimate reflects a broader valuation of implicit subsidies, such as below-market credit and land, as well as an extrapolation of listed-firm subsidy rates to the wider corporate sector.⁷

What can be said with confidence is that, in absolute terms, state support has risen as China's economy and financial capacity have expanded. The more telling metric is the share of GDP or of total spending, which shows the changing composition of support: Direct subsidies are plateauing, while tax incentives and financialized tools, such as guidance funds and procurement, have grown in relative importance.

acknowledging limited disclosure of their scale. See OECD, "The State of Play of Industrial Subsidies as of 2023," June 23, 2025e; Garcia-Macia, Kothari, and Tao, 2025; and World Bank, *China Economic Update: June 2025*, 2025.

⁶ Gerard DiPippo, Ilaria Mazzocco, and Scott Kennedy, *Red Ink: Estimating Chinese Industrial Policy Spending in Comparative Perspective*, Center for Strategic and International Studies, May 2022.

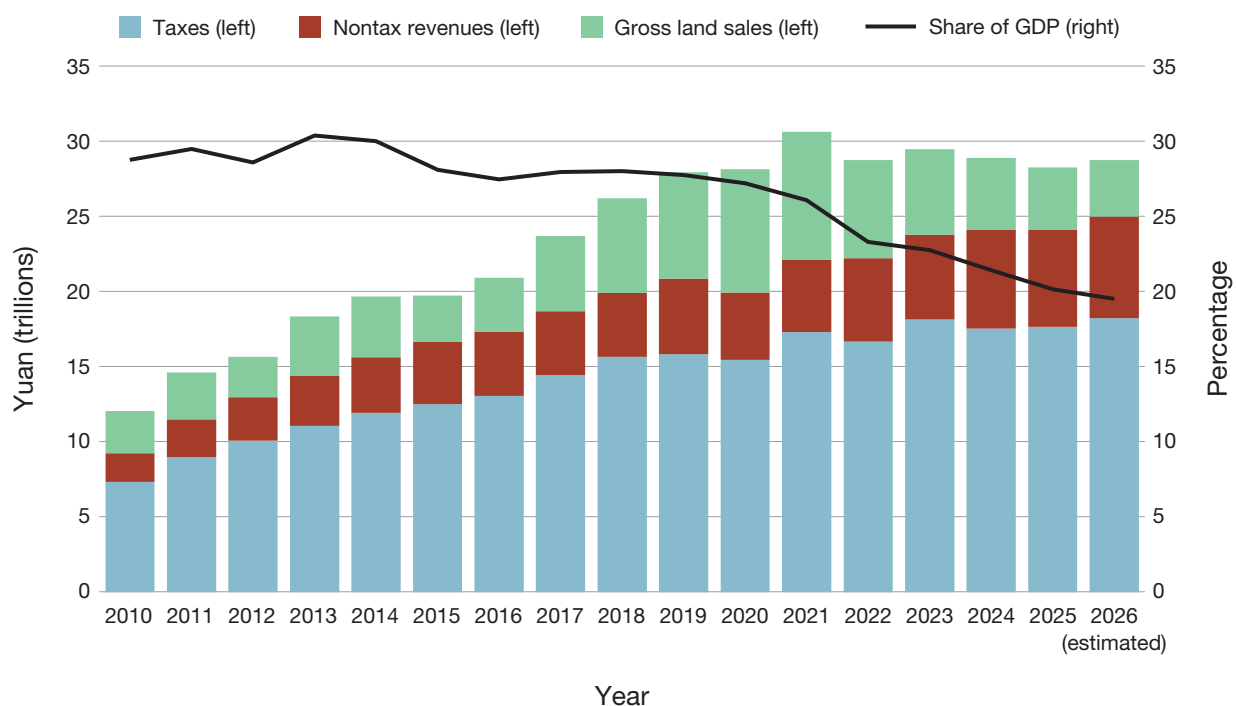
⁷ Garcia-Macia, Kothari, and Tao, 2025. Both the CSIS and IMF studies quantify only a subset of measurable support. The CSIS study includes nine instruments—direct subsidies, R&D tax incentives, other tax incentives, below-market credit and land, government R&D funding, state investment funds, SOE net payables, and debt-equity swaps—while the IMF study focuses on four: cash subsidies, tax benefits, subsidized credit, and discounted land. Both exclude large areas of industrial policy that cannot be easily quantified in terms of subsidies, such as government procurement preferences, regulatory protection, and localization mandates. The IMF estimate is higher mainly because it values implicit subsidies and extrapolates more broadly than the CSIS study. The latter is based on observable budgetary outlays to listed and state firms, excluding implicit support. The IMF instead measures the fiscal-equivalent cost of cash, tax, credit, and land support using firm- and land-level microdata through 2023. Two assumptions drive most of the gap: (1) assigning market-value estimates to below-market credit and land and (2) extrapolating listed-firm subsidy rates to unlisted firms, assuming comparable access to support across the corporate sector. However, a recent study using both the Annual Survey of Industrial Firms and listed-firm databases during their 2007–2013 overlap period found that listed firms accounted for only approximately 30 to 50 percent of subsidies received by all industrial firms, suggesting that listed firms may not be representative of the wider corporate sector and that extrapolating their subsidy rates, especially as estimated by the IMF, could overstate aggregate support. See Wenyin Cheng, Tao Liang, Bo Meng, and Hongyong Zhang, "Industrial Subsidies Along Domestic Value Chains and Their Impacts on China's Exports," Research Institute of Economy, Trade and Industry, RIETI Discussion Paper Series 25-E-119, December 2025.

Fiscal Constraints Contributing to Changes in Industrial Policy

Although China probably spends more on industrial policy as a share of its GDP than any other major economy, its ability to do so is not limitless. China is already facing increasing fiscal pressures; fiscal revenues have flatlined in nominal terms and declined relative to the economy since 2020 (Figure 5.1). Meanwhile, China's consolidated budget deficit climbed to 9 percent of GDP in 2025 (Figure 5.2). The Chinese government, especially at the local level, is reducing fiscal expenditures, including those for industrial policy, out of necessity.

Beijing can still mobilize large-scale borrowing for industrial policy. The central government and policy banks retain substantial capacity because Beijing can expand bond issuance, and national commercial or policy banks raise trillions of yuan annually in quasi-sovereign debt to finance strategic sectors. By contrast, local governments are squeezed; bond quotas are capped, land sale revenue has fallen, and off-balance sheet LGFV borrowing is under regulatory pressure. The most important immediate trigger was Beijing's "three

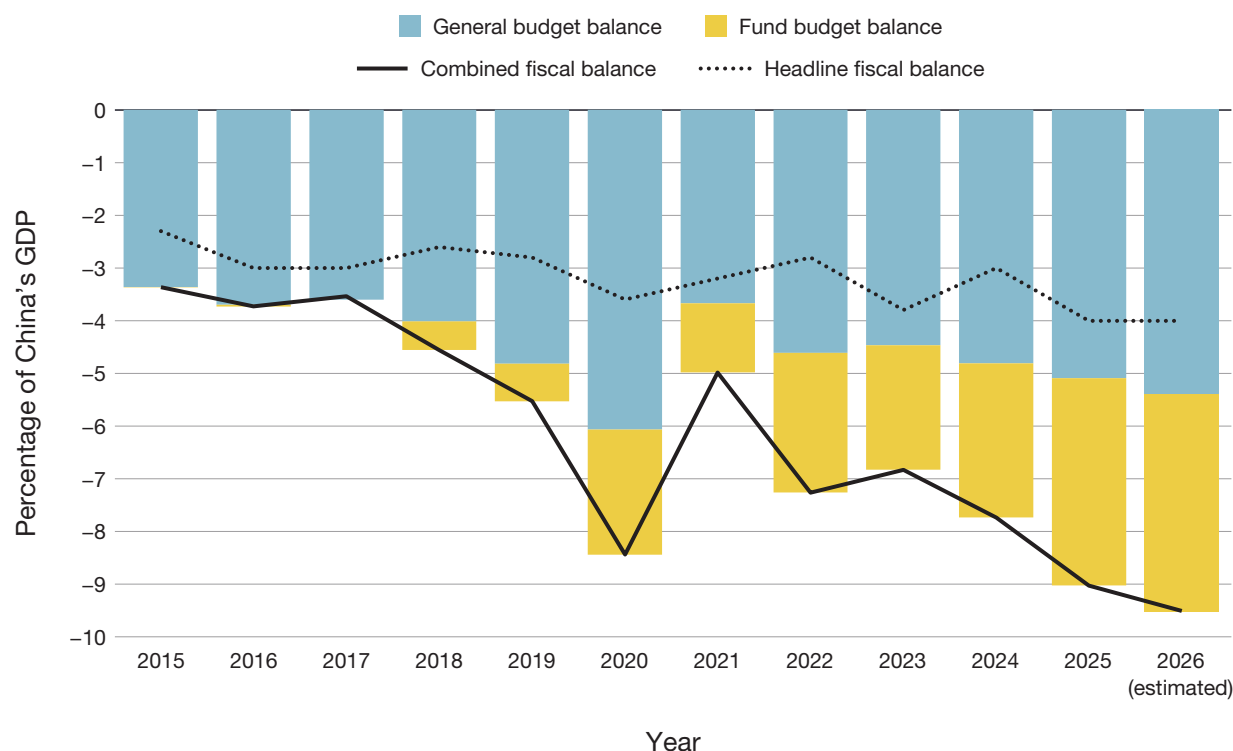
FIGURE 5.1
Stagnant Fiscal Revenues Strain Resources for Industrial Policy Spending



SOURCE: Features data from the MOF and National Bureau of Statistics via CEIC, undated.

NOTE: This figure includes General Public Budget, Government Fund Budget, and State Capital Operations Budget data. Estimates for 2026 are based on official budget targets and a 10-percent decline in land sales.

FIGURE 5.2
China's Consolidated Fiscal Deficit Has Increased



SOURCES: Features data from the MOF and National Bureau of Statistics via CEIC, undated; and Li Qiang, "Report on the Work of the Government," delivered at the Fourth Session of the 14th National People's Congress, State Council of the People's Republic of China, March 5, 2026.

NOTE: Estimates for 2026 are based on official budget targets.

red lines" policy introduced in 2020, which sharply curtailed developer borrowing and land sales. Because land sales to developers had long been a major source of revenue for local governments, the resulting property downturn led directly to a collapse in land sale proceeds and fiscal capacity at the local level. A correction was likely inevitable because of the structural exhaustion of a land finance model that relied on perpetually rising property values to fund local growth. Therefore, localities are less able to cofinance subsidies or guidance funds.

Policy is shifting as a result. Centrally directed (or funded) credit and investments remain important. Meanwhile, traditional cash subsidies and local incentives play a reduced role. Government procurement has fallen relative to GDP. Instead, Beijing is leaning on lower-cost directives and mandates (including procurement requirements, regulatory obligations, and selective credit guidance) that require less fiscal outlay but effectively spread the costs of industrial policy across firms and institutions required to comply.

Dialing Back Cash Subsidies

These fiscal constraints have had concrete implications for how Beijing pursues industrial policy. Among the most visible changes has been a retrenchment in the use of direct fiscal subsidies. Those on-budget cash grants to firms were once a hallmark of local industrial promotion. In the early 2010s, such subsidies were widespread, discretionary, and often tied to tax rebates or investment commitments. As fiscal pressures intensified and regulatory scrutiny increased, Beijing moved to rein in these practices. It has narrowed

the scope of permissible subsidies and shifted toward more centralized allocation mechanisms governed by formal eligibility criteria and program designations.

Early in the Xi era, the State Council's 2014 Circular 62 ordered a nationwide cleanup of ad hoc rebates, "first collect then refund" (先征后返) arrangements, and fiscal rewards tied to tax payments, requiring that unlawful or WTO-inconsistent subsidies be canceled unless centrally approved.¹ By 2019, the Government Investment Regulations codified which forms of direct fiscal support remained permissible, such as capital injections, investment subsidies, and interest subsidies, under tighter budget rules.²

A series of accompanying reforms further narrowed subsidy channels. The Fair Competition Review System (公平竞争审查制度) required ex ante screening of preferential fiscal support.³ The 2018 rollout of comprehensive budget performance management brought enterprise subsidies under feasibility checks and ex post audits.⁴ Crackdowns on local government debt and hidden liabilities curtailed rebate-style investment incentives,⁵ while reforms consolidated dozens of open-ended grant programs into competitive, audited, project-based awards.⁶ In March 2026, the State Council announced that it would introduce a national "negative list" for local fiscal subsidies, explicitly defining categories in which local governments are prohibited from offering them, in the most direct attempt yet to shut down the subsidy races that have fueled overcapacity and price wars in such sectors as electric vehicles (EVs) and solar energy.⁷

The best available data on China's direct fiscal subsidies come from the firms that receive them. Many beneficiaries—especially large, state-linked, or strategically important enterprises—are listed on China's stock exchanges and are required to disclose government grants or subsidies in the "non-operating income" or "other income" sections of their annual reports. Using the Wind Financial Terminal (WFT),⁸ we aggregated reported subsidies for all nonfinancial A-share listed firms to approximate total subsidy flows at the firm level. However, these data only cover listed firms, which accounted for roughly 9 percent of China's value added in 2023, or, if counting only manufacturing firms, roughly 25 percent of manufacturing rev-

¹ State Council of the People's Republic of China, "Notice on Cleaning Up and Standardizing Preferential Policies on Taxation, etc.," State Council Document No. 62 of 2014, December 9, 2014b.

² State Council of the People's Republic of China, "Government Investment Regulations," State Council Decree No. 712, April 14, 2019.

³ State Council of the People's Republic of China, "Opinions on Establishing a Fair Competition Review System in the Construction of the Market System," State Council Document No. 34 of 2016, June 14, 2016; State Administration for Market Regulation, National Development and Reform Commission, Ministry of Finance, Ministry of Commerce, and Ministry of Justice, "Detailed Rules for the Implementation of the Fair Competition Review System," State Administration for Market Regulation Anti-Monopoly Regulations Document No. 2 of 2021, June 29, 2021.

⁴ Communist Party of China Central Committee and State Council, "Opinions on Comprehensively Implementing Budget Performance Management," September 1, 2018.

⁵ State Council of the People's Republic of China, "Opinions of the State Council on Strengthening the Management of Local Government Debt," State Council Document No. 43 of 2014, October 2, 2014a.

⁶ Ministry of Finance and Ministry of Science and Technology of the People's Republic of China, "Notice on the Issuance of the 'Measures for the Administration of Funds for the National Key R&D Program,'" Finance, Science and Education Document No. 113 of 2016, March 3, 2017; Ministry of Science and Technology, Ministry of Finance, and National Development and Reform Commission, "Notice on the Issuance of the 'Performance Evaluation Standards for Central Government Science and Technology Plans (Special Projects, Funds, etc.) (Trial Implementation),' " State Science and Technology Commission Document No. 165 of 2020, June 19, 2020.

⁷ Ethan Wang and Kevin Yao, "China to Tighten Oversight of Local Government Subsidies," Reuters, March 13, 2026.

⁸ Wind, Wind Financial Terminal, data platform, undated.

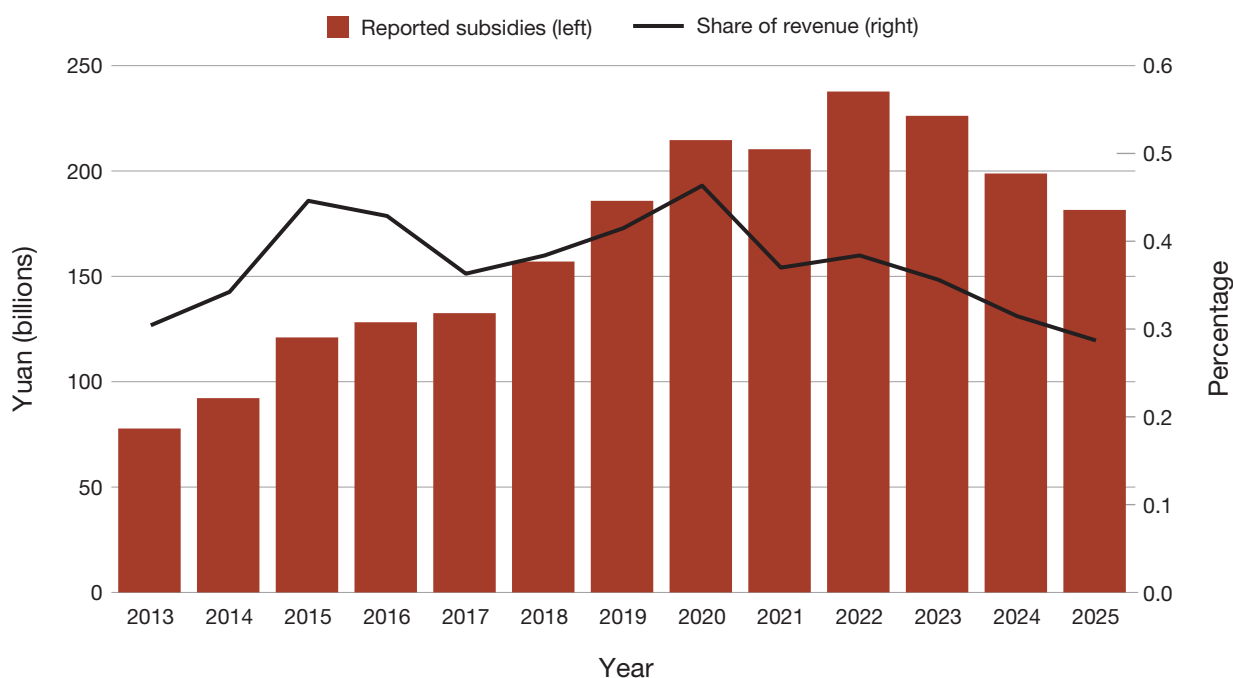
enues as reported by the official industrial survey.⁹ Total direct fiscal subsidies to all firms in China are much larger, but listed firms probably receive a disproportionate share given their size and higher profile.

The data show that total disclosed subsidies rose steadily from about ¥80 billion in 2013 to a peak of about ¥240 billion in 2022, before declining to about ¥182 billion in 2025 (Figure 5.3). When scaled by firms' revenue, subsidy intensity peaked at roughly 0.46 percent in 2020 and has since trended down to roughly 0.29 percent as of 2025. From 2020 to 2024, China's nominal GDP cumulatively grew by 30 percent. China's direct subsidies are likely to be shrinking in both absolute and relative terms, and fiscal constraints are a significant reason.

At the sectoral level, the data indicate that subsidies declined the most for industrial and IT firms in 2024 (Figure 5.4). These two sectors had experienced the largest increases in subsidies after 2018, while subsidies to other industries were generally steadier, suggesting that industrial and IT firms were the obvious targets for fiscal retrenchment. The drop in industrial sector subsidies in 2024 could also reflect the Chinese government's awareness that manufacturing investment had surged after the COVID-19 pandemic and that excess capacity and competition problems were emerging.

In recent years, subsidy intensity relative to firms' net profits has been highest in industries facing either financial distress or strategic-priority status, based on data from listed firms (Figure 5.5). The first group includes agriculture, retail, and real estate and consists of sectors with low profitability that were further hit by pandemic disruptions and weak consumer demand. In these cases, subsidies function essentially as fiscal support to stabilize employment and prevent broad losses. The second group (consisting of electronics and semiconductors, computers and software, and automobiles) maintains healthier margins yet continues to receive sizable subsidies. These industries are central to Beijing's drive for technological self-reliance and reflect the state's willingness to sustain capital- and R&D-intensive sectors through direct transfers. At the

FIGURE 5.3
Subsidies to Nonfinancial Listed Chinese Firms Are Falling



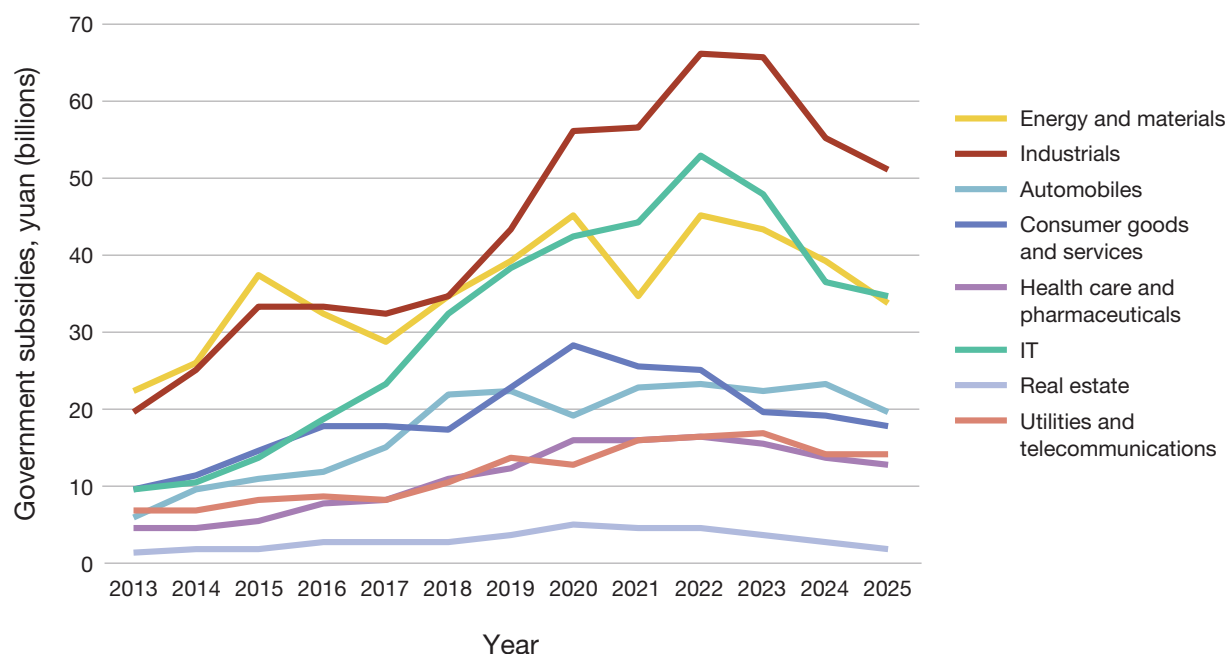
SOURCE: Authors' calculations using data from company disclosures compiled in WFT.

NOTE: This figure is based on reported disclosures from 5,377 firms listed on China's stock markets.

⁹ Garcia-Macia, Kothari, and Tao, 2025.

FIGURE 5.4

China's Direct Subsidies to Listed Industrial and Technology Firms Have Been Cut Back



SOURCE: Authors' calculations using data from company disclosures compiled in WFT.

NOTE: This figure is based on WFT Industry (2024) classifications for 5,377 firms listed on China's stock markets.

other end of the spectrum, low-value-added industries—such as food and beverage and upstream resource sectors (e.g., coal, mining, and metals)—are far less reliant on direct fiscal support because their profitability depends more on market cycles than on policy funding.

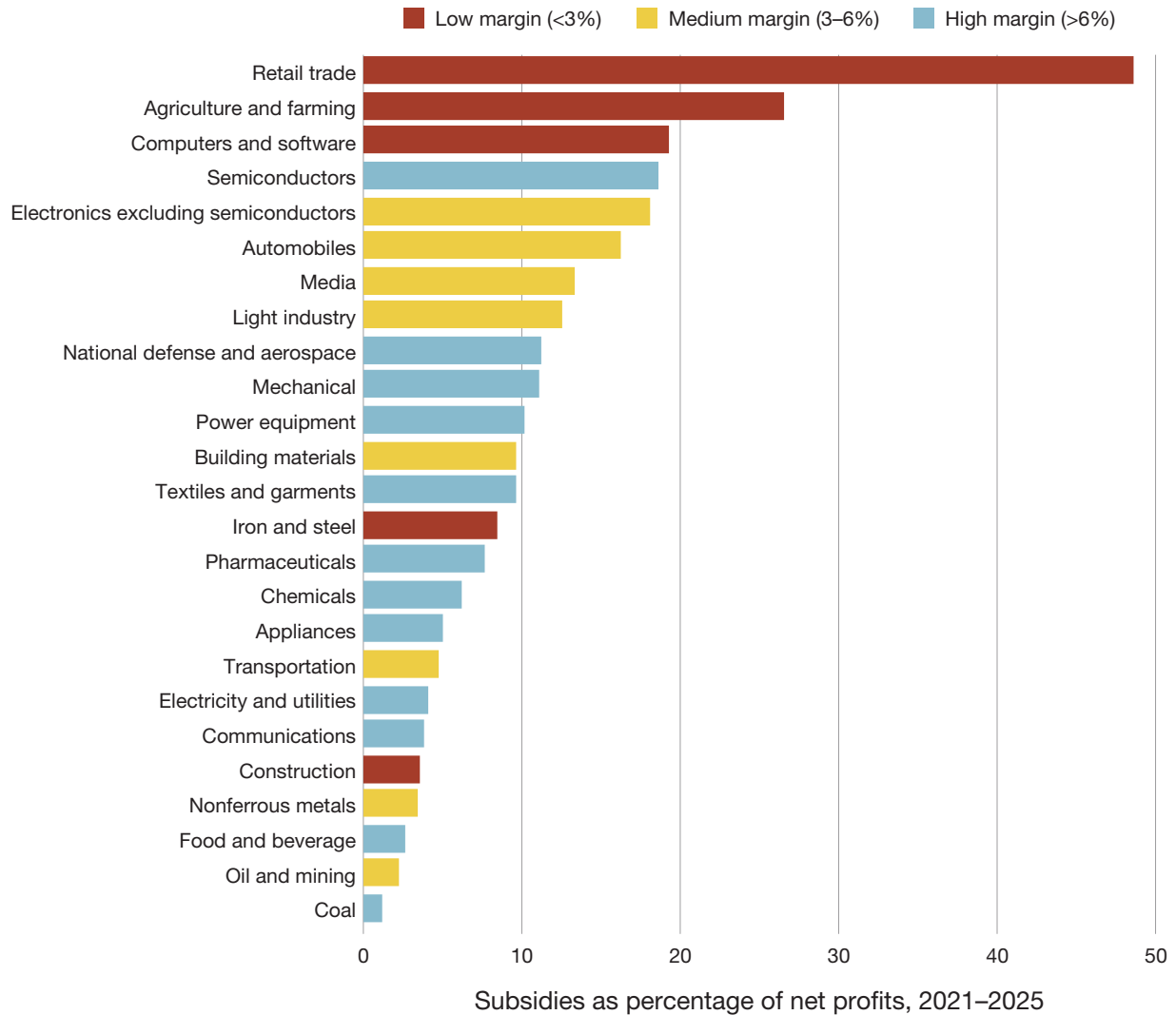
Macroeconomic policy and fiscally funded industrial policy have merged with the trade-in and equipment renewal “Two New” (两新) programs. These are state-funded subsidies that encourage households and firms to replace old consumer goods and industrial equipment with newer models, boosting domestic demand while supporting targeted industries. Officially framed as consumption support amid weak economic growth and the property sector downturn, these subsidies are designed to boost demand for specific goods and support those industries, especially new-energy vehicles, efficient appliances, digital devices, and equipment tied to the property sector.¹⁰ In effect, these trade-ins serve as a limited, demand-side supplement to industrial policy. Because these subsidies are paid to consumers at the point of sale and reimbursed to retailers or dealers by local governments, they are likely not captured in the firm-level subsidy data from WFT discussed above. They are essentially indirect fiscal subsidies to producers.

Central government funding for consumer goods trade-ins doubled from about ¥150 billion in 2024 to ¥300 billion in 2025, financed through ultra-long special treasury bonds.¹¹ Funds are distributed to provinces on a quota basis and co-shared at roughly a nine-to-one central-to-local ratio, although this varies by region.¹² Fiscal constraints have, however, affected implementation. When central funds run out, local

¹⁰ National Development and Reform Commission and Ministry of Finance, “Notice on Strengthening and Expanding the Implementation of Large-Scale Equipment Renewal and Consumer Goods Trade-In Policies in 2025,” National Development and Reform Commission Circular No. 13 of 2025, January 8, 2025.

¹¹ Casey Hall, “How China Is Doubling Down on Trade-Ins to Boost Sluggish Consumption,” Reuters, March 17, 2025.

¹² MOF, “Transcript of the Press Conference on Fiscal Support for Promoting High-Quality Development,” July 31, 2024a.

FIGURE 5.5**Government Subsidies to Listed Firms Support Struggling and Strategic Industries**

SOURCE: Authors' calculations using data from company disclosures compiled in WFT.

NOTE: This figure is based on reported disclosures from 5,377 firms listed on China's stock markets.

governments must cover extra subsidies, which has led to temporary program suspensions in some cities.¹³ The 2026 Government Work Report earmarked ¥250 billion for the program's third year, a reduction from 2025, while expanding its scope to include digital and smart products, such as smartphones, tablets, and AI-enabled products.¹⁴

¹³ Casey Hall and Zhang Yan, "Some Chinese Cities Pause Car-Buying Subsidies as Funds Run Out," Reuters, June 18, 2025; "Guangxi: Consumer Goods Trade-In Subsidy Program to Be Suspended, Resumption Date Yet to Be Determined," *Daily Economic News*, June 19, 2025; "The Autonomous Region's 2025 Consumer Goods Trade-In Subsidy Program Will Be Suspended on June 15," *Tianshan Net-Xinjiang Daily*, June 12, 2025.

¹⁴ Li, 2026.

Increased Tax Support for Favored Industries

Xi-era fiscal policy has leaned on tax relief to support favored industries. The authorities have rolled out large-scale value-added tax (VAT) cuts and refunds, preferential corporate income tax rates, and sector-specific exemptions that lower costs in priority industries.¹⁵ The 2019 VAT overhaul cut the headline manufacturing rate from 16 percent to 13 percent, and a 2022 program refunded ¥2.46 trillion in accumulated input VAT credits in a single year, equal to roughly 2 percent of GDP.¹⁶ “High and new technology enterprises” (高新技术企业) pay a 15-percent corporate income tax rate, compared with 25 percent for other firms, while integrated circuit and software firms receive multiyear tax holidays.¹⁷ Other measures include accelerated depreciation and one-off expensing of new equipment through 2027, plus extended exemptions on new-energy vehicle purchase taxes.¹⁸

Using the IMF’s methodology for estimating below-statutory tax support, we calculated that corporate tax breaks for nonfinancial listed firms have risen sharply during the Xi era, peaking in 2022 (Figure 5.6). The approach measures the tax breaks (or “tax subsidies”) as the difference between the statutory 25-percent corporate income tax rate and firms’ effective rates, multiplied by pretax profits.¹⁹ Our estimates are based on reported financials from listed firms in WFT and show that aggregate tax benefits nearly tripled from 2013 to 2022, reaching roughly ¥367 billion in 2022, before easing to about ¥308 billion in 2025. As a share of revenue, tax breaks also declined after 2022, reflecting weaker profitability and reduced fiscal space for new incentives.

The distribution of tax breaks across industries suggests that relief has been concentrated in low-margin or strategic sectors (Figure 5.7). Agriculture, media, and IT firms received the largest effective tax savings relative to profits, while capital-intensive or rent-heavy sectors (such as real estate) show low or negative values, indicating effective tax rates of more than 25 percent. This likely reflects both tighter enforcement, including through tax digitization, and the exclusion of property-related income from preferential corporate income tax treatment.

S&T and R&D Support to Spur Innovation

China’s S&T budget is the most visible line item in the government accounts tied to techno-industrial policy. It captures an important sliver of China’s industrial policy spending, which funds much of the scientific and institutional base underpinning industrial upgrading. The S&T budget includes appropriations for basic and applied research, technology development, national labs and research institutes, and grant programs. The

¹⁵ Ministry of Finance, State Taxation Administration, National Development and Reform Commission, and Ministry of Industry and Information Technology, “Announcement on Corporate Income Tax Policies to Promote High-Quality Development of the Integrated Circuit Industry and the Software Industry,” Announcement No. 45 of 2020, December 11, 2020; “Tax Breaks to Lend IT Sector Helping Hand,” *China Daily*, December 19, 2020.

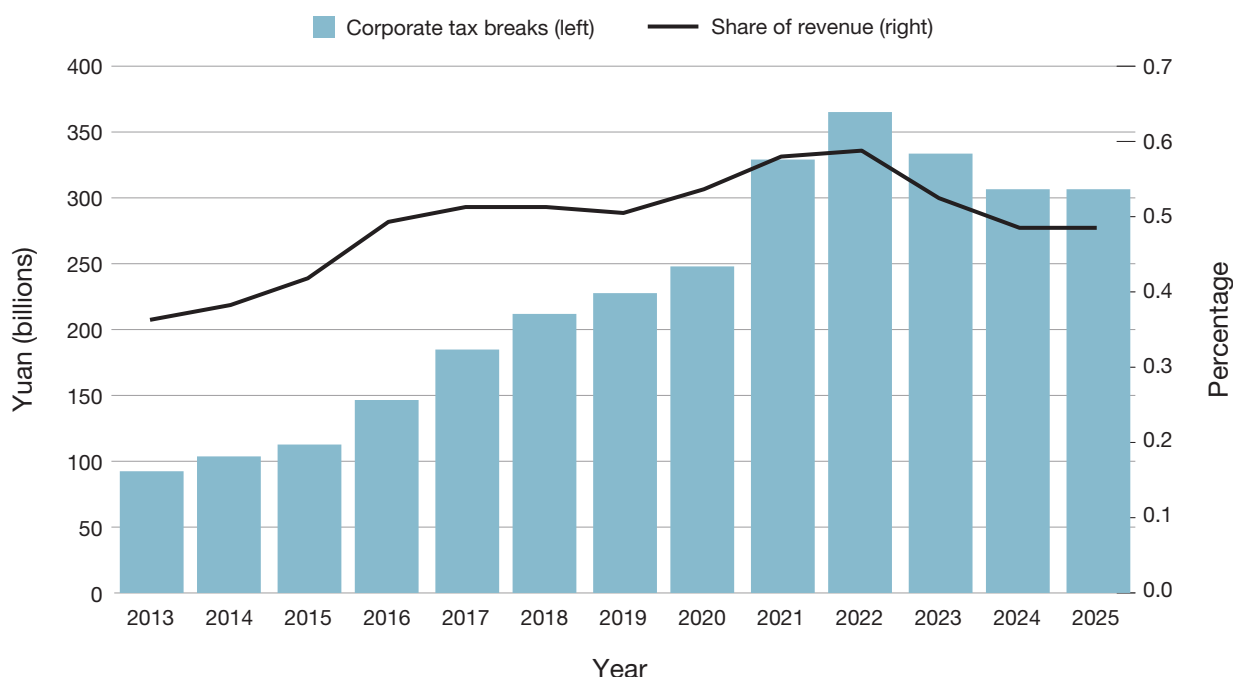
¹⁶ “China Announces Detailed VAT Reform Measures,” *Xinhua*, March 22, 2019; Liz Lee and Ellen Zhang, “China Approves Value-Added Tax Law, Taking Effect in 2026,” *Reuters*, December 25, 2024; MOF, *Report on the Implementation of China’s Fiscal Policy in 2022*, November 29, 2023b.

¹⁷ Qian Zhou, “What Are the Tax Incentives in China to Encourage Technology Innovation? (Updated),” *China Briefing*, March 29, 2023.

¹⁸ Ministry of Finance and State Taxation Administration, “Announcement on Relevant Corporate Income Tax Policies Regarding Deductions for Equipment and Instruments,” Announcement No. 37 of 2023, August 18, 2023; “China Extends Preferential Purchase Tax Policy for NEVs,” *Xinhua*, June 21, 2023.

¹⁹ Garcia-Macia, Kothari, and Tao, 2025.

FIGURE 5.6
Tax Breaks to Nonfinancial Listed Firms Peaked in 2022



SOURCE: Authors' calculations using data from company disclosures compiled in WFT.

NOTE: Tax breaks are calculated as the difference between taxes paid and the amount due if firms had paid the 25-percent statutory corporate income tax rate. This figure is based on reported disclosures from 5,377 firms listed on China's stock markets.

S&T budget rose sharply in the late 2000s and has been fairly steady as a share of GDP in the Xi era, at 0.9 percent of GDP in 2025 (Figure 5.8).²⁰

Meanwhile, total R&D spending in China has expanded even faster, from less than 2 percent of GDP in 2013 to 2.8 percent in 2025. As of 2022, China's R&D expenditure was approaching the OECD average and far above that of other middle-income economies.²¹ Within this total, the composition has shifted. The share financed directly by the government declined to 17 percent as of 2023, while business financing and especially tax-based support have grown.²²

China has increased tax-based support for enterprise innovation. The R&D super deduction was first set at 150 percent in 2008, raised to 175 percent in 2018, extended to 200 percent for manufacturing firms in 2021, and then expanded to 200 percent for all qualifying firms in 2023.²³ These changes lifted China's implied tax subsidy rate on R&D spending to more than 32 percent by 2023, compared with an OECD aver-

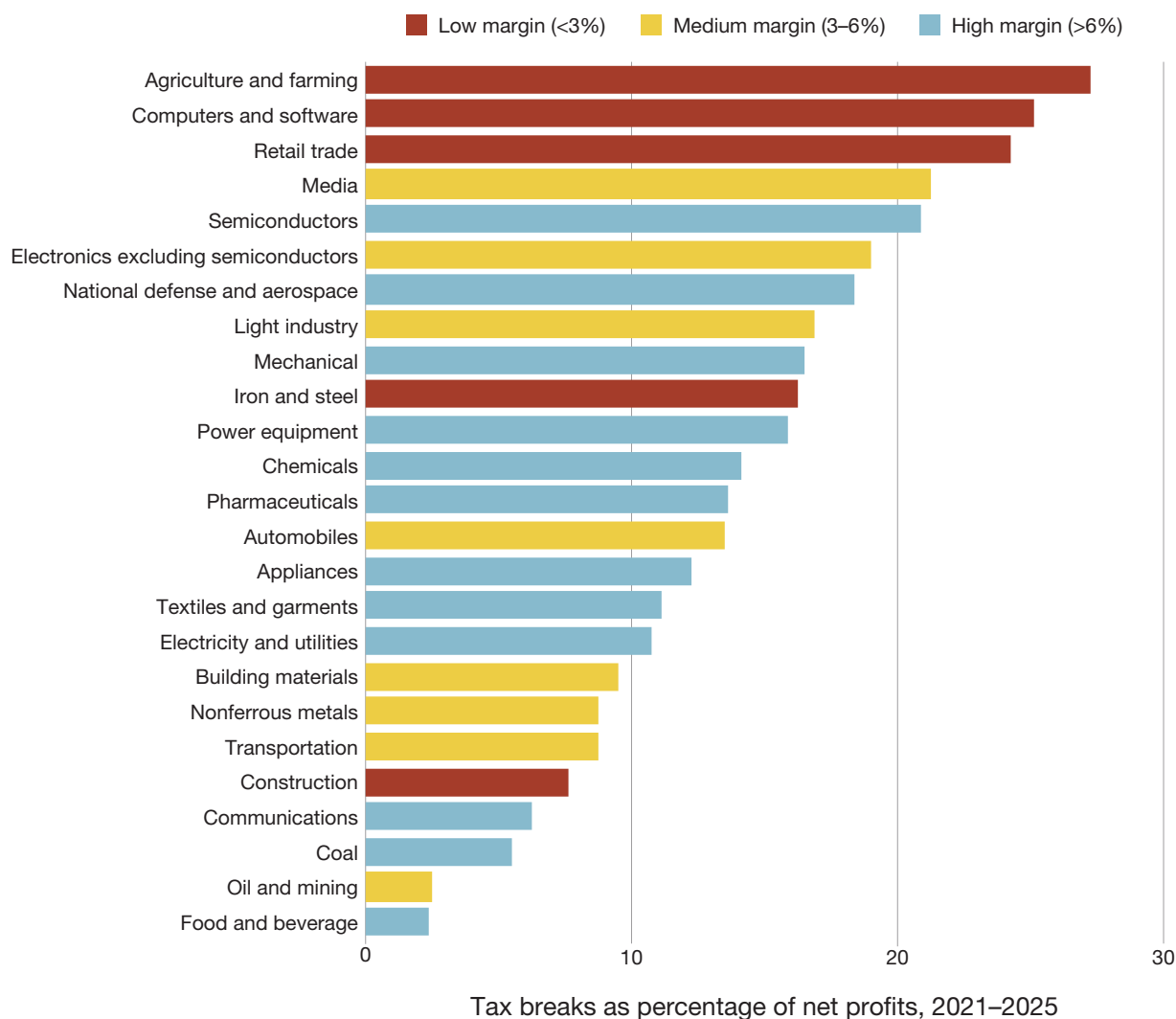
²⁰ MOF data via CEIC, undated.

²¹ National Science Board, "Global R&D and International Comparisons," in *Discovery: R&D Activity and Research Publications*, NSB-2025-7, July 23, 2025.

²² National Bureau of Statistics data via CEIC, undated.

²³ The evolution of China's R&D super-deduction scheme is well documented. See Global Trade Alert, "China: Domestic R&D Firms Eligible for 150% Deductions on Pre-Tax Expenses," December 10, 2008; Baker McKenzie, *China Tax Update*, June/July 2018, p. 5; KPMG, "75% R&D Super Deduction Extended, Raised to 100% for Manufacturers," *China Tax Alert*, No. 7, May 2021; Arendse Huld, "Pre-Tax Super Deduction of R&D Expenses in China—An Explainer," China Briefing, May 10, 2023; Global Trade Alert, "Establishment of R&D Super-Deduction Rate for All Qualifying Enterprises," March 26, 2023.

FIGURE 5.7
Tax Breaks Generally Support Strategic Industries



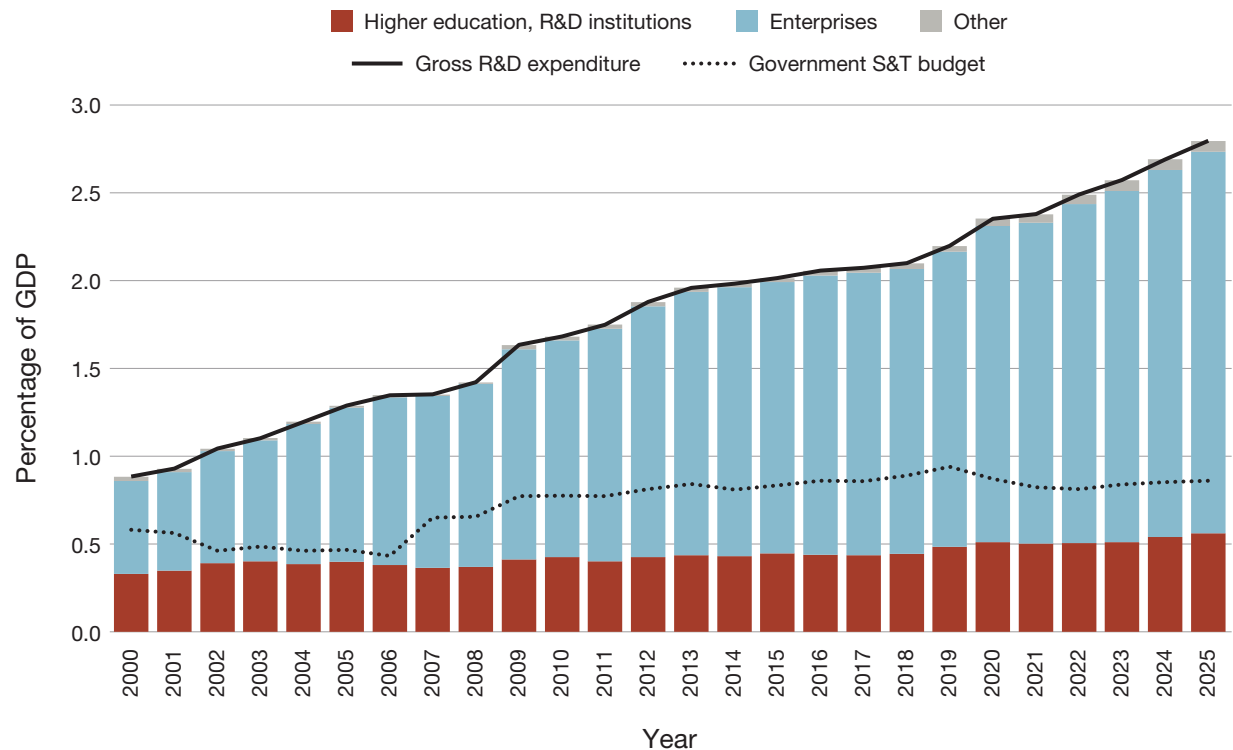
SOURCE: Authors' calculations using data from company disclosures compiled in WFT.

NOTE: Tax breaks are calculated as the difference between taxes paid and the amount due if firms had paid the 25-percent statutory corporate income tax rate. This figure is based on reported disclosures from 5,377 firms listed on China's stock markets.

age of 15 percent and just 3 percent in the United States.²⁴ Over the same period, OECD indicators show that direct government funding of business R&D stagnated or even declined as a share of GDP to 0.04 percent in 2024, while the value of indirect support through tax incentives surged from 0.06 percent in 2013 to 0.24 percent in 2022 (Figure 5.9). The OECD average in 2022 was 0.09 percent of GDP for direct government funding

²⁴ The OECD defines the implied R&D tax subsidy rate as $1 - B$ -index, where the B-index measures the before-tax income needed by a representative firm to break even on \$1 of R&D outlays. See indicator codes ITSR_LGEP_I ("Implied Tax Subsidy Rates on R&D Expenditures, Large Firms, Profitable Scenario"), BE_GVF_XGDP, and RDTAX_XGDP ("Indirect Government Support Through R&D Tax Incentives, % of GDP") in OECD, undated.

FIGURE 5.8
China's R&D Spending Is Rising, Supported by the S&T Budget



SOURCES: Features data from the MOF and National Bureau of Statistics via CEIC, undated; and OECD, "China," webpage, INNOTAX database, undated.

of business R&D and 0.13 percent for indirect support through tax incentives.²⁵ This makes China's R&D tax regime among the most generous globally.

Procurement Remains Important but Has Fallen

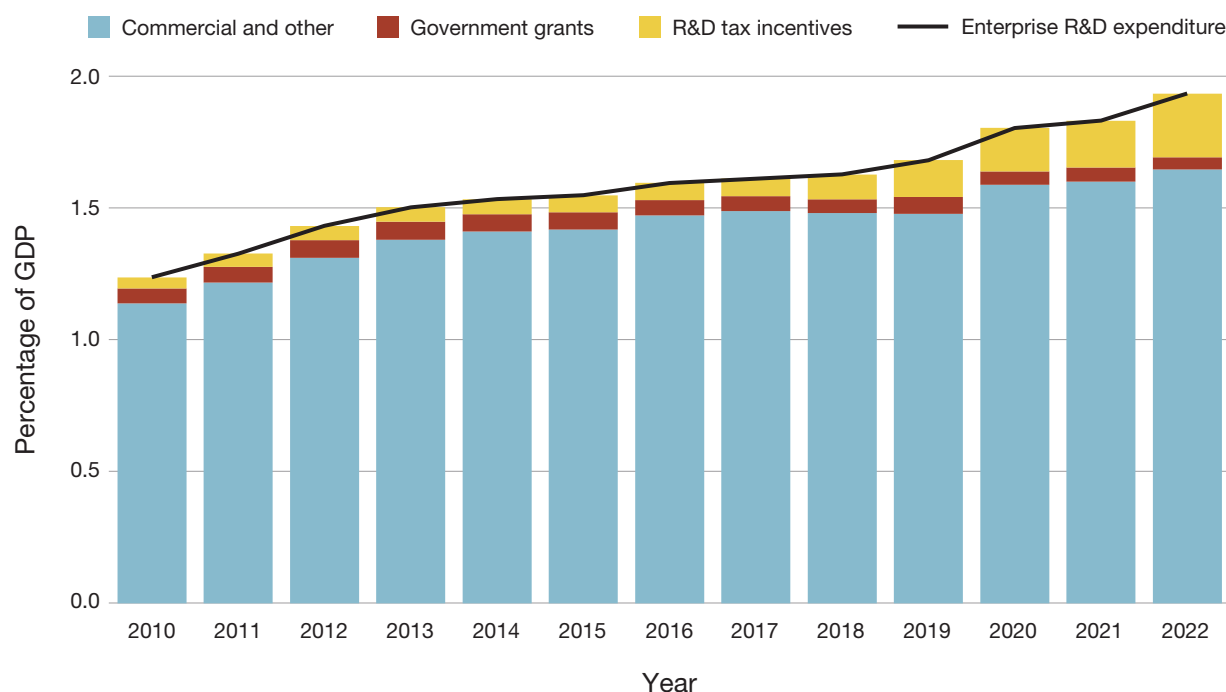
Government procurement in China, as defined by the Government Procurement Law and excluding SOE procurement,²⁶ grew quickly until the mid-2010s but has since plateaued and declined as a share of GDP. Spending peaked at 4.1 percent of GDP in 2016 and dropped to 2.5 percent by 2024 (Figure 5.10). This trend is likely due to tighter fiscal conditions, particularly at the local government level. Although official data do not clearly distinguish procurement tied to industrial policy, such purchases as ICT hardware, software meeting "secure and controllable" (安全可靠) standards, and environmentally friendly products often advance industrial policy objectives.

However, this official procurement figure understates the public sector's broader role. SOE procurement is much larger, totaling about ¥38 trillion in 2022, compared with ¥3.5 trillion in government procurement. Central SOE procurement has remained stable at around ¥13 trillion since 2021. The lack of growth at the

²⁵ OECD, undated.

²⁶ Standing Committee of the National People's Congress, The Government Procurement Law of the People's Republic of China, June 29, 2002.

FIGURE 5.9

Tax Incentives Are Increasingly Important for China's Enterprise R&D Spending

SOURCES: Features data from the MOF and National Bureau of Statistics via CEIC, undated; and OECD, undated.

central SOE level suggests stricter oversight and cost controls, although SOEs still channel significant, policy-driven demand.²⁷

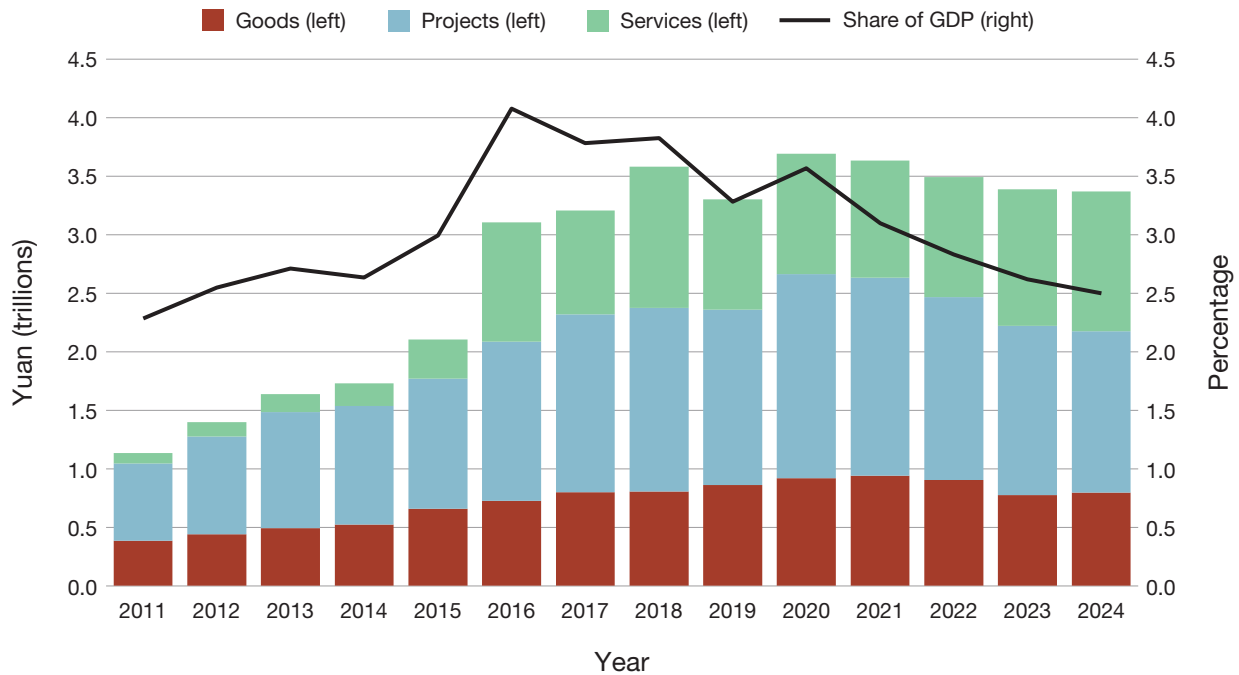
Despite procurement's smaller share of GDP, it has become more targeted, focusing on strategic sectors in which China aims to develop domestic technologies and reduce reliance on foreign suppliers (Table 5.1). Since 2017–2018, the approach has shifted from broad “buy Chinese” preferences to more specific, standards-based policies. For example, the “information technology application innovation” (信创) initiative requires government agencies to procure domestic operating systems, databases, servers, and personal computers that meet Beijing's “secure and controllable” (安全可靠) standards, steering purchases away from foreign ICT products.²⁸ New methods also allow agencies to co-fund R&D and make initial purchases of innovative technologies, while “first-set/first-batch” (首台套/首批次) policies push SOEs to trial newly developed domestic equipment, backed by a government insurance–compensation scheme in which manufacturers purchase comprehensive quality and liability coverage and receive central subsidies for most of the premium. In practice, such mandates function as partially funded obligations imposed on SOEs and public buyers, transferring the risks and costs of technology adoption from the state budget to the procuring entities.

Additionally, SASAC has strengthened procurement guidelines for SOEs, urging them to favor domestic suppliers, small and medium-sized enterprises (SMEs), and innovative products. The 2024 Action Plan for Large-Scale Equipment Renewal directs procurement budgets toward the adoption of advanced domestic

²⁷ Liu Zhiqiang, “Last Year, the Total Procurement Amount of Central State-Owned Enterprises Exceeded 13 Trillion Yuan: Its Influence Has Driven the Development of More Than Two Million Enterprises of Various Types,” *People's Daily*, April 26, 2022a; Zhang Xiaozhe, “Central Enterprises' Procurement Management Shows Significant Improvement,” State-owned Assets Supervision and Administration Commission of the State Council, December 26, 2019.

²⁸ “Secretive Chinese Committee Draws Up List to Replace U.S. Tech,” Bloomberg, November 16, 2021.

FIGURE 5.10

Government Procurement Has Fallen Along with Fiscal Revenues

SOURCES: Features data from MOF, "National Government Procurement Scale Reached 1.39777 Trillion Yuan," July 24, 2013; MOF, "Report on 2013 National Government Procurement Scale," July 16, 2014; MOF, "Brief Overview of National Government Procurement in 2014," July 30, 2015; MOF, "Brief Overview of National Government Procurement in 2015," August 12, 2016; MOF, "Brief Overview of National Government Procurement in 2016," August 25, 2017b; MOF, "Brief Overview of National Government Procurement in 2017," September 30, 2018; MOF, "Brief Overview of National Government Procurement in 2018," September 6, 2019; MOF, "Brief Overview of National Government Procurement in 2019," August 27, 2020b; MOF, "Brief Overview of National Government Procurement in 2020," September 3, 2021b; MOF, "Brief Overview of National Government Procurement in 2021," June 29, 2023a; MOF, "Brief Overview of National Government Procurement in 2022," December 25, 2023c; MOF, "Brief Overview of National Government Procurement in 2023," September 14, 2024c; MOF, "Brief Overview of National Government Procurement in 2024," November 12, 2025; and National Bureau of Statistics annual GDP data via CEIC, undated.

NOTE: The jump in services procurement starting in 2016 is due to the inclusion of the shantytown renovation program and rail construction services.

machinery and green upgrades in energy, transport, and industry.²⁹ In September 2025, the State Council introduced standards that grant domestically manufactured goods a 20-percent price advantage in bidding for government contracts, further supporting self-reliance.³⁰

²⁹ State Council of the People's Republic of China, "Notice on Issuing the 'Action Plan for Promoting Large-Scale Equipment Renewal and the Trade-in of Consumer Goods,'" State Council Document No. 7 of 2024, March 13, 2024a.

³⁰ Shen Cheng, "The Ministry of Finance Seeks Public Comment on Domestic Product Standards and Implementation Policies in Government Procurement," Xinhua, December 6, 2024; General Office of the State Council, "Notice on Implementing Domestic Product Standards and Related Policies in Government Procurement," General Office of the State Council Document No. 34 of 2025, September 30, 2025b.

TABLE 5.1

Select Government Procurement Policies to Support Industrial Policy

Year	Policy	Why It Matters
2002	Government Procurement Law	Created China's Government Procurement Law regime; mandates preference for domestic goods/services, forming baseline legal support for techno-industrial targeting. ^a
2006	MLP (2006–2020), with supporting policies	Called for using procurement to promote “indigenous innovation”; piloted “first-buy/ordering” mechanisms. ^b
2011	Notice decoupling government procurement from “indigenous innovation” catalogues	Formally ended direct linkage of government procurement preferences to “indigenous innovation” certifications after WTO pressure but left indirect tools. ^c
2017	Interim Measures for the Management of Government Procurement of Government Information Systems	Embedded security/multi-level protection scheme and data-sharing into ICT procurement; pivot to “secure and controllable.” ^d
2021	Regulations on the Security Protection of Critical Information Infrastructure	Imposed strict security obligations on critical information infrastructure operators (many SOE-run sectors); procurement required localized, certified ICT/operational technology. ^e
2021	Government procurement demand-management measures	Professionalized demand formation and life-cycle evaluation; mechanisms to embed industrial/green/security criteria in tenders. ^f
2023–2024	MOF and MIIT ICT demand standards	Codified “secure and controllable” specifications in tenders; effectively mandated domestic ICT stacks in government IT. ^g
2023–2024	“First-set/first-batch” adoption	Required SOE and public tenders to accept newly developed domestic equipment on equal footing with established products; created an insurance scheme to cover financial risks from early adoption. ^h
2024	Action plan for large-scale equipment renewal and trade-ins	Multiyear capital expenditure push across industrial, energy, and transportation sectors; mobilized SOEs and public bodies to buy high-end, green digital equipment. ⁱ
2025	Standards and implementation policies for domestic products in government procurement	Defines “domestic products” and a 20-percent evaluation price advantage for domestic products in government tenders. ^j

^a Standing Committee of the National People's Congress, 2002.

^b State Council of the People's Republic of China, “Notice of the State Council on Issuing Several Supporting Policies for the Implementation of the ‘National Medium- and Long-Term Science and Technology Development Plan Outline (2006–2020),’” State Council Document No. 6 of 2006, February 7, 2006c.

^c MOF, “Notice on Ceasing the Implementation of Three Documents Including the ‘Measures for Budget Management of Government Procurement of Independently Innovative Products,’” Finance and Budget Document No. 85 of 2011, June 23, 2011.

^d MOF, “Notice on Issuing the ‘Interim Measures for the Management of Government Procurement of Government Information Systems,’” Finance and Budget Document No. 210, 2017a.

^e State Council of the People's Republic of China, “Regulations on the Security Protection of Critical Information Infrastructure,” State Council Decree No. 745, July 30, 2021.

^f MOF, “Notice on Issuing the ‘Measures for the Management of Government Procurement Demand,’” Finance and Budget Document No. 22 of 2021, April 30, 2021a.

^g Ministry of Finance and Ministry of Industry and Information Technology of the People's Republic of China, “Notice on Issuing the ‘Standards for Government Procurement of Desktop Computers (2023 Edition),’” Finance and Budget Document No. 29 of 2023, December 16, 2023.

^h Ministry of Industry and Information Technology, National Development and Reform Commission, and State-owned Assets Supervision and Administration Commission of the State Council, “Guiding Opinions on Supporting Equal Participation of Enterprises in Bidding and Tendering Activities for the First (Set) of Major Technical Equipment,” Ministry of Industry and Information Technology Joint Heavy Equipment Document No. 127 of 2023, August 16, 2023.

ⁱ State Council of the People's Republic of China, 2024a.

^j General Office of the State Council, 2025.

Human Capital as a Techno-Industrial Policy Instrument

Under Xi, China has sought to transform its human capital policy from a patchwork of elite talent schemes into a more coordinated techno-industrial instrument. The goal has been to direct fiscal support to individuals while aligning selection, evaluation, and career incentives with national self-reliance goals.³¹ This shift was formalized in the 20th Party Congress report in 2022, which called for building a “world-important talent center” (世界重要人才中心), strengthening “strategic talent forces” (战略人才力量), and accelerating progress toward “high-level S&T self-reliance and self-strengthening” (高水平科技自立自强).³²

China's talent programs rest on a national architecture that emerged after a period of fragmentation. Prior to the Xi era, talent policy was dominated by ministry-run and locally duplicated programs that relied heavily on credentials and produced small, elite cohorts with uneven outcomes. The 2016 Central Committee's “Opinions on Deepening the Reform of Talent Development Systems and Mechanisms” and subsequent evaluation reforms curbed the “four-only” (破四唯) metrics, which pared back duplication, standardized programs, and tied incentives more to mission delivery.³³

Within this restructured system, national-level support has been consolidated into three core channels. The National High-Level Talent Special Support Plan (“Ten-Thousand Talents”), launched in 2012 and refined since 2017, functions as the primary vehicle for supporting high-end innovators, entrepreneurs, and teams while absorbing and replacing overlapping schemes.³⁴ The Ministry of Education's Changjiang Scholars program, established in 1998, provides centrally financed bonuses within academia and embeds positions in nationally prioritized disciplines, with a formal exit mechanism added in 2018.³⁵ Finally, the National Natural Science Foundation of China administers talent-linked grants, such as the Distinguished Young Scholars program, which provide resources for elite researchers to conduct work in China.³⁶

Career advancement, residency perks, and administrative treatment have been standardized to reinforce these national channels. Since 2017, policy has committed to easing professional title appointments through direct hiring and evaluation, facilitating household registration for high-level talent, and offering service packages for spouse employment and children's schooling.³⁷ For foreign high-end talent, China uni-

³¹ *Talent* here refers to high-level S&T and professional recruits targeted by national and local talent programs, not the education system writ large. See Central Committee of the Communist Party of China, “Opinions on Deepening the Reform of the Talent Development System and Mechanism,” Xinhua, March 21, 2016.

³² Xi Jinping, “Hold High the Great Banner of Socialism with Chinese Characteristics and Unite in the Struggle to Build a Modern Socialist Country in All Respects—Report to the 20th National Congress of the Communist Party of China (October 16, 2022),” Xinhua, October 25, 2022.

³³ 破“四唯” literally means break the four “onlys,” a policy drive to reduce rigid reliance on papers (论文), professional titles (职称), academic degrees (学历), and awards (奖项) in talent evaluation, shifting toward contributions, innovation value, and problem-solving. It is a nationwide reform slogan that has been used since 2018. See Central Committee of the Communist Party of China, 2016.

³⁴ Central Committee of the Communist Party of China, 2016.

³⁵ Historic standards of centrally financed bonuses are ¥200,000 per year for chair professors and monthly stipends for visiting chairs. Exit mechanisms are formal rules to terminate titles or funding if performance, ethics, or compliance fall short. See Ministry of Education of the People's Republic of China, “Notice on Issuing the ‘Implementation Measures of the Yangtze River Scholars Program,’” Ministry of Education Document No. 10 of 2011, December 15, 2011; Party Group of the Ministry of Education of the CPC, “Notice on Issuing the ‘Management Measures for the Yangtze River Scholars Program,’” Party and Education Department Document No. 51 of 2018, September 21, 2018.

³⁶ Distinguished Young Scholars typically receive about ¥4 million over five years, while the Overseas Excellent Young Scientists fund provides ¥1–3 million over three years to attract Ph.D.-trained researchers back to China.

³⁷ Central Committee of the Communist Party of China, 2016.

fied the work-permit visa residence chain in 2017 with national platforms and guidance to reduce processing frictions.³⁸

Local governments implement this framework through complementary incentives. They provide cash, housing and family support, and household registration (*hukou*, 户口) benefits to attract and retain talent aligned with national priorities. Shenzhen's Peacock Plan, for example, offers one-time allowances of between ¥1.6 and ¥3 million to top-tier recruits, alongside start-up packages and service guarantees.³⁹ Provinces such as Fujian match central Ten-Thousand Talents awards, earmarking at least half for work expenses and up to half for living support.⁴⁰ Beijing districts likewise provide one-off supplements, priority project access, and commercialization support tied to service-year commitments.⁴¹

Recent reforms have tightened quality and discipline. Beijing's 2018 talent evaluation overhaul targets the "breaking the four-onlys" problem (excessive reliance on papers, titles, diplomas, and awards) and shifts selection and incentives toward innovation value and contribution.⁴² Long-running programs, such as the Changjiang, have formalized exit pathways, while overlapping schemes have been consolidated.⁴³ Finally, to bind funding to national missions, talent programs increasingly ride on mission-oriented instruments. For example, the "challenge-based leadership" (揭榜挂帅) mechanism, which assigns teams to "key core technologies," appears explicitly across S&T and decarbonization policy.⁴⁴

³⁸ State Administration of Foreign Experts Affairs, Ministry of Human Resources and Social Security, Ministry of Foreign Affairs, and Ministry of Public Security, "Notice on the Full Implementation of the Work Permit System for Foreigners in China," Foreign Experts Document No. 40 of 2017, March 28, 2017.

³⁹ Hongchuangwei Enterprise Management Consulting Co., Ltd., "Shenzhen Overseas High-Level Talent ('Peacock Plan') Recognition Measures and Subsidies," Hongchuangwei Innovation Enterprise Service Platform, undated.

⁴⁰ Fujian Provincial Department of Human Resources and Social Security, "Application Handbook for the 'Fujian Province High-Level Talent Recognition and Support Measures (Trial),' " July 7, 2020.

⁴¹ Shunyi District People's Government Office, "Notice on Forwarding the 'Shunyi District Postdoctoral (Young Talents) Innovation Practice Base Management Measures (Trial),' " Shunzhengbanfa No. 6, March 28, 2012.

⁴² "The General Office of the CPC Central Committee and the General Office of the State Council Issued the 'Opinions on Deepening the Reform of Project Review, Talent Evaluation, and Institutional Assessment,'" Xinhua, July 3, 2018.

⁴³ For example, the leading talents (领军人才) of the "Hundred, Thousand and Ten Thousand Talents Project" program (百千万人才工程) were integrated into the national Ten-Thousand Talents framework. See General Office of the Ministry of Science and Technology, General Office of the Organization Department of the CPC Central Committee, and General Office of the Ministry of Human Resources and Social Security, "Notice on Organizing the 2018 Selection for the Innovation Talent Promotion Program and the National 'Ten Thousand Talents Program'—Science and Technology Innovation Leading Talents and Science and Technology Entrepreneurship Leading Talents," Ministry of Science and Technology Document No. 53 of 2018, 2018.

⁴⁴ For the "challenge-based leadership," the state publishes a task list for bottleneck technologies. Qualified teams then pick a list and sign time-bound commitments before receiving flexible funding. For examples of these task lists, see General Office of the Ministry of Industry and Information Technology, "Notice on Organizing the 2023 Future Industry Innovation Task Challenge," Ministry of Industry and Information Technology Office Letter No. 235 of 2023, August 28, 2023; Ministry of Finance and Ministry of Science and Technology of the People's Republic of China, "Notice on Issuing the 'Administrative Measures for Funds of the National Key R&D Program,'" Finance and Education Document No. 2 of 2025, January 27, 2025.

Mobilizing Finance for Techno-Industrial Policy

In the Xi era, Beijing has sought to use the state-owned financial sector and state-linked investors as the dominant channels of state support for industries. As tighter fiscal conditions and central efforts to curb discretionary local support narrowed the scope for cash subsidies and ad hoc incentives (such as tax rebates and preferential land and utility deals), policymakers sought mechanisms to discipline capital allocation, sharpen incentives, and sustain support for priority sectors without expanding public budgets. This shift marks the emergence of an “investor state,” in which the Party-state acts less as a direct spender and more as a capital allocator using banks, funds, and markets to pursue strategic outcomes.¹

As Vice Minister of Science and Technology Qiu Yong observed in May 2025, “The emphasis is on shifting from a ‘fiscal mindset’ to a ‘financial mindset.’”² In practice, that means leveraging credit, equity, and bond markets. Finance is a way to direct resources toward priorities through channels that carry market-like features but remain subject to political direction.

Raising the (Equity) Stakes: China’s Investor State

Under Xi, the Party-state has evolved into an “investor state.” Initiated with the 2013 Third Plenum’s directive to “manage capital,” Beijing seeks to direct resources into strategic sectors via state-backed guidance funds, equity market interventions, and “golden share” stakes in private firms. This approach embeds political oversight into corporate governance, positioning the Party-state as fund sponsor, holding company, market stabilizer, shareholder, and overseer.

The guidance fund model became the central innovation. Guidance funds are state-backed investment vehicles in which governments provide seed capital and set priorities but delegate day-to-day management to professional fund managers, often alongside private co-investors. The number of government guidance funds surged from fewer than 100 in 2014 to more than 2,100 by 2024.³ Most famously, the 2014 National Integrated Circuit Industry Investment Fund (Big Fund) (大基金, 国家集成电路产业投资基金) raised ¥139 billion in its first phase, expanded to ¥200 billion in 2019, and launched a third ¥344 billion round in 2024 targeting semiconductor equipment and design. By 2025, total commitments to guidance funds exceeded ¥13 trillion, with roughly half that amount actually invested.⁴ These funds employ professional managers and often seek

¹ Meg Rithmire and Hao Chen, “The Rise of the Investor State: State Capital in the Chinese Economy,” *Studies in Comparative International Development*, Vol. 55, No. 3, September 2020.

² State Council Information Office of the People’s Republic of China, “Press Conference on Policies for Science and Technology Finance,” transcript, May 22, 2025.

³ Xuan Li and Cornel Ban, “Financing Technological Innovation in China: Neo-Developmental Financial Statecraft Through Government Guidance Funds,” Boston University Global Development Policy Center, GCI Working Paper 041, April 2025.

⁴ Li and Ban, 2025.

commercial returns, but they operate under a policy-oriented mandate in which investment direction and performance are evaluated at least as much against strategic objectives as against financial returns.⁵

Another innovation is state capital investment and operation companies: wholly state-owned holding companies that pool SOE balance sheets and invest directly on behalf of SASAC. These companies represent the institutional shift of SASAC from “managing enterprises” (管企业) to “managing capital” (管资本). Created after the 2013 Third Plenum, such pilots as China Chengtong and China Reform Holdings were tasked with restructuring state-owned assets, acquiring equity stakes, and reallocating capital across SOEs.⁶ By 2025, the two central state capital operation platforms managed ¥1.577 trillion in assets, and 21 central SOEs had been designated as capital investment or capital operation companies.⁷ Starting from about ¥248 billion combined in 2016, when the pilots were launched, their asset base has grown more than sixfold.⁸

Finally, the state began buying “golden shares” in private companies to solidify political oversight. In 2013, Beijing developed the idea of “special management shares” (特别管理股). State entities have taken 1- to 2-percent equity stakes with special governance rights in major private platforms, including ByteDance, Alibaba, Tencent, and Weibo. Though financially trivial, these stakes grant regulators veto power, board seats, and control over content licenses, symbolically embedding Party authority into corporate governance. Proliferating after the 2020–2022 platform crackdown, golden shares illustrate how the investor state extends beyond financing into political oversight of digital and data-intensive firms.⁹

By the early 2020s, the limitations of guidance funds became increasingly clear. Corruption probes into the Big Fund (大基金) in 2022 exposed widespread rent-seeking, with senior managers accused of steering billions into favored firms and embezzling state money. Other scandals, such as the collapse of Wuhan Hongxin Semiconductor, highlighted how political pressure and weak oversight allowed unqualified firms to receive large equity injections that were later squandered.¹⁰

⁵ Guidance fund performance is not uniformly transparent, and systematic fund-level internal rate of return data are limited. However, available indicators suggest uneven performance. A 2023 study of a dataset of 1,849 government guidance funds found that, by 2021, only 26 percent of funds had met their target capital size, and only about one-third had made at least one investment, implying widespread fundraising and deployment shortfalls. The same study reports that some flagship national funds performed better. The National Integrated Circuit Industry Investment Fund’s first phase invested in listed firms and reported a 125-percent return on investment over 2014–2019, making it an outlier success case. See Yifan Wei, Yuen Yuen Ang, and Nan Jia, “The Promise and Pitfalls of Government Guidance Funds in China,” *China Quarterly*, Vol. 256, December 2023.

⁶ State Council of the People’s Republic of China, “Implementation Opinions of the State Council on Advancing the Pilot Program of the Reform of State Capital Investment and Operation Companies,” State Council Document No. 23 of 2018, trans. by LawInfoChina, July 14, 2018.

⁷ As of April 2025, China Reform Holdings reported total assets of ¥980 billion, and as of March 2025, China Chengtong reported total assets of ¥597.17 billion, for a combined total of ¥1.577 trillion. See Sun Jie, “Three ‘National Teams’ Stepped in to Stabilize the Market, with China Guoxing Increasing Its Holdings by 80 Billion Yuan in the First Batch,” *Beijing Daily Client*, April 8, 2025; and China Lianhe Credit Rating Co., Ltd., “Credit Rating Report on China Chengtong Holdings Group Co., Ltd. 2025 Third Tranche of Medium-Term Notes,” August 8, 2025. For the designation of 21 central SOEs as state capital investment or operation companies, see “A New Batch of 11 Central SOEs’ State-Owned Capital Investment Companies Have Been Launched as Pilot Projects,” *People’s Daily*, January 8, 2019.

⁸ The increase is calculated from the 2016 pilot-baseline assets of China Reform (¥140 billion, SASAC) and China Chengtong (¥108.226 billion, Lianhe ratings table) to 2025 assets. Both firms were designated state capital investment and operation company pilots in February 2016. See China Lianhe Credit Rating Co., Ltd., “China Chengtong Holdings Group Co., Ltd.’s 2019-1 Debt-for-Equity Swap Bonds,” May 28, 2019; China National New and Opening-Up Corporation Limited, “China State-Owned Assets Supervision and Administration Commission (SASAC) Was Awarded an A-Level Rating in the Performance Evaluation of Central Enterprise Leaders,” July 19, 2020.

⁹ Lingling Wei, “China’s New Way to Control Its Biggest Companies: Golden Shares,” *Wall Street Journal*, March 8, 2023.

¹⁰ Wei, Ang, and Jia, 2023; Ngor Luong, Zachary Arnold, and Ben Murphy, *Understanding Chinese Government Guidance Funds: An Analysis of Chinese-Language Sources*, Center for Security and Emerging Technology, March 2021.

Even without outright fraud, guidance funds often proved too risk-averse to serve as patient capital. Studies show that more than half of local funds used “valuation adjustment mechanisms” that required start-ups to buy back equity if revenue or IPO targets were missed, effectively turning equity into disguised loans. Typical fund horizons were only six to seven years, far shorter than the 12–15 years many hard technology firms require. Local officials, judged on short-term performance, had little incentive to tolerate failures or wait for long-term payoffs.

Structural constraints compounded these issues. Local guidance funds are contractually required to invest within their own jurisdictions, often channeling money into weak local firms rather than the best national candidates.¹¹ Poorer provinces lacked viable high-tech companies to absorb capital, producing duplication and waste as cities competed to fund the same semiconductor fabrication plants or EV projects.¹² Although guidance funds boosted patent filings and improved access to capital for start-ups, they also exacerbated overcapacity in such sectors as EV batteries and photovoltaics.¹³

Fiscal stress further eroded the model. Guidance funds relied heavily on land-sale revenues and LGFVs, both of which collapsed after 2020 as the property market slumped. By 2023, fewer than one-third of funds had met their fundraising targets, and many subfunds were inactive. Local governments often pledged capital they could not deliver, leaving headline fundraising totals inflated but liquidity constrained.

Beijing recognizes these challenges and is working to centralize and professionalize the system. In January 2025, the State Council issued new guidelines that distinguish industrial from venture funds and aim to build patient capital by attracting long-term investors, tightening fiscal rules, and raising management standards. On paper, national funds will set direction, and provinces must consolidate funds, align investments with capacity data, and reduce duplication. New rules shift performance evaluation toward policy outcomes and call for a “fault-tolerance mechanism” (容错机制) that tolerates normal investment risk, avoids judging funds on single-project or single-year profit and loss, and can exempt managers from liability for losses when due diligence and compliance requirements are met. The guidelines also expand exit channels and ban hidden debt or forced contributions.¹⁴ A proposed national venture capital guidance fund, announced in March 2025, could reach ¥1 trillion for frontier technologies.¹⁵ Central SOEs and Shanghai’s ¥10 billion “futuristic industry fund” (with a 15-year horizon) reflect this strategic shift.

In January 2026, Beijing issued a trial cross-ministerial regulatory framework governing government guidance funds. Jointly released by the NDRC, MOF, Ministry of Science and Technology, and MIIT, the rules establish unified national planning, positive and negative investment lists, and formal performance evaluations. They bar funds from creating disguised local government debt, engaging in speculative activity,

¹¹ For binding local-investment requirements, see, for example, Jiangsu Provincial People’s Government Office, “Jiangsu Provincial Government Investment Fund Administrative Measures,” September 30, 2022. This evidence suggests that government guidance funds can relax financing constraints and raise innovation inputs, but they may also amplify overinvestment when local officials target the same sectors. For a broader understanding of government guidance funds and their effects, see Luong, Arnold, and Murphy, 2021.

¹² A 2025 guidance from the State Council directs local funds to align investments with their industrial base and warns against overcapacity, duplication, and idle capital in regions lacking viable projects. See General Office of the State Council, “Guiding Opinions of the General Office of the State Council on Promoting the High-Quality Development of Government Investment Funds,” General Office of the State Council Document No. 1 of 2025, January 7, 2025a.

¹³ Econometric work shows that guidance funds ease financing constraints and raise innovation output (Guangyu Li, Jun Zhang, Yiming Wang, and Zijian Zhao, “Can Government-Guided Funds Promote Corporate Technology Innovation? Evidence from the New Energy Automobile Industry,” *International Review of Economics & Finance*, Vol. 96, Part B, November 2024).

¹⁴ General Office of the State Council, 2025a.

¹⁵ Ziyi Tang and Ryan Woo, “China to Set Up National Venture Capital Guidance Fund, State Planner Says,” Reuters, March 6, 2025.

or investing in nonstrategic listed equities, while steering capital toward national priorities and early-stage “hard technology” (硬科技).¹⁶ The shift from broad guidance toward a more compliance-linked oversight regime reflects Beijing’s assessment that uneven local execution by officials overseeing these funds is the key problem to address. The 2026 trial measures aim to “harden” control by pairing explicit prohibitions with mandatory registration and annual investment-direction evaluations whose results are circulated and can trigger remedial steps, such as interviews, warning letters, and adjustments to government capital contributions and management fees.

If implemented consistently, these reforms could push China’s investor state toward a more centrally supervised and procedurally disciplined model, although it is too early to know whether they are improving capital allocation in practice.

Credit Policy: From Expansion to Strategic Targeting

Xi-era credit reforms have followed a familiar pattern seen across techno-industrial policy: First impose financial discipline, then build tools to steer the remaining credit toward strategic sectors. Early in Xi’s tenure, Beijing sought to rein in the explosive debt growth that followed the 2008–2010 stimulus while also harnessing the country’s vast banking system to support strategic sectors. This created a dual mandate for state-owned banks: to impose credit discipline and to follow state guidance. Policymakers cracked down on shadow banking and emphasized “quality over quantity” in finance.¹⁷ A 2016 *People’s Daily* “authoritative person” interview, attributed to Liu He, warned against “flood-like stimulus” and launched a deleveraging campaign that sharply reduced nonbank (shadow bank) lending.¹⁸

Beijing developed mechanisms intended to target credit allocations more precisely. The People’s Bank of China (PBOC) announced new structural lending facilities and regulatory mandates to steer bank credit toward advanced manufacturing, green investment, and SMEs. These tools represent a significant expansion of directed credit, although their effectiveness in reaching genuinely innovative firms (as opposed to incumbents with existing banking relationships) is difficult to assess given the limited availability of public data on loan-level allocation. For example, the Carbon Emission Reduction Facility (2021) refinanced 60 percent of qualifying green loans at a concessional rate, while the Equipment Upgrade Relending Facility (2022) capped borrower rates to encourage investment in industrial modernization.¹⁹ In 2024, these pilots were consolidated into a ¥500 billion Tech Innovation and Equipment Upgrade Relending Facility, explicitly linking central bank credit to innovation-oriented firms.²⁰ Regulatory guidance reinforced these instruments. The

¹⁶ NDRC, “Notice on Issuing the ‘Working Measures (Trial) for Strengthening the Planning and Investment Direction Guidance of Government Investment Funds,’” National Development and Reform Commission Finance and Economics Regulation Document No. 1752 of 2025, January 12, 2026.

¹⁷ People’s Bank of China, China Banking and Insurance Regulatory Commission, China Securities Regulatory Commission, and State Administration of Foreign Exchange, “Guiding Opinions on Regulating the Asset Management Business of Financial Institutions,” People’s Bank of China Document No. 106 of 2018, *State Council Gazette*, No. 26, April 27, 2018.

¹⁸ Gong Wen, Xu Zhifeng, and Wu Qiuyu, “Asking about the General Trend at the Beginning of the First Quarter—Authoritative People Talk About the Current Chinese Economy,” *People’s Daily*, May 9, 2016.

¹⁹ PBOC, “People’s Bank of China Officials Answer Reporters’ Questions on Carbon Emission Reduction Support Tools,” November 8, 2021; PBOC, “The Relevant Person in Charge of the People’s Bank of China Answered Reporters’ Questions on the Establishment of a Special Re-Loan for Equipment Renewal and Transformation,” September 28, 2022.

²⁰ Ministry of Science and Technology of the People’s Republic of China, “The People’s Bank of China, Together with the Ministry of Science and Technology and Other Departments, Has Established a Refinancing Program for Scientific and Technological Innovation and Technological Transformation,” April 18, 2024.

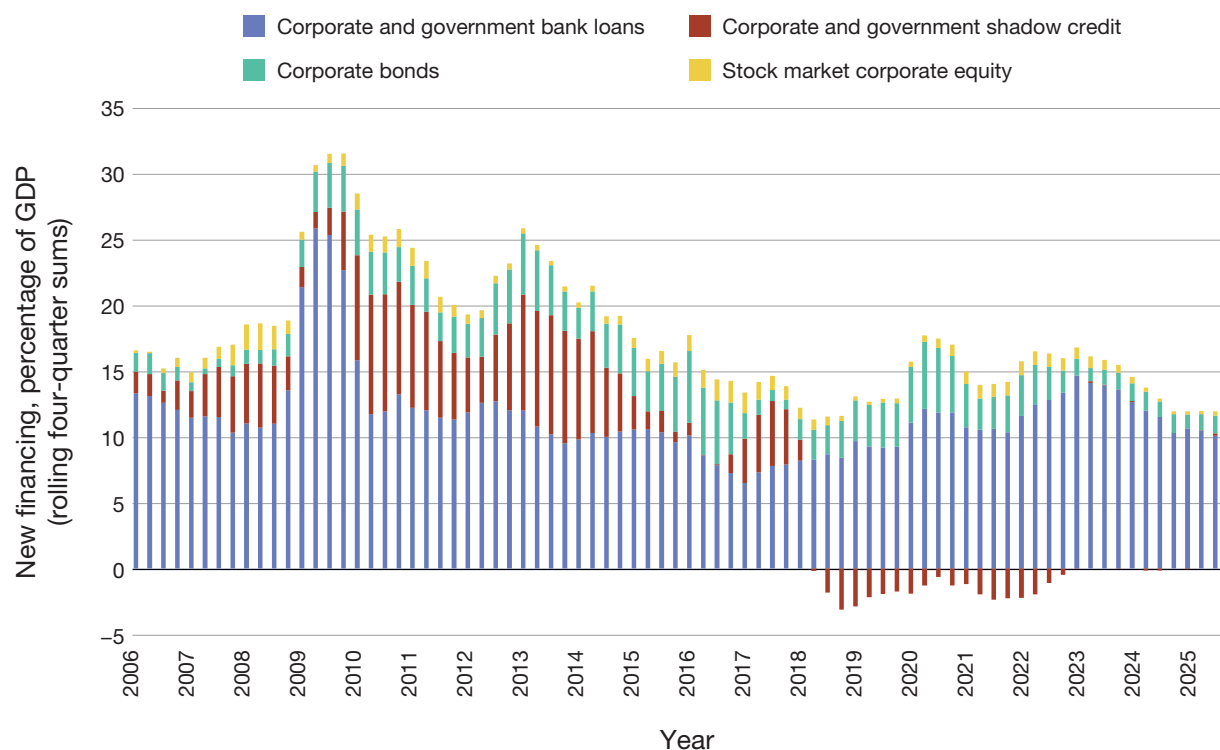
2019 Loan Prime Rate reform improved the transmission of policy rates into corporate borrowing costs, and subsequent quotas required banks to increase lending to manufacturing at double-digit rates.²¹

In China, corporate and industrial credit remain dominated by bank loans rather than by bonds or equity issuance (Figure 6.1). Bonds and stock market equity have expanded somewhat since the late 2010s, but they remain small relative to the country's loan book of more than ¥270 trillion. Within bank lending, however, most credit has not flowed to manufacturing, probably because manufacturers typically rely far more on retained earnings than credit and only need loans when in distress or seeking to expand capacity rapidly (Figure 6.2). Until the COVID-19-era stimulus, the bulk of corporate loans went to services, infrastructure, and real estate development. Manufacturing's share rose temporarily after the 2008–2010 stimulus and then again after 2020, when Beijing urged banks to support “new infrastructure” (新基建) and “new quality productive forces” (新质生产力), and export demand surged.²²

The main lenders shaping these outcomes are the big state-owned commercial banks and national joint-stock banks that supply the bulk of corporate loans. Policy banks, such as the China Development Bank (CDB), finance large infrastructure and urban projects but play a limited role in everyday industrial finance.

Preferential credit is visible at scale in firm-level data. The OECD's MAGIC database estimates that the subsidy-equivalent value of Chinese firms' below-market borrowings exceeded 2 percent of revenue in 2023,

FIGURE 6.1
Bank Loans Remain the Backbone of Corporate Finance Under Xi



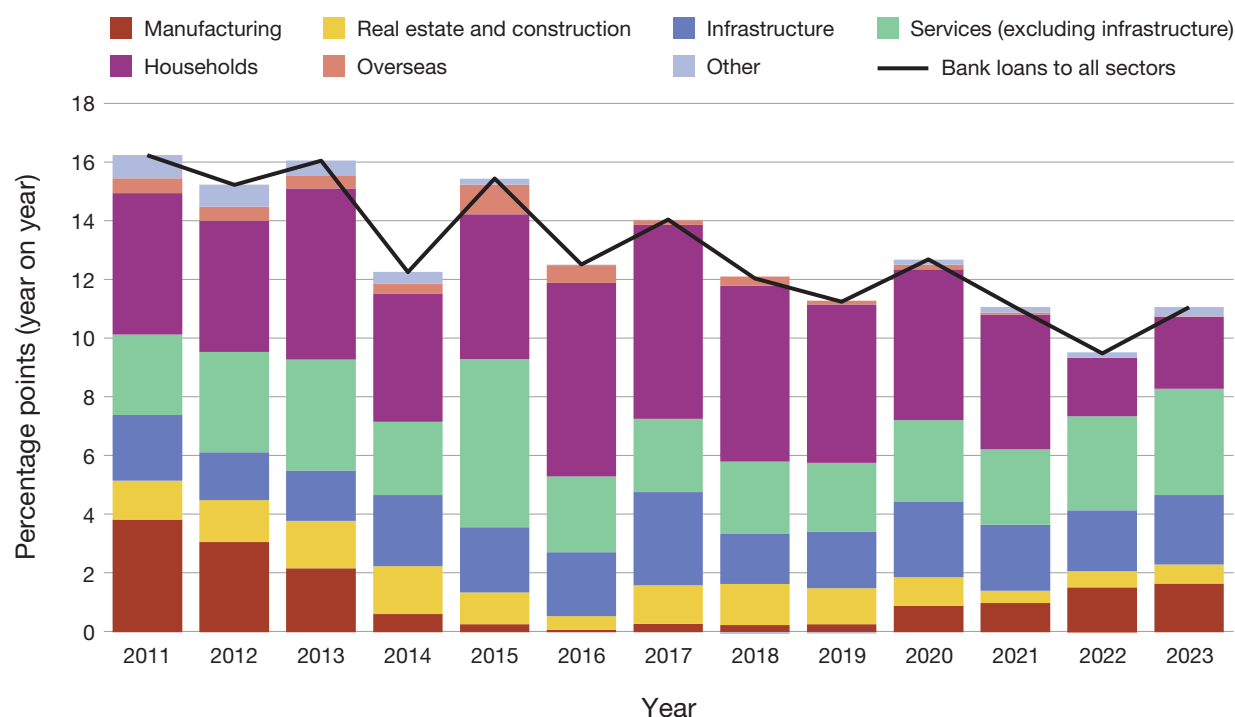
SOURCE: Features data from the PBOC and National Bureau of Statistics via CEIC, undated.

NOTE: *Corporate and government* refers to nonfinancial enterprises and government agencies, including LGFVs. *Shadow credit* includes entrusted loans, trust loans, and banker's acceptances. Latest observation is partial 2026.

²¹ “New Mechanism for Forming Loan Market Benchmark Interest Rate Is in Operation,” *People's Daily*, August 21, 2019.

²² People's Bank of China Monetary Policy Analysis Group, *China Monetary Policy Implementation Report, Fourth Quarter 2021*, February 11, 2022.

FIGURE 6.2

Industrial Lending Was a Minor Share of Bank Loan Growth Until the COVID-19–Era Push

SOURCE: Features data from the PBOC via CEIC, undated.

NOTE: This figure shows year-on-year change in banks' total local and foreign currency loans. *Infrastructure* refers to loans for the production and supply of electricity, gas, and water; water conservancy and environmental management; and transport, storage, and postal services.

a ratio that is higher than in any surveyed region.²³ This is by far the largest measured component of support for China-based firms within the MAGIC database, consistent with these firms' greater reliance on bank loans than on bonds.²⁴

Many commercially oriented firms fund new investment mainly from retained earnings rather than external borrowing. Bank loans are more critical for capital-intensive heavy industry and state-linked firms with weaker balance sheets, which helps explain why lending often reinforces incumbents rather than channeling resources to new entrants.²⁵ In practice, credit is most effective for expanding capacity, modernizing equipment, or stabilizing employment, not for financing long-horizon, highly uncertain innovation.

At the same time, credit has remained essential for infrastructure and energy deployment, where projects are long term and capital-intensive. Bank and policy-bank lending has financed everything from urban transit to power grids and renewable generation and, more recently, the expansion of EV charging infrastructure

²³ In the OECD MAGIC methodology, "below-market borrowings" are estimated as the annual interest savings from debt priced below a market benchmark for a comparable borrower. OECD's 2025 update reports that, across its covered firm sample, the mean gap between actual and benchmark borrowing rates rose from about 2.4 percentage points in 2020 to 3.2 percentage points in 2023 (OECD, 2025e). Figure 5 of that report shows lower bond share and higher below-market borrowing among Chinese firms.

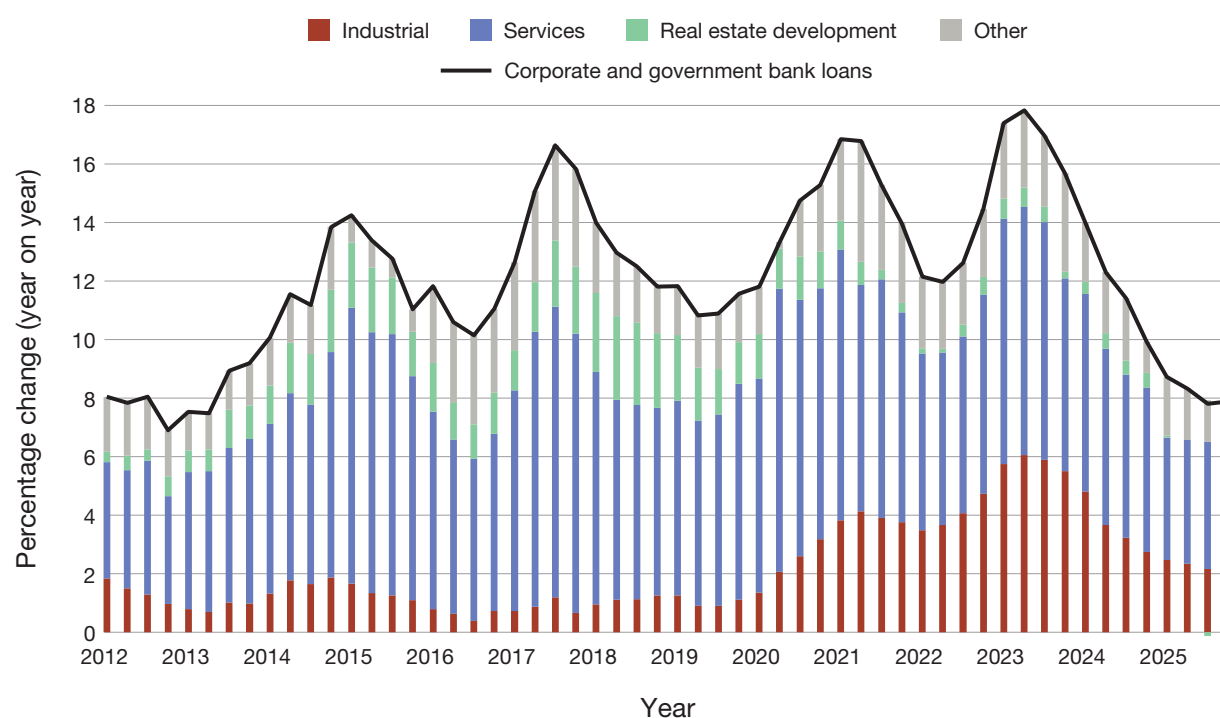
²⁴ OECD, *The Market Implications of Industrial Subsidies*, TAD/TC(2025)7/FINAL, May 28, 2025d.

²⁵ Ying Xu, *Industrial Policy and State Ownership: How Do Commercial Banks Allocate Credit in China?* International Monetary Fund, WP/24/262, December 23, 2024.

and clean energy supply chains.²⁶ These activities are not always categorized as “industrial policy” in the narrow sense. Still, they shape the environment in which manufacturing firms operate and are increasingly tied to Beijing’s climate and “new quality productive forces” (新质生产力) agenda.

The surge in industrial loans appears to be tapering off (Figure 6.3). This likely reflects retrenchment after several years of booming manufacturing investment and falling profitability. Although it is difficult to know how many of those loans are strictly due to industrial policy, the broad direction of policy is clear; Beijing is trying to tackle falling profits and prices in key industrial sectors with its “anti-involution” (反内卷式竞争) campaign to rein in excessive competition. Local governments and state-owned banks have responded, and new industrial loans are no longer expanding at the previous pace. As a result, China’s credit-driven industrial policy is likely slowing down.

FIGURE 6.3
The Surge in China’s Industrial Loans Is Likely Coming to an End



SOURCE: Features data from the PBOC via CEIC, undated.

NOTE: Figure includes medium- and long-term loans going to the corporate and government sectors. Industrial loans are primarily manufacturing but also include power-related infrastructure and mining. Services include other types of infrastructure.

²⁶ General Office of the State Council, “Guiding Opinions of the General Office of the State Council on Further Building a High-Quality Charging Infrastructure System,” General Office of the State Council Document No. 19 of 2023, June 19, 2023.

Capital Markets for Technology Development

Beijing has turned domestic capital markets into a channel for techno-industrial policy, focusing on technology listings and lower-cost debt for strategic sectors.²⁷ In China, corporate bonds and equity in the stock market remain far less important to corporate finance than bank loans.²⁸ Still, capital markets matter for firms that cannot rely solely on retained earnings or policy-directed lending, especially young technology companies that need equity capital and advanced manufacturing firms seeking thematic bond financing.

The Shanghai Science and Technology Innovation Board (STAR Market) launched in 2019 with a registration-based IPO system and rules tailored to R&D-intensive firms, including tolerance for pre-profit listings. Listing reviews are exchange-led with China Securities Regulatory Commission registration, and the Shanghai Stock Exchange's guidance sets S&T attribute indicators to screen for "hard technology" (硬科技).²⁹

In 2020, ChiNext, a board of the Shenzhen Stock Exchange created for smaller, fast-growing firms, shifted to a registration-based IPO system, replacing heavy regulatory approval with disclosure requirements and market-driven pricing.³⁰ This reform made it easier for innovative, high-growth companies in high-tech sectors to raise funds domestically, providing them with a listing venue similar to Nasdaq in the United States.³¹

To further broaden this multi-tier structure, in 2021 Beijing created the Beijing Stock Exchange (BSE) by upgrading the "select tier" of China's over-the-counter (non-exchange) system for stock trading into a national exchange dedicated to innovation-oriented SMEs.³² The BSE uses a registration-based system with more inclusive listing thresholds than the STAR Market or ChiNext and targets specialized, "Little Giant" firms in such sectors as advanced manufacturing, new energy, biomedicine, and AI.³³ By mid-2025, nearly 300 companies were listed on the BSE, with aggregate market capitalization at more than ¥900 billion and a surge in IPO applications from small-cap technology firms, making the exchange an increasingly important equity-financing channel for China's "new quality productive forces" (新质生产力), even if it still trails the main boards in overall scale (Figure 6.4).³⁴

²⁷ China Securities Regulatory Commission, "China Securities Regulatory Commission's Eight Measures on Deepening the Reform of the Science and Technology Innovation Board to Serve Technological Innovation and the Development of New Productivity," June 19, 2024.

²⁸ As of August 2025, the value of corporate bonds was about ¥33.5 trillion, equity financing for nonfinancial enterprises totaled ¥12 trillion, and corporate bank loans stood at ¥183 trillion, according to PBOC statistics in CEIC, undated.

²⁹ China Securities Regulatory Commission, "Trial Measures for the Registration Management of Initial Public Offerings of Stocks on the Science and Technology Innovation Board," Order No. 153, March 1, 2019; Shanghai Stock Exchange, "Science and Technology Innovation Board: Issuance Conditions and Listing Requirements," webpage, undated. As of August 2025, companies listed on the STAR Market had raised nearly ¥1.1 trillion in equity capital, compared with ¥4.6 trillion and ¥2.9 trillion on the normal Shanghai and Shenzhen markets, respectively, during the same period (data from Beijing Stock Exchange, Shanghai Stock Exchange, and Shenzhen Stock Exchange via CEIC, undated).

³⁰ China Securities Regulatory Commission, "Trial Measures for the Registration Administration of Initial Public Offerings of Stocks on the ChiNext Board," Order No. 167, June 12, 2020.

³¹ China Securities Regulatory Commission, 2020.

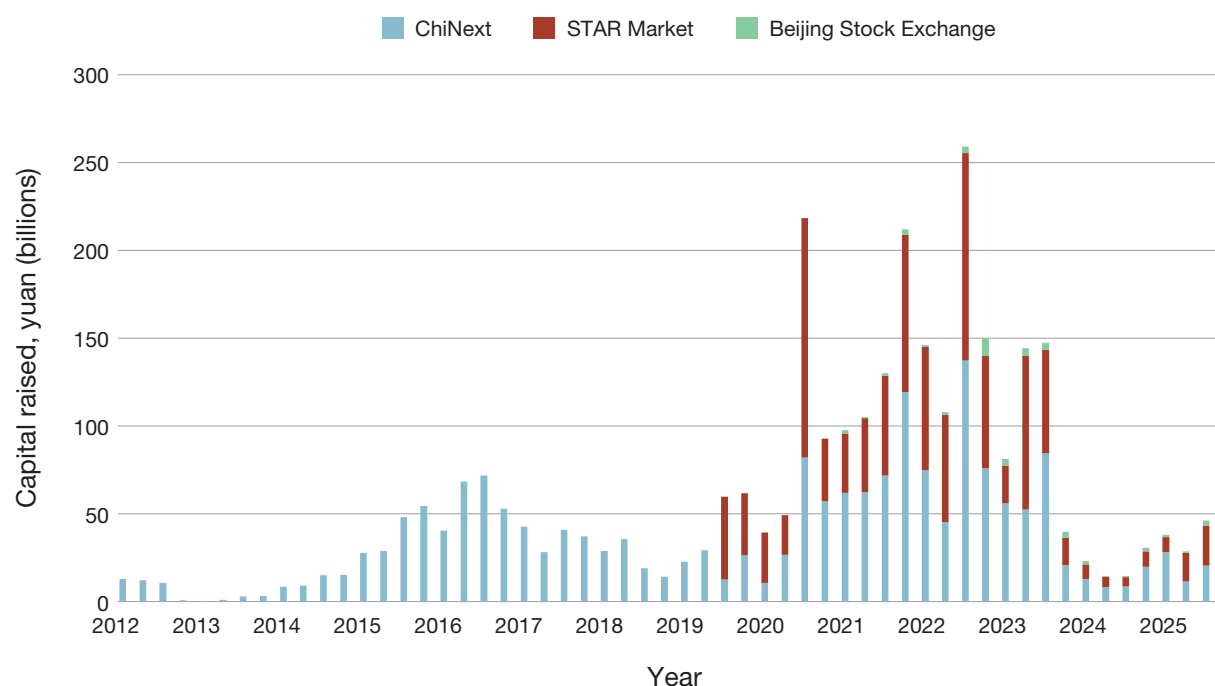
³² Tom Daly, "Xi Says China to Set Up Beijing Stock Exchange for SMEs," Reuters, September 2, 2021; Lerong Lu and Jiujing Ye, "Beijing Stock Exchange and Multi-Tier Capital Markets: How China Alters Share Listing and Trading Rules to Promote SME Finance?" *Company Lawyer*, Vol. 43, 2022.

³³ Policy and media framing consistently link the BSE to "little giants" and to "new quality productive forces." Recent *People's Daily* coverage notes that a large share of BSE firms operate in high-end manufacturing, new energy, biomedicine, and AI. See Wang Yi, "Steady Growth of BSE Firms Highlights China's Growing Innovation Vitality," *Global Times*, November 12, 2025.

³⁴ By September 2025, the BSE hosted around 274 companies (versus 2,284 in the Shanghai Stock Exchange and 2,871 in the Shenzhen Stock Exchange), with total market capitalization exceeding ¥920 billion and the average fundraising size for new listings climbing from roughly ¥191 million in 2023 to ¥333 million in 2025. See "Four Years Since Establishment: Beijing

FIGURE 6.4

Chinese Firms Are Raising Less Capital on Specialized Stock Markets Than in 2020–2023



SOURCE: Features data from the Beijing Stock Exchange, Shanghai Stock Exchange, and Shenzhen Stock Exchange via CEIC, undated. Latest observation is partial 2026.

In 2023, China implemented registration across all boards, cementing the shift from approval to disclosure-centered issuance.³⁵ The 2024 “New Nine Rules” (新“国九条”) refocused on quality over quantity by tightening IPO gates and improving delisting and investor protection while reiterating support for financing that serves technology self-reliance and “new quality productive forces” (新质生产力).³⁶

On the debt side, regulators stood up product lines that map to industrial policy goals. In 2021, Beijing issued a Unified Green Bond Catalogue, which notably removed “clean coal” from the list of eligible projects and aligned green definitions across markets.³⁷ The interbank market introduced “carbon neutral bonds” (碳中和债) for projects in clean energy, clean transport, and industrial decarbonization.³⁸ In 2022–2023, bond regulators rolled out a work plan to boost “sci-tech corporate bonds,” including a fast track for qualified

Stock Exchange’s Total Market Cap Exceeds 920 Billion Yuan,” People’s Government of Beijing Municipality, September 22, 2025; William Sandlund and Haohsiang Ko, “‘No Longer a Pet Project’: Tiny Beijing Stock Exchange Becomes IPO Hub,” *Financial Times*, July 17, 2025.

³⁵ “The Full Implementation of the Stock Issuance Registration System Has Been Officially Put into Effect,” *Xinhua*, February 17, 2023.

³⁶ State Council of the People’s Republic of China, “Opinions on Strengthening Supervision, Preventing Risks, and Promoting High-Quality Development of the Capital Market,” State Council Document No. 10 of 2024, April 12, 2024b.

³⁷ People’s Bank of China, National Development and Reform Commission, and China Securities Regulatory Commission, “Notice on Issuing the ‘Catalogue of Green Bond-Supported Projects (2021 Edition),’” People’s Bank of China Document No. 96 of 2021, April 2, 2021.

³⁸ National Association of Financial Market Institutional Investors, “Carbon Neutral Bonds,” webpage, undated-a.

issuers and measures to deepen liquidity.³⁹ Regulators also stress channeling bond issuance to “specialized, refined, distinctive, and innovative” (SRDI; 专精特新) SMEs (firms recognized for their focus, quality, and innovation), as well as to advanced manufacturing companies.⁴⁰

Despite these reforms, capital markets remain a secondary, though increasingly supportive, tool of techno-industrial policy. These markets expand access to high-tech listings and thematic bonds, but many SMEs still struggle with listing requirements or the cost of debt. Bank finance remains the dominant source of funding.

Recognizing these limits, in mid-2025, Beijing issued a framework to accelerate development of a “science and technology financial system.” The plan integrates venture capital, credit, bonds, and insurance products to channel long-term funding into high-tech development. It expands tech-themed credit guarantees, creates a bond market “Sci-Tech Board,” and broadens data-based credit scoring for innovative SMEs. Taken together, these measures are meant to build a unified financial infrastructure that supports China’s “new quality productive forces” and reduces the system’s reliance on bank-led financing.⁴¹

³⁹ National Association of Financial Market Institutional Investors, “Science and Technology Innovation Notes (Purpose-Specific),” webpage, undated-b.

⁴⁰ Tianjin Securities Regulatory Bureau, “Notice on Promoting the Financing of Science and Technology Innovation Corporate Bonds,” May 15, 2023.

⁴¹ Ministry of Science and Technology, People’s Bank of China, State Financial Regulatory Administration, China Securities Regulatory Commission, National Development and Reform Commission, Ministry of Finance, and State-owned Assets Supervision and Administration Commission of the State Council, “Several Policy Measures to Accelerate the Construction of a Science and Technology Finance System to Strongly Support High-Level Scientific and Technological Self-Reliance and Self-Strengthening,” May 13, 2025.

Below-Market Inputs—Real Economy Levers: Inputs and Industrial Zones

Under Xi, below-market inputs persist but have been more tightly disciplined and redirected toward strategic sectors. Parks and zones have moved from export-processing platforms to innovation- and manufacturing-oriented ecosystems. This shift tracks China's broader development path, as the economy has moved from catch-up growth and processing to higher-value-added, more-innovative production.

Discipline and Redirection of Input Cost Advantages

Chinese localities continue to transfer land for industrial development at prices lower than for other uses. Most land sales in China by volume are for industrial uses—even though residential land sales accounted for 75 percent of land sales by value during 2021–2025 (Figure 7.1).¹ Using China's national land registry (covering about 1.6 million land transactions between 2010 and 2023), the IMF estimated the implicit subsidy by comparing the prices of plots sold to manufacturing firms with those sold to nearby nonmanufacturing buyers in the same time frame and within a 1-kilometer radius. The analysis found that industrial land sold to manufacturing firms was priced roughly two-thirds below comparable nonmanufacturing land.² Industrial land prices were relatively flat over time compared with residential prices, suggesting that the implicit land subsidy for industry was larger during China's residential property boom but that it is now declining along with residential prices.

Since the mid-2000s, Beijing has required industrial land to be supplied via auctions and at a price above a published national minimum transfer price.³ In practice, provinces now channel discounts to priority projects by listing favored industries in official catalogues and permitting lower reserve prices when land is auctioned to those projects. At the same time, policy explicitly allows discounted reserve prices (down to 70 percent of the standard) for provincially prioritized projects, which is how land discounts are now channeled toward selected industries.⁴ Even as Beijing seeks to curb land giveaways, fiscal incentives that link local revenue to industrial activity (value-added taxes and corporate income taxes) help ensure that underpriced industrial land remains the norm.

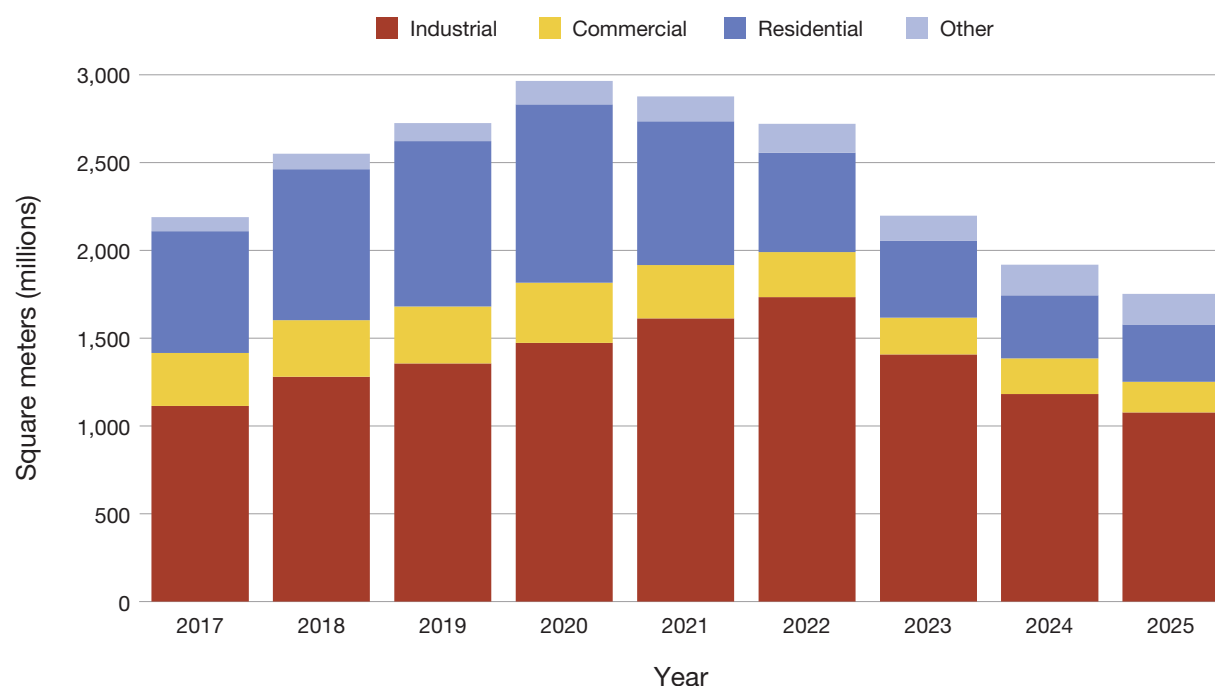
¹ China Real Estate Information Website data via CEIC, undated.

² Garcia-Macia, Kothari, and Tao, 2025.

³ Ministry of Land and Resources of the People's Republic of China, "Regulations on the Transfer of State-Owned Construction Land Use Rights by Tender, Auction and Listing," revised September 21, 2007.

⁴ Ministry of Land and Resources of the People's Republic of China, "Notice on Adjusting the Implementation Policy of the National Minimum Price Standard for Industrial Land Transfer," National Development and Reform Commission, May 16, 2017.

FIGURE 7.1
Most Land Sales by Volume Are for Industrial Uses



SOURCE: Features data from the China Real Estate Information Website via CEIC, undated.

Electricity pricing has moved in the same direction, exposing industry to market signals while tightening curbs on energy-intensive users. In October 2021, the NDRC allowed electricity prices to fluctuate much more widely and removed protections that had capped prices for energy-hungry firms while moving industrial and commercial users away from fixed government tariffs and into market-priced power contracts.⁵ In parallel, Beijing overhauled time-of-use pricing, instructing provinces to widen peak-valley differentials and introduce peak rates to encourage firms to shift load rather than seek bespoke discounts.⁶

Beijing has also sought to shut the door on special deals for emblematic “two-high” (两高) users, a policy term for high-energy-consuming, high-emission projects. Since 2021, multiple central notices have sought to curb the blind expansion of “two-high” projects as part of China’s carbon-peaking agenda. In August 2021, regulators forbade preferential tariffs and bespoke market sessions for electrolytic aluminum, tightened the sector’s tiered pricing, and directed provinces to cancel any existing discounts.⁷ Subsequent provincial notices reiterate the ban and embed enforcement in annual reviews.⁸

⁵ “China Deepens Market-Oriented Reform for On-Grid Price of Coal-Fired Power Generation,” National Development and Reform Commission, October 14, 2021; NDRC, “Notice on Further Deepening the Market-Oriented Reform of On-Grid Electricity Prices for Coal-Fired Power Generation,” National Development and Reform Commission Price Document No. 1439 of 2021, October 12, 2021d.

⁶ NDRC, “Notice on Further Improving the Time-of-Use Electricity Pricing Mechanism,” National Development and Reform Commission Price Document No. 1093 of 2021, July 29, 2021b.

⁷ NDRC, “Notice on Improving the Tiered Electricity Pricing Policy for the Electrolytic Aluminum Industry,” National Development and Reform Commission Price Document No. 1239 of 2021, August 27, 2021c.

⁸ Shaanxi Provincial Development and Reform Commission, “Forwarding the National Development and Reform Commission’s ‘Notice on Improving the Tiered Electricity Pricing Policy for the Electrolytic Aluminum Industry,’” September 17, 2021.

The state now promotes strategic loads through targeted locations and contracts rather than broad subsidies. Green power trading, launched in 2021 and standardized nationally in 2024, allows large users to secure renewable energy through bundled electricity and certificates, supporting decarbonization without tariff breaks.⁹ Provinces, such as Gansu, Guizhou, and Inner Mongolia, are cutting power costs for data centers using domestic AI chips to offset higher energy needs, encouraging local semiconductor use.¹⁰ The East-Data, West-Computing program directs data-center growth to renewable-rich regions by linking favorable power and land terms to digital infrastructure goals rather than providing universal discounts.¹¹

Industrial Parks as Innovation Platforms

China's parks and zones have shifted from export bases to platforms for high-tech manufacturing. Before Xi, local governments used land, tax breaks, and plug-and-play infrastructure to attract FDI. Under Xi, parks have been recast as innovation manufacturing hubs aligned with AI, biopharmaceuticals, and green energy. Financing is increasingly routed through local special-purpose bonds and LGFVs, alongside park-based service platforms that bundle intellectual property support, testing, and pilot-line access.¹² In Wuhan's East Lake High-Tech Zone, for example, Wuhan Zhongguancun Hard Space Technology operates a one-stop platform that combines pilot testing, policy consultation, and intellectual property services, with on-site labs that enable firms to iterate on products within the park.¹³

After the first pilot free trade zone in Shanghai in 2013, free trade zones rolled out nationwide and were later capped by the 2020 Hainan Free Trade Port plan.¹⁴ In parallel, Beijing designated national "advanced manufacturing clusters" to concentrate supply chains and research assets and, in 2024, launched a large-scale equipment renewal drive that pushes public buyers and SOEs to refresh capital stock with domestic technologies.¹⁵

As of late 2023, China counted 178 national high-tech zones whose firms produced ¥12.33 trillion in output in the first three quarters alone, highlighting their weight in advanced manufacturing.¹⁶ These are just the most successful subsets of zones. More broadly, according to the most recent catalogue, China has 2,543 zones at the national and provincial levels (municipal and county-level zones are not included). Since 2022, the composition of local special-purpose bond issuance has shifted away from municipal and indus-

⁹ Colleen Howe, "China Issues Guidelines on Green Power Trading," Reuters, August 23, 2024.

¹⁰ Zijiang Wu and Eleanor Olcott, "China Offers Tech Giants Cheap Power to Boost Domestic AI Chips," *Financial Times*, November 3, 2025.

¹¹ NDRC, "An Official from the High-Tech Department of the NDRC: The 'Eastern Data, Western Computing' Initiative Has Fully Launched, with Eight Hubs Powering New Data Vitality," March 21, 2022.

¹² MOF, "Notice on Accelerating the Issuance and Use of Local Government Special Bonds," Finance and Budget Document No. 94 of 2020, July 27, 2020a; "China's Number of National High-Tech Zones Increases to 178: Official," Xinhua, December 12, 2023.

¹³ Wuhan East Lake High-Tech Development Zone Administrative Committee, "Optics Valley Has Established 58 Service Stations for Small and Medium-Sized Enterprises (SMEs), with 'Enterprise Service Specialists' Making Regular Visits to These Businesses," September 4, 2023.

¹⁴ "China Releases Master Plan for Hainan Free Trade Port," Xinhua, June 1, 2020.

¹⁵ "MIIT Announces 45 National-Level Advanced Manufacturing Clusters, Boosting High-Quality Dev't of Industry," Xinhua Silk Road Information Service, December 1, 2022.

¹⁶ "China's Number of National High-Tech Zones Increases to 178: Official," 2023.

trial park infrastructure and toward refinancing and swaps, indicating that fewer new park projects are being financed.¹⁷

Geographic and industry focuses are shifting, with concentration in university cities and second-tier hubs. Such zones as Hefei and Changzhou combine land, labs, and procurement to reduce transaction costs and accelerate growth but face duplication and debt risks.¹⁸ Beijing is responding by tightening regulations on special bonds and project approvals.¹⁹

Trade and Foreign Investment: Selective Opening for Capacity-Building

China has shifted trade and investment policies to selectively open sectors that help build domestic capabilities, especially in manufacturing. In 2024, the national Foreign Investment Negative List was cut to 29 items, removing all manufacturing restrictions.²⁰ Officials call this a “high-standard opening” (高水平开放), balancing openness to lower costs with caution in areas involving data and security.²¹ The 2026 Government Work Report emphasizes incremental reductions to service-sector negative lists and controlled digital-sector opening.²²

Beijing has trimmed the negative list since 2018 and dismantled automobile ownership caps on a published timetable, ending passenger car limits in 2022.²³ That sequence enabled Tesla to establish a wholly owned plant in Shanghai, often cited as a “catfish effect” (鲶鱼效应)—a Chinese business metaphor for introducing a competitive outsider to spur incumbents—in which a leading foreign entrant pressures domestic firms to upgrade.²⁴

China also uses tariff and export rebate policies as targeted industrial policy tools rather than as a general mechanism for lowering trade costs. Tariff reductions, such as the 2018 cuts on 1,585 lines (many covering inputs and capital goods), have been deployed to reduce upstream costs and support domestic production.²⁵ By contrast, export VAT rebates have increasingly been tightened to discipline supply expansion in sectors facing overcapacity, environmental constraints, or intense price competition. In 2021, China removed export VAT rebates for 146 steel products.²⁶ This approach broadened in late 2024, when authorities cut or elimi-

¹⁷ Bond quotas are increasingly earmarked to refinance existing debts, and tighter project vetting is weeding out duplicative or low-quality park builds.

¹⁸ “Along Low-Carbon Pathways, How This Chinese City Thrives on New-Energy Ambitions,” Xinhua, February 15, 2024; Arch Daily, “NIO House Hefei Xinqiao Industrial Park / SHL,” webpage, May 21, 2024.

¹⁹ MOF’s midyear fiscal report calls for tilting quotas toward well-prepared, efficient projects and tightening debt risk control (MOF, *Report on the Implementation of China’s Fiscal Policy in the First Half of 2024*, August 30, 2024b).

²⁰ “China Fully Opens Manufacturing Sector to Foreign Investors in Landmark Opening Up Move,” Xinhua, November 1, 2024; “The New National Negative List for Foreign Investment Access Has Been Implemented, and Restrictions on Foreign Investment Access in the Manufacturing Sector Have Been Eliminated (Policy Interpretation),” *People’s Daily*, November 19, 2024.

²¹ “Xi Stresses Promoting High-Standard Opening-Up for Reform, Development,” Xinhua, October 20, 2024.

²² Li, 2026.

²³ Pei Jianru, “Foreign Ownership Restrictions on Passenger-Vehicle Manufacturing Have Been Lifted, Leading Multinational Automakers to Increase Their Investment in China,” *Securities Times*, December 22, 2022.

²⁴ Hua Ping, “Tesla’s ‘Catfish Effect’ and China’s 10-Million-Unit Milestone in NEV Production,” *People’s Daily*, December 4, 2024.

²⁵ “China to Cut Import Tariffs on 1,585 Taxable Items,” Xinhua, September 26, 2018.

²⁶ Ministry of Finance and State Taxation Administration, “Announcement on the Cancellation of Export Tax Rebates for Certain Steel Products,” Announcement No. 16 of 2021, April 26, 2021.

nated rebates for a variety of products, including metals, refined oil products, batteries, and photovoltaic goods. In January 2026, policymakers went further, eliminating export VAT rebates for photovoltaic products and initiating a phased reduction and eventual removal of rebates for batteries.²⁷ These measures indicate a shift from using rebates to support export competitiveness toward using them to manage industrial structure and external frictions.

In addition to export rebate policy, financial market access has widened since around 2022. New panda bond (renminbi-denominated bonds issued by foreign entities in China) rules took effect in early 2023. They made it easier for overseas issuers to raise renminbi onshore, while the launch of Swap Connect in 2023 gave offshore investors access to onshore interest-rate swaps to hedge China exposures.²⁸ In 2025, the central bank opened interbank bond repurchase agreement transactions to eligible overseas institutions.²⁹ These moves lower funding costs and improve risk management for firms that operate in China while keeping supervisory levers intact.

Alongside the negative list, Beijing maintains the Catalogue of Encouraged Industries for Foreign Investment, effectively a positive list of sectors and regions in which foreign investment is actively promoted. The 2025 catalogue, effective February 1, 2026, continues to prioritize advanced manufacturing nationwide while expanding support for modern services, high technology, and energy-saving and environmental protection sectors.³⁰ It also steers foreign investment toward region-specific industries in central, western, and north-eastern China, as well as Hainan, in line with local resource endowments and industrial strengths. Encouraged projects may import qualifying self-use equipment free of customs duties and can receive priority land supply. Relative to the 2022 version, the catalogue expanded further, reinforcing Beijing's push to channel foreign capital into hard-tech, greener industries and targeted regional development.³¹

Service openness remains modest by comparison. International indices still place China above the OECD average on service restrictiveness, with telecommunications among the tighter sectors (Figure 7.2).³² A separate index of FDI rules points in the same direction, placing China on the more restrictive side of the cross-country distribution relative to OECD norms.³³ The 15th FYP signals scope for selective opening in services.³⁴ The Party's recommendations call for expanding service-sector opening, with a particular emphasis on opening telecommunications, internet, education, culture, and medical services.

Data policy is a clear example of selective openness in services, in which routine, low-risk tasks are made easier while core controls remain in place. Rules issued in March 2024 outline common situations in which

²⁷ Shi Bu and Ryan Woo, "China to Scrap Export Tax Rebates for Photovoltaic and Battery Products," Reuters, January 9, 2026.

²⁸ "China Releases New Rules for Panda Bonds to Facilitate Overseas Issuers," Xinhua, December 2, 2022; Hong Kong Monetary Authority, "The People's Bank of China, Securities and Futures Commission and Hong Kong Monetary Authority to Commence Mutual Access Between the Mainland and Hong Kong Interest Rate Swap Markets," press release, May 5, 2023.

²⁹ PBOC, "Officials from the PBOC and the SAFE Addressed Press Questions About the Announcement on Further Supporting Overseas Institutional Investors in Conducting Bond Repurchase Business," press release, last updated September 26, 2025.

³⁰ National Development and Reform Commission and Ministry of Commerce of the People's Republic of China, "Catalogue of Industries Encouraging Foreign Investment (2025 Edition)," Order No. 37 of 2025, December 24, 2025.

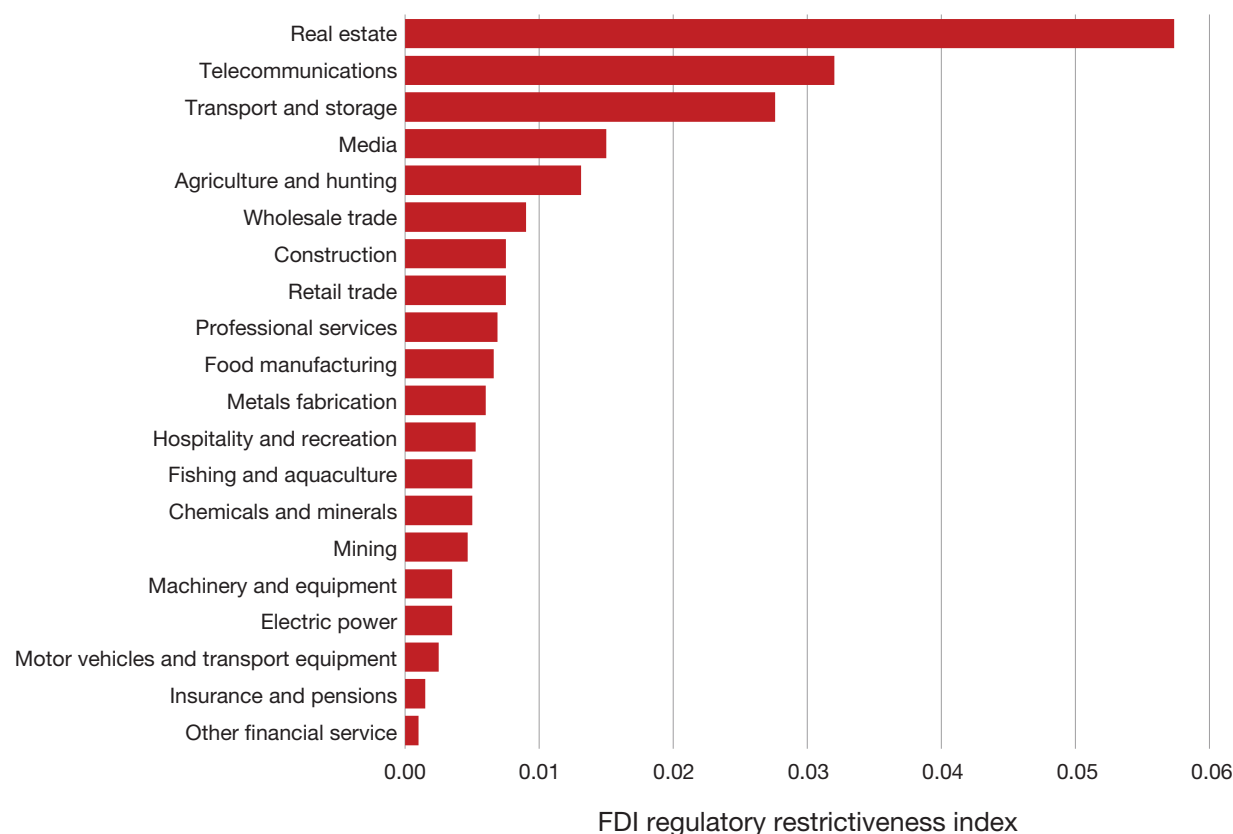
³¹ "Nation Opens Up Wider to Foreign Investors," *China Daily*, March 21, 2024; Qian Zhou, "China Released Draft Encouraged Catalogue to Boost Foreign Investment," China Briefing, December 24, 2024.

³² OECD, "Services Trade Restrictiveness Index: China," 2025b.

³³ OECD, *OECD FDI Regulatory Restrictiveness Index 2024: Key Findings and Trends*, OECD Business and Finance Policy Papers, No. 93, 2025a.

³⁴ "Outline of the Fifteenth Five-Year Plan for National Economic and Social Development of the People's Republic of China," 2026.

FIGURE 7.2

China Restricts FDI in the Service Sector More Than in Manufacturing

SOURCE: Features data from OECD, "FDI Regulatory Restrictiveness Index—Scores," dataset, 2024.

NOTE: This figure uses the OECD's revised FDI Regulatory Restrictiveness Index Scores series. Sectors are shown for 2024 (latest available). The sector labels are shortened from OECD definitions.

cross-border transfers of nonpersonal or nonsensitive data do not require a security review.³⁵ In 2025, financial regulators added a guideline that lists specific business cases for compliant cross-border use of financial data and explains the documentation that firms should prepare.³⁶ These steps give companies already in the market a clearer path to run day-to-day R&D, analytics, payments, and reporting across borders. They reduce friction but do not change who can enter the market or loosen restrictions on sensitive services, such as telecommunications and cloud computing.

³⁵ China Law Translate, "Provisions on Promoting and Regulating the Cross-Border Flow of Data," webpage, March 25, 2024.

³⁶ "Six Departments Issue Document to Promote and Regulate the Cross-Border Flow of Financial Data," Xinhua, April 17, 2025.

Integrating the State and the Market: The Party-State and Its Firms

Beijing has tried to integrate state and private enterprises into a coordinated corporate system to advance its techno-industrial strategy. SOEs remain the backbone of heavy industry, infrastructure, and “commanding heights” sectors that secure national capacity, while private firms drive technological innovation in competitive and fast-moving fields. Through political supervision, regulatory control, and selective incentives, Beijing has embedded coordination across both. The result is a managed hierarchy in which market actors operate within Party-defined boundaries to serve national technological and strategic objectives.

SOEs: From Commanding Heights to New Frontiers

Since Xi became leader in 2012, SOEs have been further integrated into China’s industrial strategy. The CCP increased political oversight through anticorruption efforts and the “two-way entry, cross appointment” (双向进入、交叉任职) system, making Party secretaries also serve as board chairs.¹ Xi emphasized SOEs as essential to national security and the economy, shifting their focus from pure commercial performance toward strategic security and resilience.² In effect, SOEs have been reinforced as the state’s preferred instrument in upstream, capital-intensive, and security-sensitive sectors, while private firms remain more important in competitive, innovation-driven fields.

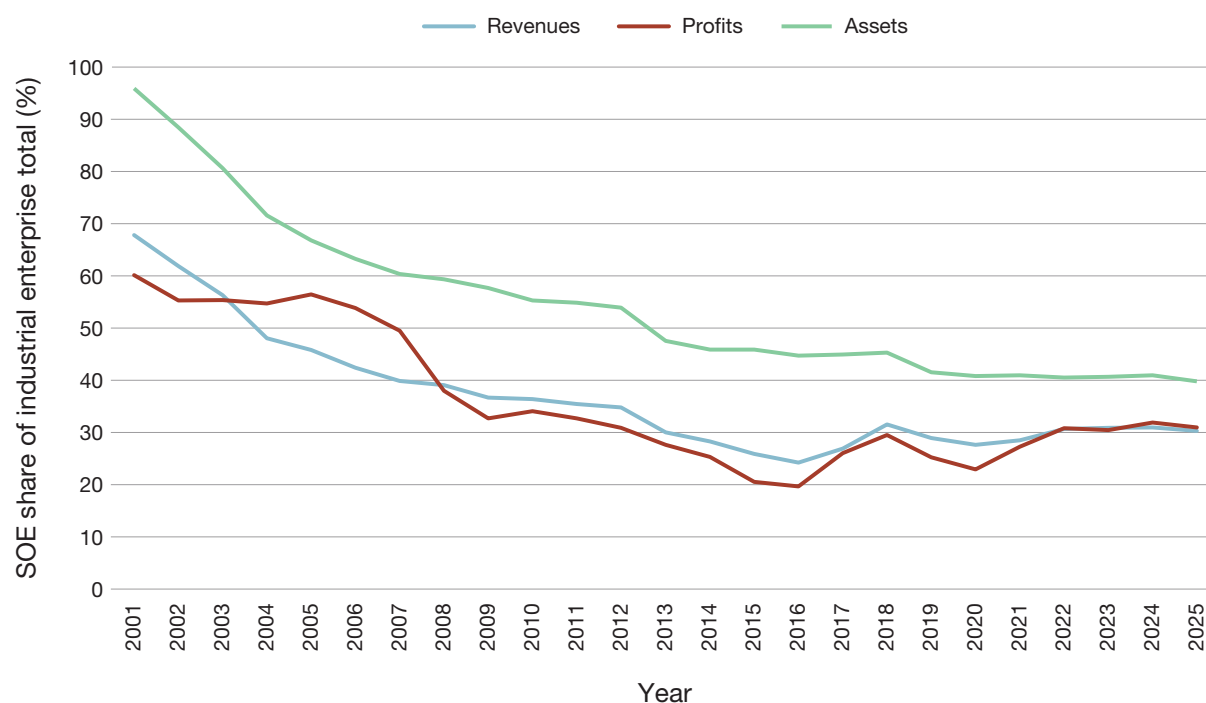
Several major reforms have reshaped SOEs. Mixed-ownership reform, announced at the Third Plenum in 2013 and piloted in 2014–2015, sought to bring in private and foreign capital to broaden funding sources, reduce debt burdens, and expose SOEs to market discipline without surrendering Party control (Figure 8.1).³

¹ The “two-way entry, cross appointment” system was introduced in 2016 by Xi as a way of embedding CCP leadership directly into the governance and management of SOEs. See Party Leadership Group of Shenhua Group Co., Ltd., “Promoting the Implementation of Strict and Comprehensive Party Governance Requirements in State-Owned Enterprises,” *Qiushi*, No. 7, March 31, 2017. For an example of Party doctrine being embedded in corporate governance, see Central Committee of the Communist Party of China, “Regulations on the Work of Primary-Level Organizations of the Communist Party of China in State-Owned Enterprises (Trial Implementation),” December 30, 2019, which mandates that SOE Party organizations be “guided by Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era” and lead ideological and political work.

² “Xi Jinping Presided over the Fourth Meeting of the Central Leading Group for Comprehensively Deepening Reforms,” *Xinhua*, August 18, 2014; Liu Zhiqiang, “Reforms Help State-Owned Enterprises Achieve High-Quality Development,” *People’s Daily*, June 13, 2022b.

³ Central authorities began pilots in 2014. SASAC selected China National Building Materials and Sinopharm as the first mixed-ownership pilots that year. Broader, batched pilots (including employee stock ownership at mixed-ownership firms) rolled out from 2016 onward. See Weng Jieming, “Actively and Orderly Promote the Mixed Ownership Reform of State-Owned Enterprises in the New Era,” *People’s Weekly*, undated; and Central Committee of the Communist Party of China, “Decision of the Central Committee of the Communist Party of China on Several Major Issues Concerning Comprehensively Deepening Reform,” *Xinhua*, November 15, 2013.

FIGURE 8.1
China's Industrial SOEs Stopped Downsizing Around 2015



SOURCE: Features data from the National Bureau of Statistics via CEIC, undated.

NOTE: Shares are calculated as SOE revenue, profits, and assets divided by total revenue, profits, and assets for industrial enterprises in the National Bureau of Statistics industrial enterprise data.

In practice, this reform recapitalized SOEs and allowed for minority private stakes, but decisionmaking remained firmly in state hands.⁴

In 2015, Beijing issued the “Guiding Opinions on Deepening the Reform of SOEs,” the blueprint for Xi-era SOE reform.⁵ This classified SOEs into either “commercial” enterprises expected to compete globally and generate profits or “public welfare” enterprises tasked with advancing national objectives, such as energy security, infrastructure, and defense. The policy entrenched SOEs as differentiated instruments of industrial policy, clarifying which were to be run like businesses and which were to function as policy arms.

The Three-Year Reform Action Plan (2020–2022) required SOEs to be formally converted into limited-liability or joint-stock companies, standardizing their legal status and governance structures, and pushed the disposal of “zombie” firms.⁶ SASAC declared the program a success by claiming that more than 98 percent of central SOEs had completed corporatization, but outside observers see limited evidence of real efficiency

⁴ A flagship case is Sinopec. In September 2014, its sales subsidiary received ¥107.094 billion in new equity from 25 domestic and foreign investors for 29.99 percent of shares (Sinopec retained 70.01 percent). See China Petroleum and Chemical Corporation, *Announcement on Capital Increase and Introduction of Investors in Sinopec Sales Co., Ltd.*, Announcement No. Lin 2014-040, September 12, 2014.

⁵ Chinese Communist Party Central Committee and the State Council, “Guiding Opinions on Deepening the Reform of State-Owned Enterprises,” LawInfoChina, August 24, 2015.

⁶ “Observations on the Three-Year Action Plan for State-Owned Enterprise Reform,” Xinhua, February 9, 2023.

gains or governance changes.⁷ The reforms largely strengthened balance sheets and created room for SOEs to play a bigger role in industrial policy rather than transform them into market-driven firms.

Beijing also pursued SOE consolidations to create bigger “national champions,” resolve chronic overcapacity, reduce loss-making duplication, and improve capital efficiency. The number of central SOEs overseen by SASAC fell from 117 in 2012 to 97 by 2022 and, after subsequent additions and reorganizations, stood at 100 as of July 2025.⁸

SOEs are explicitly charged with leading large, state-driven projects. They have constructed 5G networks (China Tower),⁹ expanded data centers (China Telecom, China Mobile),¹⁰ and invested in such sectors as aerospace, nuclear power, and advanced materials.¹¹ Internationally, SOEs—such as China Communications Construction Company, PowerChina, and China National Petroleum Corporation—have built infrastructure and secured resources, supporting such initiatives as the BRI.¹² In October 2025, SASAC launched a ¥51 billion fund backed by central SOEs to support strategic sectors, such as AI and quantum technology.¹³

Despite their size, SOEs are less efficient and innovative than private firms, mainly dominating heavy industry, energy, and infrastructure but lacking in emerging technology fields (Figure 8.2).¹⁴ Therefore, Chi-

⁷ “The Corporatization Reform of Central State-Owned Enterprises Has Been Basically Completed,” State-owned Assets Supervision and Administration Commission of the State Council, April 25, 2018. For outside observer critiques, see Barry Naughton and Briana Boland, *CCP Inc.: The Reshaping of China’s State Capitalist System*, Center for Strategic and International Studies, January 31, 2023, Chapter 2, which shows that the SOE Reform Three-Year Action Plan formalized Party “pre-decision” authority and policy key performance indicators, such that “profit and efficiency do not have a presumptive claim on enterprise decisionmaking” (p. 10). See also Daniel H. Rosen, Charles Austin Jordan, Laura Gormley, and Matthew Mingey, *Fire Sale: Prospects for SOE Privatization in China*, Rhodium Group, May 9, 2025, which finds that, since 2012, reform has waned, Party oversight has intensified, and “the advance of the state in the economy has sapped China’s productivity” (p. 3).

⁸ Publicity Department of the Central Committee of the Communist Party of China, “Press Conference on Reform and Development of State-Owned Assets and State-Owned Enterprises in the New Era,” State Council Information Office of the People’s Republic of China, June 17, 2022; “China Inaugurates Changan Auto as Central SOE,” Xinhua, July 29, 2025. Examples of these additions and reorganizations include the 2016 merger of COSCO and China Shipping to create China COSCO Shipping Corporation; the 2016 consolidation of Baosteel and Wuhan Iron and Steel to form China Baowu Steel Group, followed by Baowu’s acquisitions of Magang/Masteel (2019), TISCO (2020), Sinosteel (approved in 2022), and control of Xinyu Steel (2022), which helped Baowu become the world’s largest steel producer; the 2017 merger of Shenhua Group and China Guodian to create China Energy Investment Corporation, now among the world’s largest power companies; and the 2021 establishment of China Rare Earth Group by combining Minmetals Rare Earth, Chinalco Rare Earth, and Ganzhou Rare Earth Group to consolidate a sector critical to high-tech supply chains.

⁹ There were 412,000 5G sites added in 2024, taking cumulative 5G base stations to 2.759 million. See China Tower Corporation Limited, “2024 Annual Results,” presentation slides, March 17, 2025.

¹⁰ China Telecom reported 563,000 internet data center cabinets by the end of 2023, while China Mobile reported data centers covering all national hub nodes under “East Data, West Computing” in 2025. See China Telecom Corporation Limited, *Sustainability Report 2023*, 2023; and China Mobile Limited, *Interim Report 2025: Stride into the AI+ New Era*, 2025.

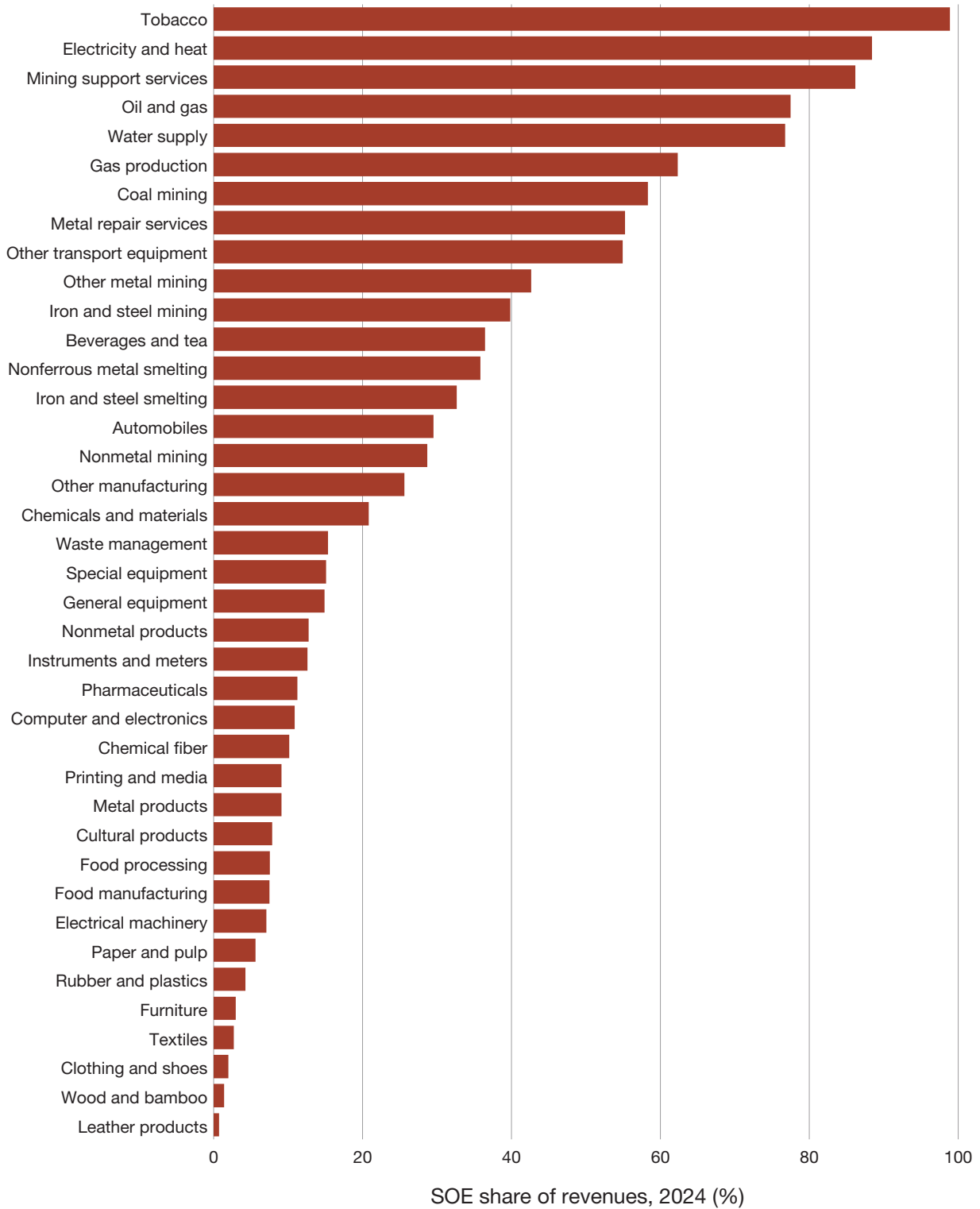
¹¹ China Aerospace Science and Technology Corporation highlights its role as the main force behind China’s human-spaceflight launches throughout the 14th FYP. See Deng Yunan, “During the 14th Five-Year Plan Period, China Aerospace Science and Technology Corporation Will Provide Full Support for the Development of the Manned Space Program,” China Aerospace Science and Technology Corporation, September 12, 2025.

¹² China Communications Construction Company Limited, “My Story with CCCC Along the Belt and Road Initiative,” webpage, undated.

¹³ Jiang Dan, “The First Phase of the Strategic Emerging Industries Fund for Central State-Owned Enterprises Is 51 Billion Yuan!” *Securities Times Online*, October 29, 2025.

¹⁴ IMF economists, using firm-level data, find that listed SOEs are systematically less productive and less profitable than private firms across sectors (Emilia Jurzyk and Cian Ruane, *Resource Misallocation Among Listed Firms in China: The Evolving Role of State-Owned Enterprises*, International Monetary Fund, WP/21/75, March 2021). World Bank research similarly shows a persistent profitability gap, with SOE returns on assets nearly 9 percentage points below those of private firms in manufac-

FIGURE 8.2

China's SOEs Generally Lead in Upstream or Heavy Industrial Sectors

SOURCE: Features data from the National Bureau of Statistics via CEIC, undated.

na's policy under Xi also actively involves private firms, including leaders in AI, robotics, drones, biotechnology, and the consumer internet. The state uses regulation, Party presence, and incentives to guide these private companies toward national strategic objectives. This dynamic is unpacked in the next section.

The result is a sectoral division of labor. SOEs dominate “commanding heights” industries, such as energy, transport, telecommunications, and defense, in which scale, stability, and security are paramount. Private firms dominate emerging industries, in which agility and rapid innovation cycles are decisive (Table 8.1). Semiconductors and EVs show partial hybrid ecosystems in which state-backed actors tend to build capital-intensive foundations (such as fabrication plants, grids, and parts of the charging network) while private firms lead many downstream segments, but coordination is uneven, and SOE advantages in credit and procurement can sideline private entrants.

TABLE 8.1
Major Private (Non-SOE) Chinese Firms

Sector/Technology	Key Firms
AI and cloud computing	Alibaba Cloud, Baidu, iFlytek, SenseTime, Tencent (Cloud)
Semiconductors (design/fabless)	Cambricon, GigaDevice, Goodix, HiSilicon (Huawei), Horizon Robotics, Montage Technology, Rockchip, VeriSilicon
E-commerce and platforms	Alibaba, ByteDance, JD.com, Kuaishou, Meituan, Pinduoduo
Robotics and industrial automation	Estun Automation, Geek+, Inovance Technology, UBTECH Robotics, Unitree Robotics
Drones	Autel Robotics, DJI
EVs	BYD, Geely, Great Wall Motor, Leapmotor, Li Auto, NIO, XPeng
EV batteries	BYD (FinDreams), CATL, EVE Energy, Gotion High-Tech, SVOLT Energy
Autonomous driving and advanced driver-assistance systems	AutoX, Momenta, Pony.ai, WeRide
Commercial space	Galactic Energy, iSpace, LandSpace, Orienspace
Solar photovoltaics (modules and materials)	Daqo New Energy, JA Solar, JinkoSolar, LONGi Green Energy, Risen Energy, Tongwei, Trina Solar
Wind turbines	Envision Energy, MingYang Smart Energy
Power electronics/inverters	GoodWe, Sungrow Power Supply
Biotechnology and life sciences	BeiGene, BGI, Innovent Biologics, WuXi AppTec, WuXi Biologics, Zai Lab
Financial technology/digital payments	Ant Group (Alipay), Tencent (WeChat Pay)
Gaming and digital media	Bilibili, NetEase, Tencent Games
Logistics and smart supply chains	Cainiao (Alibaba), JD Logistics, Meituan Delivery, SF Express

NOTE: This table lists major non-SOE (non-SASAC-controlled) Chinese firms that play central roles in strategic and high-tech sectors targeted by Beijing's industrial policy. It focuses on independent private or mixed-ownership companies that lead in innovation, exports, or market share. Many listed firms receive state guidance fund investment, procurement support, or local government backing but remain privately managed.

turing (Loren Brandt, John Litwack, Elitza Mileva, Luhang Wang, Yifan Zhang, and Luan Zhao, “China's Productivity Slowdown and Future Growth Potential,” World Bank Group, Policy Research Working Paper 9298, June 2020). Private firms in China are significantly more innovative than SOEs, with ex-SOEs performing in between (Ann Harrison, Marshall Meyer, Peichun Wang, Linda Zhao, and Minyuan Zhao, “Can a Tiger Change Its Stripes? Reform of Chinese State-Owned Enterprises in the Penumbra of the State,” National Bureau of Economic Research, Working Paper 25475, revised July 2025).

Disciplining, Steering, and Reassuring the Private Sector

Beijing employs a three-pronged approach to manage private firms: disciplining through Party-state involvement and regulatory controls, steering by elevating selected champion firms, and reassuring through policies and laws that promise fair treatment. The goal is to harness private-sector dynamism within boundaries that serve national priorities.

The CCP exerts control over private firms primarily by embedding Party cells, enacting Party-building statutes, and integrating Party members into corporate governance. State-linked ownership—including minority stakes and “special management shares” (特别管理股)—provides further leverage, enabling the Party-state to influence decisions and align private enterprise with industrial and technological goals even when the state does not hold a majority stake.¹⁵

Party organizations are widespread in private companies. The 2018 Company Law mandates that Party units “shall be established” and that companies must provide the “necessary conditions,” a requirement maintained in later revisions. The Chinese Private Enterprise Survey reported that Party units were present in half of private firms by 2018, up sharply from 2012.¹⁶

Beijing uses competition law to reinforce Party authority over dominant private firms. After Jack Ma’s 2020 criticism of regulators, authorities halted Ant Group’s IPO and forced it to restructure as a financial holding company, placing the financial technology industry under state oversight.¹⁷ In 2021, the State Administration for Market Regulation issued “Antitrust Guidelines for the Platform Economy,” penalizing Alibaba ¥18 billion for exclusivity and Meituan ¥3.4 billion for coercive contracts, ordering Tencent Music to abandon exclusive licenses, and blocking Tencent’s Huya-Douyu merger.¹⁸ DiDi was fined ¥8 billion for data-law violations after its 2021 New York IPO.¹⁹ While some cases addressed real market issues, they also curtailed private autonomy and clarified regulatory boundaries.

The second line of effort is to support and connect champion firms. The SRDI program identifies SMEs with deep technical capabilities or unique processes, designating top performers as “Little Giants” (小巨人). Initiated in 2018, the program has recognized about 14,600 Little Giants across six nationwide selection rounds by late 2024.²⁰ Nearly 90 percent operate in manufacturing, with 80 percent in SEIs.²¹ Little Giants receive targeted credit and equity support, tax incentives, and priority in public procurement. This system

¹⁵ Köncke, Erlbacher, and Schmalz, 2026.

¹⁶ Standing Committee of the National People’s Congress, Company Law of the People’s Republic of China (2018 Amendment), Order No. 15, trans. by LawInfoChina, October 26, 2018; Neil Thomas, “Party Committees in the Private Sector: Rising Presence, Moderate Prevalence,” MacroPolo Archive, Paulson Institute, December 16, 2020.

¹⁷ Tony Munroe, “China Extends Crackdown on Jack Ma’s Empire with Enforced Revamp of Ant Group,” Reuters, April 12, 2021.

¹⁸ Wei Huang, Wendy Zhou, Xiumin Ruan, and Xi Zhang, “Antitrust Guidelines for the Platform Economy in the Era of Enhanced Antitrust Scrutiny,” *CPI Antitrust Chronicle*, March 2021; Scott Murdoch and David Stanway, “China Fines Alibaba Record \$2.75 Bln for Anti-Monopoly Violations,” Reuters, April 10, 2021; Yingzhi Yang and Tony Munroe, “Chinese Delivery Giant Meituan Handed \$527 Mln Antitrust Fine,” Reuters, October 8, 2021; Joanna Tan, “China Orders Tencent to Give Up Exclusive Music Licensing Rights as Crackdown Continues,” CNBC, July 24, 2021.

¹⁹ “China Fines Didi \$1.2 Billion After Wrapping Year-Long Cybersecurity Probe,” Bloomberg, July 21, 2022.

²⁰ Hubei Provincial Department of Economy and Information Technology, “Announcement from the Ministry of Industry and Information Technology on the List of the First Batch of Specialized, Refined, Unique and Innovative ‘Little Giant’ Enterprises,” Ministry of Industry and Information Technology Enterprise Letter No. 153 of 2019, June 13, 2019; Ralph Jennings, “China’s Strategic ‘Little Giant’ Firms Quickly Rise to 14,600, Surpass 2025 Goal,” *South China Morning Post*, December 3, 2024.

²¹ Wenmin Shen, “The Number of Specialized, Refined, and Innovative ‘Little Giant’ Enterprises Has Reached 14,600,” *People’s Daily*, December 3, 2024.

creates a pool of high-tech private suppliers to fill gaps in upstream components and machinery while maintaining their formal independence from state ownership.²²

MIIT also cultivates larger, globally competitive “Single Champion” (单项冠军) manufacturers. Launched in 2016, this initiative rewards firms with at least a decade of focus on a single niche product, top three global market share, and leading production technology. By 2025, MIIT had named 1,557 Single Champions in eight national batches, mostly in the industrial machinery, materials, chemicals, and electrical equipment sectors.²³ These included recognizable firms listed in Table 8.1, such as Estun Automation in industrial robotics, CATL in lithium-ion power batteries, and LONGi Green Energy in monocrystalline silicon wafers. Like Little Giants, these champion firms benefit from priority access to national funds, provincial R&D subsidies, land and energy support, and low-cost credit.²⁴

MIIT has established more than 30 National Manufacturing Innovation Centers as of early 2025, linking private lead enterprises with SOEs, universities, and labs, with more planned. Additionally, 45 advanced manufacturing clusters host innovation centers that include both Single Champions and Little Giants. Local officials employ a “chain chief system” (链长制), in which designated officials work with lead firms that function as supply chain architects.²⁵

The third line of effort is to reassure the private sector. Beijing issued the “Opinions on Promoting the Development and Growth of the Private Economy” (the “31 Measures”) in July 2023, pledging equal treatment, improved market access, and complaint channels. In September 2023, China established an NDRC Bureau for the Private Economy to coordinate policy, monitor issues, and maintain regular dialogue with firms.²⁶ In April 2025, the National People’s Congress Standing Committee passed the Private Economy Promotion Law, a dedicated national-level “basic law” on private-economy development, which sets out cross-cutting principles and responsibilities on fair competition and equal market access, investment and financing support, innovation, and rights protection.²⁷

Finally, Beijing is trying to address long-standing late-payment problems affecting private suppliers, including overdue payments from government organs, public institutions, and large enterprises, such as SOEs. The State Council issued and revised regulations in 2020 and in early 2025 mandating that govern-

²² Alexander Brown, François Chimits, and Gregor Sebastian, *Accelerator State: How China Fosters “Little Giant” Companies*, Mercator Institute for China Studies, August 3, 2023.

²³ These are fields in which China wields global supply chain influence. See Shenzhen Science and Technology Information Platform, “A Compilation of the First to Eighth Batches of National Manufacturing ‘Single-Item Champions’: An Analysis of 1,557 Enterprises,” July 28, 2025.

²⁴ Force Distance Times, *Little Giants, Single Champions: China’s Blueprint for Asymmetric Industrial Advantage*, March 2023.

²⁵ Michael Molnar, Kelley Rogers, Susan Ipri-Brown, Joseph Long, Kimmai Tran, Frank Gayle, Clifton Ray, Albert Shih, and Douglas Thomas, *China’s Manufacturing Innovation Centers: A Benchmarking Report for the Manufacturing USA Network*, National Institute of Standards and Technology, NIST AMS 600-17, May 2025; “MIIT Announces 45 National-Level Advanced Manufacturing Clusters . . .,” 2022.

²⁶ “China Vows to Spur Growth of Private Economy,” Xinhua, July 19, 2023; “China Establishes Bureau for Private Economy Development amid Strong Policy Support,” Xinhua, September 4, 2023.

²⁷ Beijing has periodically issued major private-sector promotion policy packages, including the State Council’s 2005 “Several Opinions” on the nonpublic economy and the 2010 “Certain Opinions” on encouraging and guiding private investment, often nicknamed the “36 Articles” and “New 36 Articles.” The 2025 Private Economy Promotion Law differs in form and intended force because it is a national statute adopted by the National People’s Congress Standing Committee, described by NDRC as the first basic law specifically devoted to private-economy development. See “Law of the People’s Republic of China on the Promotion of Private Economy,” Xinhua, April 30, 2025; and Jamie P. Horsley, “China’s New Private Economy Promotion Law: Good Intentions Meet Weak Government Accountability,” *NPC Observer*, May 15, 2025.

ment agencies and large enterprises pay SMEs within 60 days.²⁸ An MIIT-led campaign requires public-sector borrowers to register and repay arrears by June 2027. As of 2025, estimates of public-sector arrears, owed by government organs, public institutions, and state-owned entities, reportedly reached ¥4.5 trillion. Separate from these, regulators have also targeted arrears owed by SOEs and LGFVs. MOF allocated ¥200 billion in special-purpose bonds and directed banks to lend against approximately ¥1.8 trillion owed by SOEs and LGFVs.²⁹

²⁸ “China Unveils New Regulations to Guarantee Payments for SMEs,” Xinhua, March 24, 2025.

²⁹ Yu Hairong, Ding Feng, Cheng Siwei, and Qing Na, “In Depth: Clearing Arrears to Businesses Tests Chinese Localities’ Resolve,” Caixin Global, October 15, 2025.

Projecting Techno-Industrial Policy Abroad

China has extended its techno-industrial policy beyond its borders, using such initiatives as the BRI to blend financial support, overseas industrial parks, digital and green infrastructure, and standards diplomacy. This comprehensive approach advances China's industrial and technological influence while aligning foreign investments with its own economic goals.

Techno-Industrial Policy Under the Belt and Road Umbrella

The BRI is Xi's flagship framework for China's economic diplomacy. Launched in 2013, it was designed to promote infrastructure connectivity, trade, and investment with China's partners. Although the BRI is not a formal industrial policy (it does not set sectoral targets or allocate subsidies), it has functioned as an indirect industrial policy by creating overseas demand for Chinese industries, such as construction, steel, rail equipment, and power generation.

Initially, Chinese policy banks funded major overseas construction projects, helping to absorb surplus capacity in heavy industries, even though China's domestic demand for those industrial goods dwarfed the scale of overseas BRI projects. Since 2019, however, new projects have slowed as Beijing has emphasized "high-quality" cooperation, and the COVID-19 pandemic disrupted economic cooperation.¹ The BRI has evolved into the main platform for China's external industrial policy, integrating state financing, technology exports, digital infrastructure, and standards promotion into a unified strategy to expand industrial and technological influence abroad (Figure 9.1).²

The Digital Silk Road (DSR), announced in 2015 as a BRI component, promotes cross-border fiber networks, 5G and core telecommunications, cloud and data centers, satellite navigation (BeiDou), smart-city systems, and digital payment rails.³ It pairs Chinese vendors with policy support and standards outreach, embedding a Chinese "digital stack" in partner markets.⁴ The DSR thus operates as an external arm of

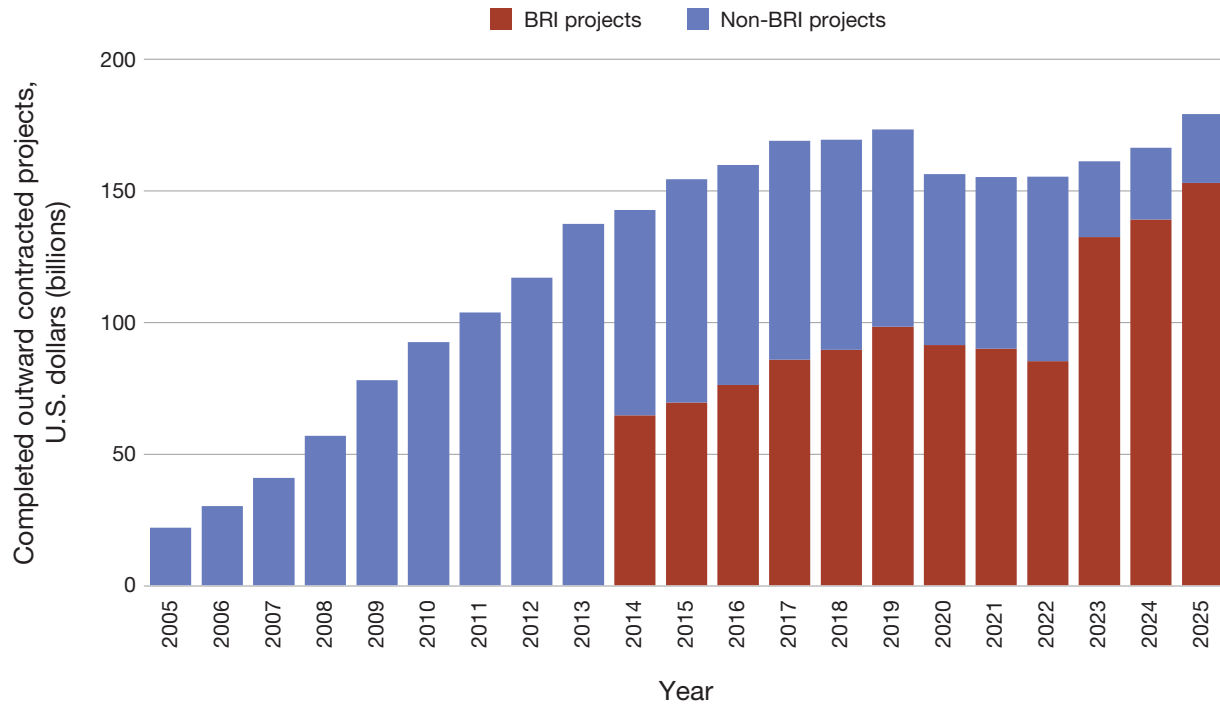
¹ Xi Jinping, "Remarks by H.E. Xi Jinping, President of the People's Republic of China, at the Press Conference of the Second Belt and Road Forum for International Cooperation," Permanent Mission of the People's Republic of China to the United Nations, April 27, 2019a; Matthew Mingey and Agatha Kratz, "China's Belt and Road: Down but Not Out," Rhodium Group, January 4, 2021.

² State Council Information Office of the People's Republic of China, *The Belt and Road Initiative: A Key Pillar of the Global Community of Shared Future*, October 10, 2023.

³ Joshua Kurlantzick, "Assessing China's Digital Silk Road Initiative: A Transformative Approach to Technology Financing or a Danger to Freedoms?" Council on Foreign Relations, January 4, 2021.

⁴ A Clingendael Institute report notes that companies assisted by the Chinese government work to promote their technical standards internationally and describes efforts to ensure interoperability on Chinese terms (Brigitte Dekker, Maaïke Okano-Heijmans, and Eric Siyi Zhang, *Unpacking China's Digital Silk Road*, Clingendael: Netherlands Institute of International Relations, July 2020).

FIGURE 9.1
China's Overseas Contracting Has Shifted Toward BRI Partners



SOURCE: Features data from the Ministry of Commerce via CEIC, undated.

NOTE: The apparent jump in 2023 BRI project values reflects expanded statistical coverage as new partner countries were added to the BRI, as well as front-loading of project completions tied to the initiative's tenth anniversary. Total overseas contracting volumes remained roughly flat, indicating a classification and timing effect rather than a surge in overall activity.

China's techno-industrial policy for the digital economy, expanding markets for its ICT firms while exporting associated technical standards and governance norms.

The most visible Xi-era shift on the energy side is the pivot away from overseas coal toward renewables, grid infrastructure, and EV-adjacent manufacturing. After Xi's 2021 United Nations pledge to end support for new overseas coal plants, Chinese firms redirected outward investment toward the clean-energy value chain, such as solar, wind, batteries, and green hydrogen.⁵ Chinese companies have announced or committed at least \$227 billion across 461 green manufacturing projects in 54 countries, with more than 88 percent of that capital committed after 2022.⁶ This "green outbound" wave aligns with China's domestic industrial strategy; it channels policy bank finance, corporate champions (CATL, BYD, LONGi), and supply chain control into global production networks. In contrast to earlier resource extraction and infrastructure projects, these ventures represent the outward expansion of China's clean technology manufacturing base and a low-carbon industrial policy exported through the BRI.⁷

Together, DSR and Green BRI instruments reorient the BRI from infrastructure toward exporting China's technology and standards (digital networks, platforms, and protocols) and creating external demand

⁵ Xi Jinping, "Bolstering Confidence and Jointly Overcoming Difficulties to Build a Better World," statement at the General Debate of the 76th Session of the United Nations General Assembly, New York, September 21, 2021.

⁶ Xiaokang Xue and Mathias Larsen, *China's Green Leap Outward: The Rapid Scale-Up of Overseas Chinese Clean-Tech Manufacturing Investments*, Net Zero Industrial Policy Lab, Johns Hopkins University, September 2025.

⁷ Xue and Larsen, 2025.

for Chinese clean technology. They are industrial policy channels operating under the BRI umbrella. Recent BRI investment reviews also describe a tilt to high-quality and smaller, more targeted deals, consistent with that shift.⁸

The Financial Arm of China's Global Strategy

The BRI and China's external industrial policy rely on state-linked financial institutions to provide loans, insurance, and export support.⁹ The CDB, Export-Import Bank of China (Exim), and Silk Road Fund form the backbone of this system.¹⁰ Commercial banks, including the Industrial and Commercial Bank of China and the Bank of China, play supporting roles, cofinancing or syndicating BRI projects once policy-bank lending and Sinosure insurance de-risk the transaction.¹¹

However, China's overseas lending slowed sharply after 2021 as banks retrenched, reflecting rising global interest rates, tighter domestic credit conditions, and Beijing's shift toward high-quality BRI projects (Figure 9.2).¹² According to China's balance of payments, total overseas credit (including loans, trade credit, and FDI debt) peaked at around \$1.9 trillion in 2021 and then plateaued before resuming modest growth in mid-2024.¹³ The Boston University Global Development Policy Center notes that new loans from CDB and Exim peaked at \$62 billion in 2016 but averaged just \$6 billion per year between 2020 and 2024.¹⁴ This reflects a pivot away from large, state-to-state infrastructure lending toward smaller, commercially viable, and technology-oriented projects, consistent with China's post-2019 emphasis on high-quality BRI cooperation.¹⁵

Sectorally, Chinese overseas lending has mainly supported industrial, mining, and construction projects, followed by energy and transport (Figure 9.3). An increasing share now targets sectors tied to China's techno-industrial objectives, such as semiconductors, robotics, biotechnology, and advanced materials. AidData reports that the share of China's cross-border acquisition lending for these sectors rose from 46 percent in 2015 to 88 percent by 2023.¹⁶ State banks are instructed to ensure that Chinese firms can finance such foreign

⁸ Two reports from 2023 document a move toward smaller, more targeted projects and describe a "small is beautiful" approach that prioritizes high-quality, targeted support over large, general lending. See Rebecca Ray, "Small Is Beautiful": A New Era in China's Overseas Development Finance? Boston University Global Development Policy Center, GCI Policy Brief 017, January 2023; and Christoph Nedopil Wang, *China Belt and Road Initiative (BRI) Investment Report 2023*, Green Finance and Development Center, Fanhai International School of Finance, Fudan University, August 1, 2023.

⁹ Beijing's 2015 "Vision and Actions" calls for standing up the Silk Road Fund and explicitly promotes "syndicated loans and bank credit" via interbank cooperation (National Development and Reform Commission, Ministry of Foreign Affairs, and Ministry of Commerce of the People's Republic of China, "Vision and Actions on Jointly Building Silk Road Economic Belt and 21st-Century Maritime Silk Road," March 28, 2015).

¹⁰ The World Bank's 2019 *Belt and Road Economics* report notes that CDB, Exim, and the Silk Road Fund provide most of the BRI financing alongside China's big four commercial banks (World Bank, *Belt and Road Economics: Opportunities and Risks of Transport Corridors*, 2019).

¹¹ Yunnan Chen and Zongyuan Zoe Liu, *Hedging Belts, De-Risking Roads: Sinosure in China's Overseas Finance and the Evolving International Response*, Overseas Development Institute, December 2023.

¹² Mingey and Kratz, 2021; Xi, 2019a.

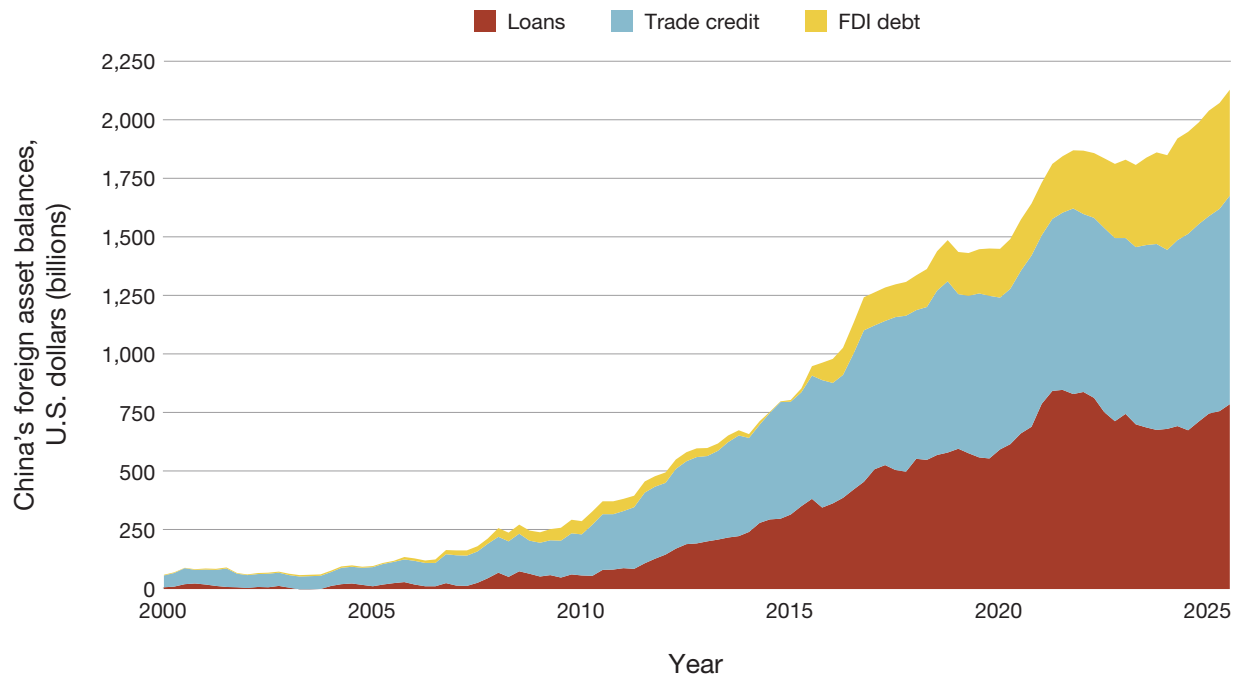
¹³ State Administration of Foreign Exchange data via CEIC, undated.

¹⁴ Boston University Global Development Policy Center, China's Overseas Development Finance Database, undated.

¹⁵ Xi Jinping, "Building an Open, Inclusive and Interconnected World for Common Development," keynote speech at the Opening Ceremony of the Third Belt and Road Forum for International Cooperation, October 18, 2023.

¹⁶ Bradley C. Parks, Sheng Zhang, Brooke Escobar, Katherine Walsh, Rory Fedorochko, Lydia Vlasto, Jacqueline Zimmerman, Julie Sickell, Emma Bury, Ameya Joshi, et al., *Chasing China: Learning to Play by Beijing's Global Lending Rules*, AidData at William & Mary, 2025.

FIGURE 9.2
China's Overseas Lending Stalled After 2021 but Increased in 2025



SOURCE: Features data from the State Administration of Foreign Exchange via CEIC, undated.

NOTE: We calculated the cumulative flow of net assets in the balance of payments. FDI debt consists of intercompany lending across borders.

acquisitions, which now include significant volumes of both brownfield mergers and new overseas manufacturing investments.¹⁷ Although these cross-border acquisition loans remain a small part of China's total overseas credit, they signal that the financial arm of China's global strategy is closely linked to its domestic industrial upgrade agenda.

Exporting the Industrial Park Model

To support industrial globalization, China has expanded overseas economic and trade cooperation zones and related BRI industrial park partnerships. These zones serve as production and logistics hubs, allowing Chinese companies to cluster suppliers, utilities, and services near target markets or resource bases. They also act as long-term anchors for Chinese products, machinery, and standards.¹⁸

China's official overseas industrial presence has grown substantially under Xi. The program began in 2006–2007 with 19 pilot zones, increasing to 75 by 2015 (with \$18 billion invested and 150,000 local jobs created) and to more than 100 zones in 49 countries by 2023 (\$73 billion invested and 500,000 local jobs).¹⁹

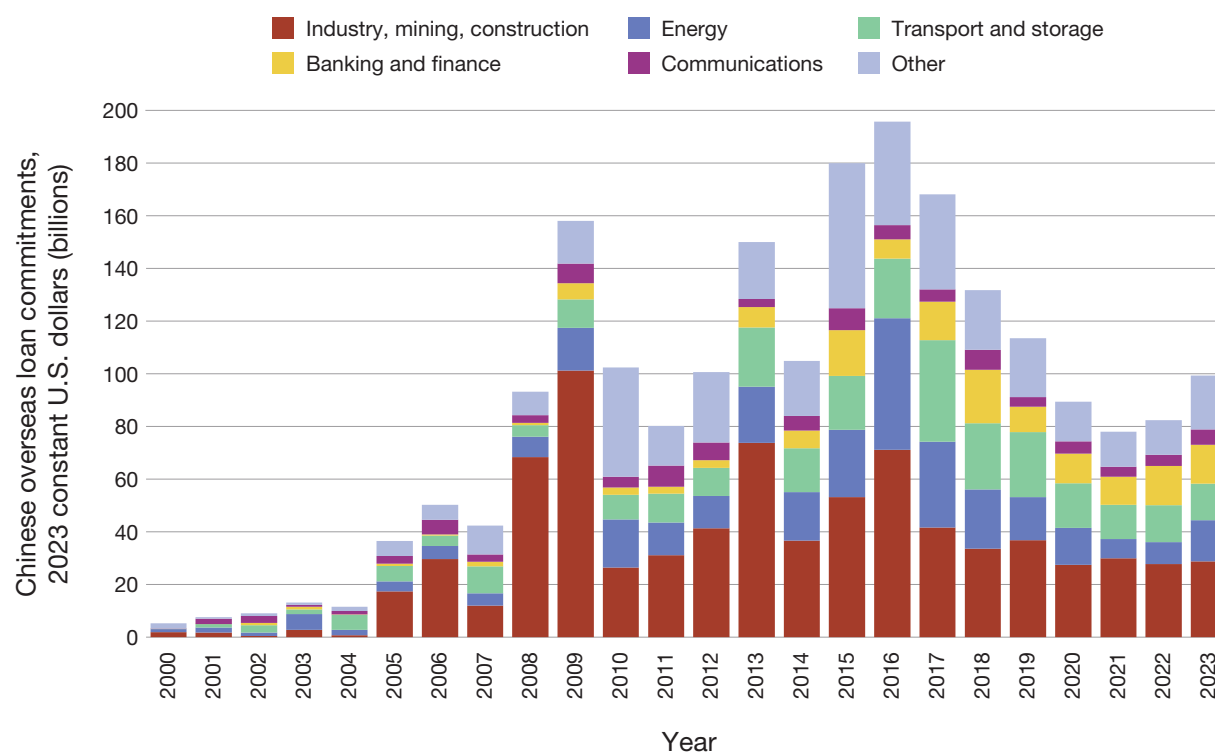
¹⁷ Parks et al., 2025.

¹⁸ Meng Guangwen, Wang Ran, and Wang Shufang, "A Review of China's Overseas Economic and Trade Cooperation Zones along the Belt and Road: Progress and Prospects," *Journal of Geographical Sciences*, Vol. 33, No. 7, 2023.

¹⁹ Chinese Institute of Certified Public Accountants, "The Construction of Overseas Economic and Trade Cooperation Zones and the 'Going Global' Strategy of Chinese Enterprises," September 18, 2008; Chinese Academy of International Trade and Economic Cooperation, Ministry of Commerce of the People's Republic of China, and United Nations Development Program China, *Belt and Road Overseas Economic and Trade Cooperation Zones for Sustainable Development: An Analysis and Practi-*

FIGURE 9.3

China's Overseas Loan Commitments Are Mostly for Industrial, Energy, and Transport Projects



SOURCE: Features data from AidData, "China's Global Loans and Grants Dataset," version 1.0, 2025.

Earlier zones focused on trade and light manufacturing; under Xi, they are more closely aligned with supply chain relocation and industrial security. New policies, such as the 2025 "Guiding Opinions on Improving the Overseas Comprehensive Service System," explicitly link these zones to supply chain security, interagency coordination, and risk management. This marks a shift from ad hoc investment promotion to systematic industrial development abroad.²⁰

This approach mirrors China's domestic development model: State-linked developers provide infrastructure, utilities, customs support, and vocational training, enabling companies to localize operations while maintaining access to Chinese technology and supply chains.²¹ This approach reflects the high-quality phase of the BRI, emphasizing smaller, commercially disciplined projects that extend manufacturing and energy supply chains rather than large sovereign loans.

cal Guide Based on an Economic, Social, and Environmental Framework, April 2019; Ministry of Commerce of the People's Republic of China, *China's Outward Investment and Economic Cooperation Report 2023*, 2023.

²⁰ Ministry of Commerce of the People's Republic of China, "The Ministry of Commerce and Four Other Departments Jointly Issued the 'Guiding Opinions on Further Improving the Overseas Comprehensive Service System,'" press release, October 17, 2025.

²¹ Angela Tritto and A. Camba, "State-Facilitated Industrial Parks in the Belt and Road Initiative: Towards a Framework for Understanding the Localization of the Chinese Development Model," *World Development Perspectives*, Vol. 28, December 2022.

Standards as an Arm of China's Industrial Strategy

China increasingly leverages its economic and technological strength to set global standards. By exporting standards through the BRI, China locks in demand for its equipment and platforms, lowers switching costs for foreign buyers, and gives Chinese firms influence over the rules governing next-generation technologies. BRI projects often bundle Chinese equipment, software, testing, and certification, with financing often provided by policy banks, making the “Chinese stack” the default for many partners.

Two main initiatives drive China's standards diplomacy. The first is the BRI's “Standards Connectivity” (标准联通) program (2015–2017), which ties standardization work to overseas projects.²² The second is the 2021 National Standardization Development Outline, which, while primarily domestic, explicitly directs China to align its domestic standards with its international engagement by 2025 and calls for deeper international cooperation.²³ China's 2023 BRI white paper reported 107 standardization agreements with 65 countries by mid-2023, creating formal channels for translating, recognizing, and codeveloping standards alongside infrastructure and trade.²⁴

China has expanded its footprint in international standard-setting bodies, including leadership roles that shape agendas and workflows. Chinese institutions held 51 of roughly 725 International Organization for Standardization (ISO) technical committee and subcommittee secretariats in 2012, rising to 76 of about 760 worldwide by 2022. At the ISO, China now ranks just behind Germany, the United States, France, and Japan in secretariat holdings. A similar pattern is visible at the International Electrotechnical Commission (IEC), where Chinese-held secretariats increased from six in 2012 to 15 by 2024, placing China among the top tier of participating countries.²⁵

The DSR exports Chinese telecommunications standards and data-governance norms. Independent audits place Huawei among the top three holders of 5G standards-essential patents, giving it significant influence in technology licensing and vendor selection.²⁶ In EV charging, China codeveloped the ChaoJi ultra-fast charging system with Japan and approved the GB/T “ChaoJi-1” standard in 2023 for rollout from 2024, positioning Chinese companies at the forefront of next-generation charging technologies.²⁷ In solar and grid technology, China's growing influence in ISO/IEC committees supports the export of Chinese photovoltaic and grid equipment.²⁸

²² Office of the Leading Group for the Belt and Road Initiative, “Action Plan on Belt and Road Standard Connectivity (2015–17),” Belt and Road Portal, updated November 22, 2017.

²³ Chinese Communist Party Central Committee and the State Council, “National Standardization Development Outline,” Xinhua, October 10, 2021.

²⁴ State Council Information Office of the People's Republic of China, 2023.

²⁵ John Jiong Gong and Vanessa Yanhua Zhang, “China's Evolution in International Standardization: From Follower to Global Leader,” World Bank, July 10, 2025; Tom Barrett, *Standards Development Organisations in an Era of Strategic Competition*, United States Studies Centre, December 16, 2024.

²⁶ Tim Pohlmann, Marco Richter, and Magnus Buggenhagen, *Who Is Leading the 5G Patent Race? 2025 Report*, LexisNexis, 2025.

²⁷ Rho Motion, “China Approves Next-Gen GB/T Charging Standard, ChaoJi-1,” October 5, 2023; CHAdeMO Association, “ChaoJi GB/T Standards Released: China's Next-Gen Ultra High Power Plug Officially Approved,” September 20, 2023.

²⁸ Gong and Zhang, 2025.

State Control over Outbound Direct Investment and Its Alignment with Industrial Priorities

Since late 2016, Beijing has tightened control over outbound investment to align capital with techno-industrial policy goals and reduce balance-of-payments pressures. The 2017 crackdown curbed risky, nonstrategic deals and redirected outbound direct investment toward manufacturing, digital infrastructure, energy, and BRI projects.

Following regulatory concerns about “irrational” outbound direct investment in 2016, the State Council issued guidelines in 2017 that classified investments as *encouraged* (e.g., advanced manufacturing, high-tech, agriculture, BRI infrastructure), *restricted* (e.g., real estate, hotels, entertainment), or *prohibited* (e.g., projects that harm national interests).²⁹ Reviews were tightened, and the 2018 NDRC “Administrative Measures for Outbound Investment by Enterprises” established a “sensitive projects” approval regime and brought off-shore special purpose vehicles under scrutiny.³⁰ In 2026, Beijing elevated outbound-investment governance into a State Council regulation through the Provisions on Outbound Investment (Order No. 837, signed May 5, released June 1, and effective July 1, 2026), adding whole-process supervision, formal national security review of outbound investments, and controls on exporting or using restricted or prohibited goods, technology, services, and related data in outbound-investment activity.³¹

These measures have worked as intended. Since 2017, total outbound investment volumes have dropped, and the mix has shifted from property and entertainment to productive sectors, supporting China’s industrial expansion logic (Figure 9.4).³² The result is outbound investment that builds manufacturing capacity, exports Chinese equipment and standards, and supports the Digital and Green BRI, turning overseas investment into a coordinated tool of China’s techno-industrial strategy. Emerging evidence suggests that corporate ownership and control, especially of R&D-intensive foreign firms, may be another channel through which Chinese outward investment internalizes technological capacity at home.³³

²⁹ General Office of the State Council, National Development and Reform Commission, Ministry of Commerce, People’s Bank of China, and Ministry of Foreign Affairs, “Guidelines on Further Guiding and Regulating Overseas Investments,” State Council Document No. 74 of 2017, August 4, 2017; National Development and Reform Commission, Ministry of Commerce, People’s Bank of China, and Ministry of Foreign Affairs, “Opinions on Further Guiding and Regulating the Directions of Overseas Investments,” August 4, 2017.

³⁰ NDRC, “Administrative Measures for Outbound Investment by Enterprises,” National Development and Reform Commission Order No. 11, December 26, 2017; United Nations Conference on Trade and Development, “China—NDRC’s Measures for the Administration of Outbound Investment by Enterprises Came into Effect,” webpage, Investment Policy Monitor, March 1, 2018.

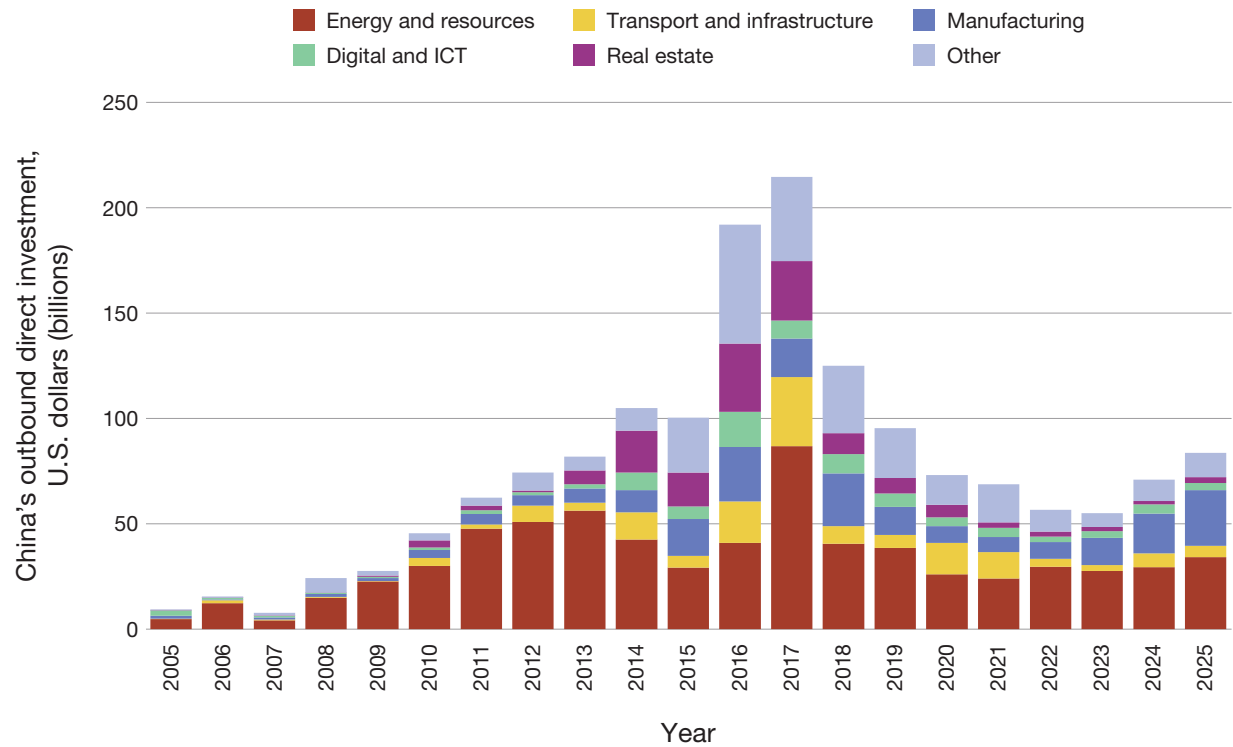
³¹ “China Unveils Regulation on Outbound Investment,” Xinhua, June 1, 2026.

³² Rhodium Group, “China Cross-Border Monitor: Industries,” database, undated.

³³ Jennie Bai, Luc Laeven, Yaojun Ke, and Hong Ru, *China’s Global Ownership*, National Bureau of Economic Research, Working Paper 35106, 2026.

FIGURE 9.4

Beijing Imposed Discipline on Outbound Investment After 2017, Redirecting Flows Toward Priority Sectors



SOURCE: Features data from Rhodium Group, undated.

Conclusion and Outlook

Reordering Priorities: Technology Leadership, Security, and Ideology

Throughout most of China's reform era, the main goal was catching up with advanced economies. This meant building up manufacturing capacity, integrating into global value chains, and absorbing foreign technology, fueled by FDI.¹ Industrial policy reemerged after the 2006 MLP and the 2010 SEI lists.² Still, China's focus remained on promoting specific sectors and export competitiveness, with "indigenous innovation" (自主创新) mostly about adopting and localizing existing technologies rather than leading at the global frontier.

Under Xi, China's techno-industrial strategy has shifted in three main ways:

- **Innovation and upgrading:** China aims to move beyond low-cost, catch-up growth to innovation-driven, higher-value production and global commercial leadership. As wages rise and productivity slows, sustaining growth requires reaching the technological frontier before becoming a high-income economy.
- **National security:** Security concerns have made technological self-reliance and supply chain resilience central state priorities. Since 2017, U.S. export controls, sanctions, and tariffs have pushed China to elevate technology self-sufficiency, now viewed as essential for development. Our policy discourse analysis ranks *self-reliance* and *technology leadership* first, followed by structural upgrading, decarbonization, and regional balancing.
- **Ideology and politics:** Xi has recast techno-industrial policy as a Party-led mission, linking economic modernization with national rejuvenation and "common prosperity" (共同富裕). Technological progress is tied to the Party's vision of governance and China's ambition for great-power status.

Building a Party-Led Techno-Industrial Policy State

Xi's approach aims to centralize techno-industrial policy as a Party instrument, prioritizing "top-level design" (顶层设计) and greater central government control. The main goals are to accelerate technological development for national power, address the downsides of fragmented local production while preserving adaptability, align financial and industrial resources with security goals, and make the most of limited fiscal capacity through coordination and centralized direction.

¹ Gerard DiPippo, Francesca Ghiretti, and Benjamin Lenain, "Beyond Tariffs: What the U.S. Can Learn from China's Industrial Playbook," commentary, RAND Corporation, April 17, 2025.

² State Council of the People's Republic of China, *The National Medium- and Long-Term Program for Science and Technology Development (2006–2020): An Outline*, 2006b.

Key institutional changes include the following:

- upgrading leading small groups into permanent Party commissions for economics, finance, and S&T
- establishing a central S&T commission above ministries and a central finance commission that steers banks and markets politically
- clarifying execution, with MIIT leading applied technology and manufacturing and the Ministry of Science and Technology overseeing basic research and S&T planning
- verticalizing bureaucracy, with central agencies such as the Central Commission for Discipline Inspection and National Financial Regulatory Administration overseeing local offices to balance local initiative with central control
- incentivizing cadres with both career rewards and disciplinary measures to align local policy with central ambitions.

Beijing has tightened oversight with increased audits, digital taxation and supervision, and expanded disciplinary activity, strengthening central influence over local actions. Efforts to unify regulatory standards and implement fair-competition reviews aim to reduce local protectionism and the patchwork of fragmented rules that have long divided the domestic market.

Overall, China's industrial governance is now more vertically managed, although local experimentation still matters. The central government's signals prompt coordinated action across officials, firms, and funds, leading to unified surges in targeted sectors. However, this system risks overcorrection, sometimes requiring repeated supply-side reforms to address excesses. China now runs a more centralized and formally structured industrial policy apparatus. Whether *disciplined* and *coordinated* are the right descriptors is less certain. Research suggests that Xi-era centralization has improved compliance with central directives but reduced the local policy experimentation that historically drove adaptation and innovation, with measurable economic costs.³ The system's strengths (strategic focus, mobilization capacity, and scale) coexist with significant vulnerabilities, including politicized capital allocation, suppressed local initiative, and mandates whose costs are borne by firms and institutions with little capacity to resist.

The bet is that these Party-state institutional changes will build on existing strengths across China's techno-industrial ecosystem while mitigating wastefulness. Yet these reforms also deepen the Party-state's direct involvement in day-to-day economic decisionmaking through tighter vertical supervision and greater organizational and financial presence within firms and funds, which runs counter to the conventional view that, as a country approaches the frontier, state guidance should be phased out.⁴

Rationalizing the Toolkit: Fiscal Discipline, Financialization, and Party Control

Xi-era policy has reshaped how the state mobilizes resources, now relying on five main channels, each increasingly shaped by fiscal constraints, with a focus on more selective support and a greater reliance on

³ Luo, Wang, and Yang, 2025.

⁴ Joe Richard Frank Studwell, *From Learning to Earning: The Transition from Manufacturing Catch-up to Competitiveness at the Global Business Frontier, as Pursued in China's Energy Equipment Sector*, dissertation, University of Cambridge, 2019; Barry Naughton, "China's Economic Policy Today: The New State Activism," *Eurasian Geography and Economics*, Vol. 52, No. 3, 2011.

mandates, standards, and state-directed financial channels that operate through market infrastructure but under political guidance:

- **Fiscal tools:** The state is moving from local, ad hoc subsidies to centrally determined eligibility standards, mandated adoption requirements, tax incentives, and targeted procurement; the effect is to shift costs from its budget to firms and institutions compelled to comply.
- **Financial tools:** The state acts as an investor, allocating capital through state funds, lending, and capital market reforms.
- **Real economy tools:** Cheap land and energy are now reserved for strategic clusters and innovation parks, not handed out broadly.
- **Party-firm coordination and private-sector mobilization:** The Party-state is relying on SOE consolidation and preferential positioning, tighter Party embedding and regulatory steering over both SOEs and large private firms, and MIIT-administered champion-designation programs that provide national ladders for scaling specialized private suppliers.
- **External tools:** The BRI now emphasizes digital, green, and standards-based industrial projects, with stricter oversight of outbound investment.

Rising fiscal pressures have been a major driver of these policy shifts. China's industrial policy spending is huge by global standards, and maintaining it is increasingly difficult as growth slows and social obligations expand. Local governments already face declining land revenues, borrowing caps, and budget deficits, straining their ability to fund industrial policy subsidies. Beijing recognizes these limits and is shifting its industrial policy away from direct fiscal spending toward regulatory, equity, and credit tools to achieve its goals without exhausting state finances.

Planning for the Next Phase of Techno-Industrial Policy

Beijing is overhauling its techno-industrial strategy to reinforce the “real economy” (实体经济) and build a “modernized industrial system” (现代化产业体系) amid tighter fiscal and geopolitical constraints. The scope of recent changes—new Party commissions, restructured ministries, centralized oversight of funds, expanded mandates for firms and local governments, and tighter political control over corporate governance—goes well beyond fine-tuning. Whether this amounts to rationalization or to a qualitatively different, more interventionist phase of industrial policy is a question that the 15th FYP period will help answer.

The 15th FYP (2026–2030), formally published after approval by the National People's Congress in March 2026, confirms this direction.⁵ The outline showcases rising confidence in the current approach and codifies the priorities signaled in the October 2025 Fourth Plenum communiqué and the recommended proposals.⁶ In the 14th FYP, S&T self-reliance was the top economic goal, with industrial development supporting that aim. The 15th FYP confirms a reordering of this sequence, placing the construction of a “modernized industrial system” (现代化产业体系) and the consolidation of the “foundation of the real economy” (实体经济根基)

⁵ “Outline of the Fifteenth Five-Year Plan for National Economic and Social Development of the People's Republic of China,” 2026.

⁶ Ministry of Foreign Affairs of the People's Republic of China, “Communiqué of the Fourth Plenary Session of the 20th Central Committee of the Communist Party of China,” October 23, 2025; “The CPC Central Committee's Proposal on Formulating the 15th Five-Year Plan for National Economic and Social Development,” Xinhua, October 28, 2025; “Full Text: Explanation of the Recommendations of the Central Committee of the Communist Party of China for Formulating the 15th Five-Year Plan for National Economic and Social Development,” Xinhua, October 28, 2025.

first. The second objective, in terms of order, is to accelerate “high-level science and technology self-reliance and self-strengthening” and develop “new quality productive forces” (新质生产力), a term for productivity growth driven by frontier technology, innovation, and advanced manufacturing rather than by labor and capital accumulation. This does not diminish the focus on self-reliance. The outline calls for “extraordinary measures” (超常规措施) to achieve “decisive breakthroughs in key core technologies in priority fields such as integrated circuits, industrial machine tools, high-end instruments, basic software, advanced materials, and biomanufacturing,” reflecting Beijing’s concerns about dependence on foreign technologies amid geopolitical tensions.⁷

This reprioritization signals that Beijing increasingly views techno-industrial upgrading less as a near-term growth strategy than as a way to reinforce the “real economy” (实体经济), enhance national power, and reduce geopolitical vulnerabilities, even if that emphasis can weigh on immediate growth. Industrial strength is the goal, while indigenous S&T serves as an enabler. Self-reliance is a means to industrial and economic power, as well as national security.

Unlike many economies, in which *self-reliance* might mean replacing superior foreign goods with inferior local ones, China’s expanding market share in key technologies has so far allowed self-reliance and commercial leadership to complement each other rather than compete. But this complementarity is not guaranteed to hold. In sectors in which China lags (advanced semiconductors, aircraft engines, specialized software), forced substitution still involves performance and cost trade-offs. And as mandated adoption requirements expand, the risk grows that political compliance rather than commercial merit will determine procurement outcomes, potentially eroding the competitive pressures that have driven China’s gains.

Regarding manufacturing, the language shifts from the 14th FYP’s “basically stable share” to “maintain a reasonable proportion,” suggesting that Beijing is comfortable with manufacturing’s share of GDP (around 25 percent in 2025) declining gradually. On consumption, the 15th FYP aims to boost consumer demand through employment, income growth, and improved supply of consumer goods and services more than through increased direct transfers. This aligns with Xi’s aversion to “welfarism” and current fiscal constraints. The plan is to lift real wages and thus consumption by raising productivity and firm profitability. The 15th FYP attempts to reconcile the need for higher consumption with a continued preference for manufacturing and techno-industrial policy.

The next phase of China’s techno-industrial strategy will face constraints and contradictions. Beijing’s recent planning documents highlight several challenges that policy will need to manage. The subsections that follow address these challenges in turn:

- **Involution** (内卷): destructive competition—such as subsidy races, price wars, duplication, and “copy-cat” capacity—that can erode profitability, quality, and long-run innovation even as it accelerates scaling.
- **Center-local dissonance**: a persistent tension between “top-level design” (顶层设计) and local growth incentives, protectionism, and experimentation. This tension raises the question of how far central coordination can go without choking off the dynamism that decentralized competition has historically provided.
- **Innovation**: sustaining genuine technological upgrading and frontier breakthroughs (not just scale), and converting large input mobilization (R&D, talent, infrastructure) into durable capabilities.
- **Weak total factor productivity (TFP) performance**: improving allocative efficiency by shifting resources away from low-efficiency capacity and enabling exit.

⁷ Jimmy Goodrich, “Beijing Calls for ‘Extraordinary Measures’ to Boost Tech Self-Sufficiency,” *Global Policy at a Glance* blog, University of California Institute on Global Conflict and Cooperation, November 20, 2025.

- **International resistance:** growing external pushback, including trade remedies and industrial policy responses abroad as China's scale, export surplus, and price-based competition intensify global tensions.

Unified National Market and Anti-Involution

China's leadership assesses that the country's fragmented industrial policy landscape has become a hindrance to development. The UNM is a core objective of the 15th FYP, with implications for techno-industrial policy.⁸ It aims to streamline implementation, reduce local barriers and local protectionism, and improve factor allocation by standardizing rules, supervision, and competition enforcement.⁹ However, the same standardization also expands Beijing's ability to discipline local investment promotion and curb duplicative projects in strategic industries, so the UNM can serve as both a market-integration mechanism and a tool for more centralized industrial coordination. Introduced in 2022 and operationalized in January 2025, the UNM is guided by NDRC principles emphasizing "five unifications, one removal" (五统一、一破除): Unify basic market rules and institutions, unify factor and resource markets, unify goods and services markets, unify high-standard connectivity for market facilities, unify fair and consistent market supervision, and remove local protectionism and market segmentation.¹⁰

For industrial policy, the UNM is meant to serve as a coordination tool. By defining standards and standardizing procurement and competition enforcement, Beijing hopes to curb duplicative provincial projects, pool national demand, and create economies of scale without larger subsidies. The UNM is designed to reduce interprovincial competition in strategic industries, lower fiscal costs, and facilitate prioritization and orderly consolidation, including mergers or exits of duplicative or nonviable projects.

Concurrently, the "anti-involution" (反内卷式竞争) agenda seeks to address harmful competition. The 15th FYP instructs authorities to "comprehensively rectify involution-style competition." A June 2025 *People's Daily* front-page article tied "involution" to subsidy races, local protectionism, copycat production, and price wars in such sectors as EVs, solar, and storage.¹¹ These issues stem partly from existing industrial policies. Remedies include stricter competition reviews, bans on discriminatory procurement and designated suppliers, prompt-payment rules in supply chains, tougher quality and safety standards, and exit mechanisms for low-efficiency or unprofitable firms. Whether exiting becomes credible in practice will depend in part on China's bankruptcy and restructuring system.¹²

The "anti-involution" (反内卷式竞争) campaign and the UNM drive are both part of Beijing's effort under Xi to more centrally guide techno-industrial policy by imposing unified market rules and administrative discipline to counter excess investment and destructive competition. A major issue, however, is that

⁸ "Outline of the Fifteenth Five-Year Plan for National Economic and Social Development of the People's Republic of China," 2026.

⁹ NDRC, "A Special Press Conference Held by the National Development and Reform Commission Introduced the Relevant Situation Regarding the In-Depth Promotion of the Construction of a Unified National Market," January 8, 2025a.

¹⁰ Chinese Communist Party Central Committee and the State Council, "Opinions of the CPC Central Committee and the State Council on Accelerating the Construction of a Unified National Market," Xinhua, March 25, 2022; NDRC, "Notice on Issuing the 'Guidelines for the Construction of a Unified National Market (Trial),' " National Development and Reform Commission Document No. 1742 of 2024, December 4, 2024.

¹¹ Jin, 2025.

¹² The country's bankruptcy and restructuring system remains underused and unevenly implemented. It is often slow to initiate and resolve, and it is historically susceptible to local intervention despite ongoing reforms and proposed revisions to the Enterprise Bankruptcy Law. See Li Jiaying, "Draft Revision of Bankruptcy Law in Review," *China Daily*, September 11, 2025.

decentralized competition, even when wasteful and duplicative, has been a core ingredient to China's rapid techno-industrial development. The precise contours of center-local discretion remain in flux, but the centralizing trend could inhibit this dynamic. If Beijing strikes the balance right, however, reforms could achieve their aims of enhancing allocative efficiency, reducing local duplication and making state support more fiscally sustainable.

Leadership frames this as both a tough battle and a long-term campaign. It calls for a “national chess-board” (全国一盘棋) approach that breaks local protectionism, restrains low-level repetitive construction and excessive homogeneous competition, and disciplines local investment promotion practices under a single set of market rules.¹³ Official guidance also links progress to changes in local incentives, including revisions to cadre evaluation and the central-local fiscal relationship.¹⁴ The key uncertainty, therefore, is whether these enforcement and incentive changes stick when they become politically costly.

The Challenge of Boosting Productivity

For sustained growth, China must raise TFP, which is the efficiency with which labor, capital, and technology are converted into output. The 15th FYP retains total labor productivity as a headline indicator, targeting growth “higher than GDP growth”—the same framing as that used in the 14th FYP.¹⁵ TFP does not appear as a formal indicator, although it is mentioned, and the Fourth Plenum proposals elevated it rhetorically as a measure of high-quality development. The gap between the rhetorical emphasis on TFP and the continued reliance on labor productivity as the operational metric is telling. TFP is difficult to measure, impossible to target directly, and poorly suited to cadre evaluation.

Industrial policies have produced mixed results for TFP. An IMF study in 2025 found that resource misallocation associated with China's industrial policy lowered domestic TFP by about 1.2 percent and GDP by up to 2 percent, with subsidies prompting excess production and trade barriers suppressing output elsewhere.¹⁶ At the firm level, research by the National Bureau of Economic Research indicates that subsidies do not systematically “pick winners”: More productive firms are less likely to get support, and recipients often see stagnant or negative TFP growth even as employment rises.¹⁷ OECD synthesis suggests that, for large manufacturers, many subsidy designs correlate weakly or negatively with productivity outcomes.¹⁸

¹³ Chinese Communist Party Central Committee and the State Council, 2022.

¹⁴ Lu Xiaocheng, “Adhering to the Principles of ‘Five Unifications and One Openness,’ We Will Further Advance the Construction of a Unified National Market,” *Guangming Daily*, July 10, 2025.

¹⁵ NDRC, *Outline of the 14th Five-Year Plan for National Economic and Social Development and the Long-Range Objectives for 2035*, March 2021a; “Outline of the Fifteenth Five-Year Plan for National Economic and Social Development of the People's Republic of China,” 2026.

¹⁶ Garcia-Macia, Kothari, and Tao, 2025.

¹⁷ Lee G. Branstetter, Guangwei Li, and Mengjia Ren, *Picking Winners? Government Subsidies and Firm Productivity in China*, National Bureau of Economic Research, Working Paper No. 30699, December 2022. Another 2022 study also used firm-level data to demonstrate that local officials' short-term promotion and stability incentives lead them to channel subsidies and tax breaks toward large, loss-making, low-productivity incumbent firms (“picking losers”), which protects their own careers but distorts industrial policy by suppressing firm entry and productivity growth. See David J. Bulman, Xun Yan, and Qiong Zhang, “Picking Losers: How Career Incentives Undermine Industrial Policy in Chinese Cities,” *Journal of Development Studies*, Vol. 58, No. 6, 2022.

¹⁸ OECD, 2025d. A 2026 OECD MAGIC analysis reaches a parallel conclusion at the market level, finding that subsidies explain around 22 percent of global market-share gains by expanding firms, rising to 60 percent for Chinese firms, while not producing significant productivity or profitability gains. See OECD, “Industrial Subsidies Reach Highest Levels Since the Global Financial Crisis,” press release, June 1, 2026.

The evidence points to a compositional problem: China is home to both low-efficiency producers and globally competitive “champions,” but resources flow excessively to the former. Within-industry reallocation explains much of the TFP issue rather than an economywide collapse in firm-level efficiency.¹⁹ The 15th FYP aims to address this by imposing stricter discipline, consolidating fiscal support, and shifting to performance-based funding. Unified standards and procurement should reduce duplication, while fiscal consolidation will make it harder for inefficient projects to survive. Success depends on whether central Party-state authorities can credibly distinguish which firms and projects merit continued support and which should lose funding or exit—and then enforce those reallocations over resistance from local governments and state investors.

The Role of Techno-Industrial Policy in China’s Technological Gains

Do China’s techno-industrial policies explain its technological rise? The balance of evidence suggests that they have been a meaningful contributor, albeit unevenly.

The gains themselves are substantial. China’s share of global manufacturing value added has been relatively stable over the past decade, climbing from 26 percent in 2015 to 28 percent in 2024.²⁰ But its technological and innovative capabilities have accelerated across multiple dimensions. In frontier research, the Nature Index ranks China ahead of the United States in high-impact natural-science publications, and the Australian Strategic Policy Institute Critical Technology Tracker finds China leading in output for 66 of the 74 key technologies that it monitors.²¹ In commercial technology, China now produces well over half of global EVs, holds dominant shares in battery and solar manufacturing, and has enabled such firms as DJI to capture the bulk of the consumer drone market, while Huawei and other companies push the frontiers in mobile and 5G technologies, and a growing number of players indigenize the semiconductor ecosystem. The resulting *electric stack* (batteries, magnets, motors, and power electronics) forms a highly modular, general-purpose ecosystem for industrial innovation.

In strategic and military domains, China has launched the world’s first quantum communications satellite (Micius), fielded operational hypersonic glide vehicle systems, completed the BeiDou global navigation satellite system, executed far-side lunar landings and sample returns, brought the Tiangong space station into regular operation, deployed J-20 fighters increasingly powered by domestically produced engines, and begun commercial service of the Commercial Aircraft Corporation of China (COMAC) C919.²² China’s rising investment in R&D, from 4 percent of global R&D in 2000 to 26 percent in 2023, is paying off.²³

¹⁹ Brandt et al., 2020.

²⁰ World Bank data via CEIC, undated.

²¹ Benjamin Plackett, “Nature Index 2025 Research Leaders: United States Losing Ground as China’s Lead Expands Rapidly,” *Nature Index*, June 11, 2025; Jenny Wong-Leung, Stephan Robin, and Linus Cohen, “ASPI’s Critical Technology Tracker: 2025 Updates and 10 New Technologies,” *The Strategist*, Australian Strategic Policy Institute, December 1, 2025.

²² Sheng-Kai Liao, Wen-Qi Cai, Wei-Yue Liu, Liang Zhang, Yang Li, Ji-Gang Ren, Juan Yin, Qi Shen, Yuan Cao, Zheng-Ping Li, et al., “Satellite-to-Ground Quantum Key Distribution,” *Nature*, Vol. 549, August 9, 2017; U.S. Department of Defense, *Military and Security Developments Involving the People’s Republic of China 2024: Annual Report to Congress*, 2024; Missile Defense Project, “DF-17,” webpage, Center for Strategic and International Studies, last updated April 23, 2024; State Council Information Office of the People’s Republic of China, *China’s BeiDou Navigation Satellite System in the New Era*, November 4, 2022; Huizhong Wu, “A Chinese Lunar Probe Returns to Earth with the World’s First Samples from the Far Side of the Moon,” Associated Press, June 25, 2024.

²³ Davide Bonaglia, Lorena Rivera León, and Sacha Wunsch-Vincent, “End of Year Edition—Against All Odds, Global R&D Has Grown Close to USD 3 Trillion in 2023,” *Innovation Insights* blog, World Intellectual Property Organization, December 18, 2024; National Science Board, 2025.

These policies appear to have played a role in accelerating scale, diffusion, and system integration once technologies became commercially viable (if not yet profitable) and, in some cases, in shaping the speed and direction of sectoral development. Party-state support has reduced financing constraints, compressed time to scale, and enabled ecosystem formation in capital-intensive, learning-by-doing sectors (EVs, batteries, solar manufacturing, telecommunications equipment, and drones) where production scale and cost curves are central to competitiveness. In these domains, policy has helped overcome coordination failures and early-stage risks that may otherwise have slowed or fragmented development.

However, policy has been more effective at expanding inputs (R&D spending, infrastructure, and talent pipelines) than at reliably improving allocative efficiency. Breakthroughs have occurred, but often alongside duplication, incrementalism, and uneven quality, reflecting both information constraints and incentive distortions. Overall, China's policies have reshaped industrial trajectories by privileging scale, speed, and resilience over near-term efficiency, producing genuine technological capabilities while embedding structural costs.

Assessing the causal impact with precision is difficult. These policies do not operate as discrete, isolable interventions but as part of a complicated and changing ecosystem, as described throughout this report. Policy selection bias is substantial. Beijing often intensifies support in sectors already showing momentum, so observed success may reflect reinforcement rather than creation. Empirical evaluation is further constrained by data opacity, particularly for unlisted firms and local governments, which limits firm-level or counterfactual analysis. China's techno-industrial gains are themselves multidimensional, spanning commercial scaling, frontier research, system integration, and manufacturing execution, with each dimension responding differently to policy tools, complicating aggregate judgments.

A full assessment of effectiveness is beyond the scope of this analysis, which focuses on how the system operates rather than how well it works. But the system as described here contains a core tension that any such assessment would need to confront: The same centralization and mandate-driven approach that Beijing is adopting to address the acknowledged failures of the earlier, more fragmented model may also suppress the competitive dynamics and local adaptation that generated many of the gains documented above.

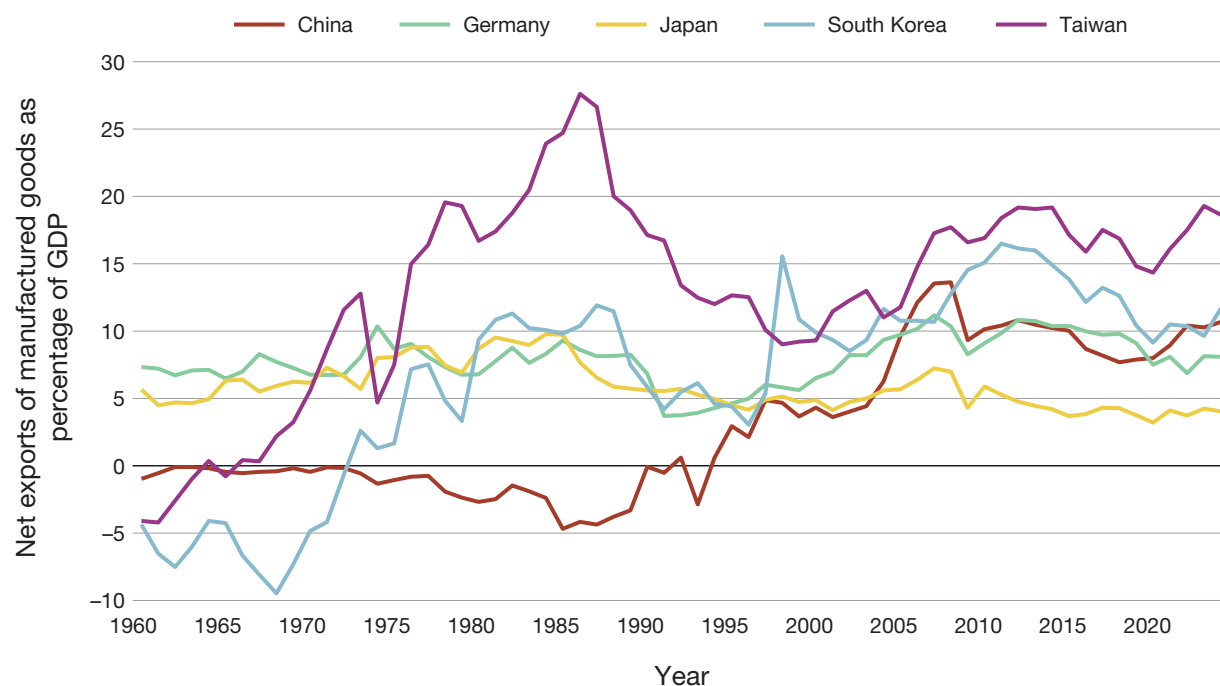
In this sense, much of what Beijing now labels *waste* (duplicative local investment, subsidy races, excess entry, price wars) may be better understood as the cost side of a model whose benefits the leadership is keen to preserve. Decentralized competition generated redundancy, but it also generated the speed, scale, and cost compression that made Chinese firms globally competitive in EVs, solar equipment, and batteries. The policy challenge is not simply to eliminate waste but to do so without also eliminating the competitive intensity that produced the gains. That may prove to be a harder separation than Beijing's current reforms assume.

Techno-Industrial Might, Global Tensions

China's ambitions in techno-industrial policy are unprecedented. While Japan, South Korea, and Taiwan followed similar state-led paths to industrialization and scaled back intervention as they reached the technological frontier, China is intensifying its efforts at an unmatched scale. If Beijing seeks analogies, it finds them in manufacturing-oriented economies, such as Germany, Japan, South Korea, and Taiwan, all of which maintain manufacturing trade surpluses and rely on industrial exports to remain competitive. China's manufacturing surplus (about 10 percent of GDP) is comparable to that of its peers (Figure 10.1). However, China's economy is larger than the combined economies of these countries, and its manufacturing trade surplus as a share of global GDP exceeds that of any of them (Figure 10.2). The world's production and demand balances now hinge on Beijing's choices more than those of any other economy.

However, China's ability to keep expanding manufacturing output is ultimately limited by demand, not by techno-industrial policy. China's policies remain powerful in mobilizing resources and sustaining priority

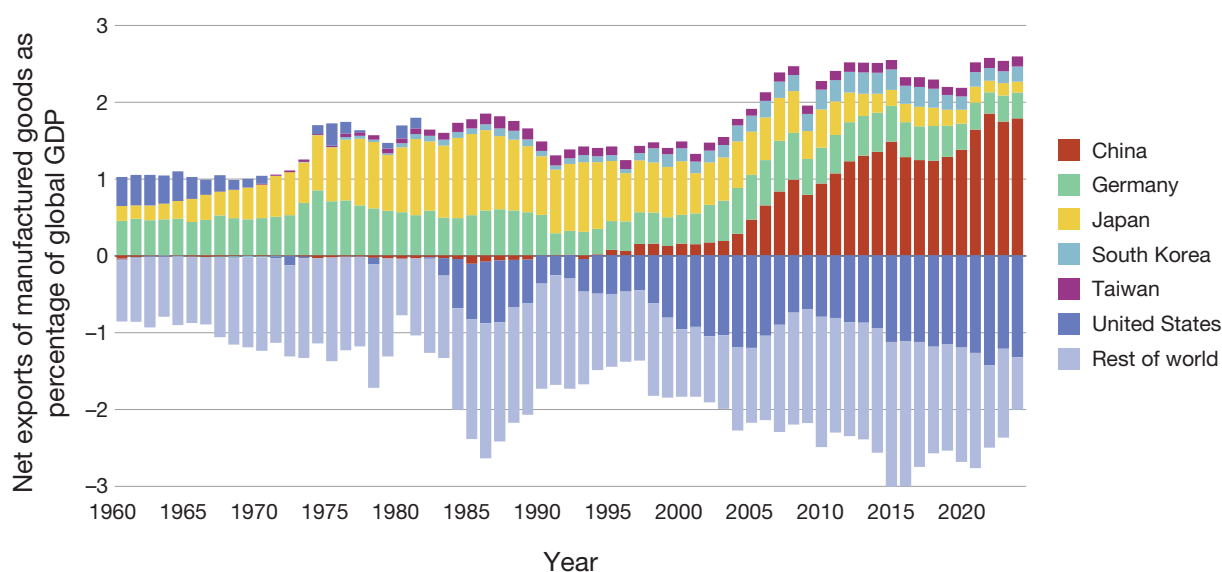
FIGURE 10.1

China's Manufacturing Trade Surplus Is Not Uniquely Large Relative to Its Economy

SOURCES: Data for China, Germany, Japan, and South Korea are from the World Bank via CEIC, undated; data for Taiwan are from the Directorate-General of Budget—Accounting and Statistics, MOF, and IMF via CEIC, undated.

NOTE: Manufacturing goods' share of Taiwan's trade is estimated based on historical research for years prior to 1998.

FIGURE 10.2

China's Manufacturing Trade Surplus Is Historically Large Relative to the Global Economy

SOURCES: Data for China, Germany, Japan, South Korea, United States, and rest of world are from the World Bank via CEIC, undated; data for Taiwan are from the Directorate-General of Budget—Accounting and Statistics, MOF, and IMF via CEIC, undated.

sectors. At the same time, as our findings show, Beijing is increasingly dialing back the most fiscally intensive tools amid tighter budget constraints and growing concern about wasteful duplication. Industrial policy can shape incentives, coordinate actors, and prop up targeted industries, but it cannot by itself create sufficient end demand to absorb ever-growing production.

Domestic demand imposes the first constraint. China's broader economy remains weak, burdened by the property downturn, subdued income growth, and employment pressures outside manufacturing. Even as household consumption recovers in absolute terms, its composition is shifting toward services, as is typical for an upper-middle-income economy. This structural shift limits the pace at which manufacturing demand can grow, reinforcing Beijing's apparent acceptance that manufacturing's share of GDP can gradually decline while remaining central to the economy.²⁴

External demand imposes a second constraint. The rest of the world can absorb only so much additional Chinese manufacturing output, both economically and politically. Although Beijing has emphasized diversifying toward emerging and developing economies (EMDEs), most final demand for high-value manufactured goods remains in advanced economies. EMDEs lack the purchasing power to offset slower absorption in the United States, Europe, and other advanced economies, even if China continues to expand its export footprint in these regions.²⁵ Sustained increases in China's manufacturing surplus would therefore require either greater tolerance for import displacement abroad or escalating trade frictions.

This creates structural and geopolitical friction. The 15th FYP remains focused on building a “modernized industrial system” (现代化产业体系) and cultivating “new quality productive forces” (新质生产力), with a priority on manufacturing and technology-intensive production. Yet, Chinese consumers, like those in other maturing economies, allocate more of their spending to such services as health care, education, and leisure.²⁶ If manufacturing continues to outpace domestic consumption, the surplus must be absorbed by the rest of the world, inviting greater resistance through tariffs, trade remedies, and industrial policy responses.

For Beijing, the objectives extend beyond economic growth or productivity. Techno-industrial policy also serves as a security strategy and a means to achieve “self-reliance” (自力更生) and “comprehensive national power” (综合国力). Reliance on foreign demand may be acceptable if China can dominate critical technologies and secure supply chains, thereby bolstering its autonomy and influence. In this respect, China's strategy is already yielding results, having fostered global champions in EVs, batteries, solar equipment, and industrial machinery, with the state acting as planner, investor, and regulator.²⁷

²⁴ “The CPC Central Committee's Proposal on Formulating the 15th Five-Year Plan for National Economic and Social Development,” 2025.

²⁵ From 2015 to 2025, advanced economies' (AEs') share of global GDP (excluding China) fell only 1.4 percentage points to 74.3 percent, proxying the size of the markets to which China can export. The diversification of China's export markets from AEs to EMDEs has been only incremental based on final demand for China's manufacturing value added rather than direct gross exports, which capture China's exports to offshore manufacturing hubs dependent on AE demand. From 2016 to 2022, following the first U.S.-China trade war, foreign final demand for Chinese manufacturing value added in EMDEs shifted only 0.8 percentage points to 38.2 percent. The biggest shift was within AEs, as U.S. importers shifted away from Chinese goods and European importers increased their purchases. EMDEs' share of final demand may have accelerated since 2022. Chinese export data indicate that Southeast Asia is the top destination for increased exports since then. This likely reflects the region's role as the preferred offshoring hub for Chinese manufacturers rather than growing final demand in that region. Smaller increases in Chinese exports to Africa and Latin America, however, could reflect increased final demand (World Bank data via CEIC, undated; OECD, 2025; General Administration of Customs of the People's Republic of China data via CEIC, undated).

²⁶ The share of Chinese urban per capita expenditures going to services increased from 42 percent in 2014 to 48 percent in 2024. The growth of services' share accelerated before the COVID-19 pandemic, declined in 2020–2022 because of the lockdowns and travel restrictions, and then bounced back in 2023–2024 (National Bureau of Statistics of China data via CEIC, undated).

²⁷ “US Efforts to Contain Xi's Push for Tech Supremacy Are Faltering,” Bloomberg, October 30, 2024.

For China's trading partners, the available direct responses broadly fall into two categories. The first is targeted trade remedies, such as tariffs, countervailing duties, and investment screening.²⁸ These can offset specific competitive distortions but are blunt instruments that also raise costs for importers and risk fragmentation of the trading system.

The second channel is the exchange rate. China's real effective exchange rate (a measure of the price of Chinese goods and services relative to those of its trading partners, adjusted for inflation differentials) depreciated roughly 13 percent from 2022 to 2025, driven primarily by low domestic inflation relative to trading partners, combined with Beijing's preference for bilateral exchange rate stability against the U.S. dollar (Figure 10.3). This depreciation has compounded the competitive advantages of Chinese exporters, generated by industrial policy, economies of scale, and supply chain integration. The IMF's 2025 Article IV assessment, published in February 2026, found China's external position "stronger than implied by fundamentals and desired policies" and estimated the yuan's real effective exchange rate to be undervalued by 12 to 21 percent. The IMF recommended a package of consumption-oriented fiscal expansion (partially funded by cutting industrial policy spending), higher social spending, and greater exchange-rate flexibility that, if

FIGURE 10.3
China's Exchange Rate Depreciation Has Contributed to Its External Surpluses



SOURCE: Features data from the Bank for International Settlements, General Administration of Customs, and State Administration of Foreign Exchange via CEIC, undated.

NOTE: The real effective exchange rate index is normalized so that 2020 = 100.

²⁸ On March 11, 2026, the U.S. Trade Representative initiated Section 301 investigations into structural excess capacity and production in manufacturing sectors; China was among the economies subject to the investigations. See Office of the United States Trade Representative, "Section 301—Structural Excess Capacity and Production in Manufacturing Sectors," webpage, undated.

implemented, would reflate the economy, produce real appreciation, and narrow China's trade surplus.²⁹ In practice, the IMF's recommended rebalancing package is unlikely to materialize. The 2026 Government Work Report maintains Beijing's commitment to keep the renminbi "generally stable at an adaptive, balanced level," and consumption support remains focused on the supply side, without fiscal resources to back up a household consumption push.³⁰

Chinese authorities contest the IMF's assessment. In the Article IV consultations, they attributed China's export competitiveness primarily to economies of scale, integrated supply chains, and a large, well-educated labor pool, with real effective exchange rate depreciation and industrial policy playing only a "secondary role." But this framing understates the interaction between the two: Industrial policy lowers unit costs and expands capacity, while a weakened real exchange rate makes that capacity cheaper in foreign-currency terms. Neither alone fully accounts for the scale of China's manufacturing surplus. Together, they are mutually reinforcing.³¹

At the macroeconomic level, a real appreciation of the renminbi is the most direct path to easing the trade imbalances that China's techno-industrial policies amplify. A stronger currency would reduce the price advantage of Chinese exports, partially allay international concerns about industrial policy subsidies and overcapacity, and help shift domestic incentives toward consumption and services. To deny that the exchange rate matters for China's manufacturing export competitiveness is akin to denying that prices matter. China would retain substantial competitive advantages and would continue to pursue the industrial policies documented in this report. But a revalued currency would mean that China's producers (rather than the rest of the world) absorb more of the adjustment costs when domestic demand fails to keep pace with capacity. Without it, Beijing is in effect asking trading partners to accommodate a structural surplus driven in part by policy choices that depress the real exchange rate.

Whether revaluation occurs through Beijing's own policy shift or through external pressure is a central open question. The IMF's recommended package would achieve it as a byproduct of domestic rebalancing. But as noted, Beijing shows no inclination to pursue that path in the near term. Trading partners can press the issue through multilateral channels or bilateral negotiations or turn to the blunter tools of tariffs and countervailing duties. None of these external instruments can compel Beijing to adjust course, but they can raise the cost to China of the status quo.

In 1971, U.S. Treasury Secretary John Connally famously told foreign finance ministers that the dollar was "our currency, but your problem."³² He was describing a world in which one country's domestic policy choices imposed adjustment costs on everyone else. Today, although the United States still anchors global finance, China occupies an analogous position in global manufacturing. Beijing's techno-industrial policy is designed for national strength, self-reliance, and development. But the combination of ambitious industrial policy atop a weak currency produces a structural surplus that the rest of the world must either absorb, resist, or negotiate away. Like Connally's dollar, China's techno-industrial might is Beijing's strategy, but, increasingly, the world's problem.

²⁹ IMF, *People's Republic of China: 2025 Article IV Consultation—Press Release; Staff Report; and Statement by the Executive Director for the People's Republic of China*, IMF Country Report No. 26/44, February 2026.

³⁰ Li, 2026.

³¹ François de Soyres, Ece Fisgin, Mike Liu, and Eva Van Leemput, "China's Trade Dominance and the Role of Industrial Policies," Board of Governors of the Federal Reserve System, March 23, 2026.

³² For the quoted line and context, see Paul Volcker and Toyoo Gyohten, *Changing Fortunes: The World's Money and the Threat to American Leadership*, Times Books, 1992.

Select Chinese Terms and Policy Concepts

This appendix defines Chinese policy terms, slogans, and institutional phrases used in the report. The entries are working definitions tailored to how the terms function in Xi-era techno-industrial policy. They are not meant to be exhaustive translations of the broader Chinese policy lexicon.

Chinese terms are listed under the English formulation used in the report. Where several official variants matter, they are grouped together in a single entry.

Historical Foundations

Independent and relatively complete industrial system (独立的比较完整的工业体系). A Mao-era formulation for building a national industrial base that could produce the essentials of modern industry inside China. The phrase signals breadth and autonomy, not just output. In this report, it marks the early ambition to reduce structural dependence on foreign powers.

Self-reliance (自力更生). Literally relying on one's own efforts. Historically, it referred to building industry and defense with limited external support and a willingness to substitute domestic mobilization for foreign technology or capital. In this report, it is the antecedent to later language about technological autonomy.

Walking on two legs (两条腿走路). A Mao-era slogan calling for simultaneous development through large modern industry and smaller local or rural industry. In practice, it legitimated parallel strategies and redundancy. It captures a long-standing Chinese preference for mobilizing multiple production channels at once.

Third Front (三线建设). The Mao-era campaign to relocate factories, research institutes, and infrastructure to China's interior for security and wartime resilience. It fused industrial policy and national defense. In this report, it matters as an early model of security-driven industrial geography.

Reform and opening (改革开放). The post-1978 shift toward market reform, foreign investment, trade integration, and pragmatic experimentation. The phrase does not mean simple liberalization. It denotes a state-managed opening in which the Party retained the right to steer development.

Socialist market economy (社会主义市场经济). The formula adopted in the 1990s to justify using markets under continued Party-state direction. It treats market mechanisms as instruments of socialism, not as an alternative to it. The phrase remains central to how Beijing explains state-guided development.

Pillar industries (支柱产业). Sectors designated as strategic supports of national development, usually large upstream or infrastructure-linked industries, such as automobiles, petrochemicals, and telecommunications. The term signals preferential treatment and long planning horizons. It helped organize sectoral targeting before Xi.

Grasp the large, release the small (抓大放小). The 1990s SOE reform line that consolidated state control over large strategic firms while many smaller state firms were privatized, merged, or closed. It did not shrink the state so much as refocus it. The approach laid the institutional basis for today's national champions.

Indigenous innovation (自主创新). A mid-2000s slogan urging China to shift from importing and adapting foreign technology to building domestic innovation capacity. In practice, it blended local R&D, foreign technology absorption, and state support for commercialization. It is a key bridge between reform-era upgrading and Xi-era self-reliance.

Strategic emerging industries (战略性新兴产业). The sectoral package launched around 2010 to promote industries expected to shape long-run growth and national competitiveness, such as biotechnology, new energy, and next-generation IT. The label signaled concentrated planning, subsidies, and credit support. It was a precursor to Xi-era techno-industrial policy.

Made in China 2025 (中国制造2025). The 2015 industrial plan aimed at upgrading manufacturing, reducing dependence on foreign technology, and increasing domestic capacity in advanced sectors. It became politically sensitive abroad because it made China's ambitions unusually explicit. In this report, it marks the move toward a more systematic and security-linked industrial policy.

Supply-side structural reform (供给侧结构性改革). Xi-era language for restructuring the production side of the economy through deleveraging, capacity reduction, destocking, cost cutting, and later, industrial upgrading. The phrase is broader than industrial policy alone. In practice, it provided political cover for both discipline and selective support.

Xi-Era Strategy and Ideology

Science and technology self-reliance and self-strengthening (科技自立自强, 高水平科技自立自强). The current core slogan for building indigenous S&T capabilities, especially in areas vulnerable to foreign choke-points. Self-strengthening adds the idea of durable internal capacity, not just import substitution. When qualified as high level, the phrase points to frontier technologies and systemwide capability.

Independent and controllable (自主可控). A standard used for technologies, systems, and supply chains that China can operate, maintain, and secure without unacceptable foreign dependence. The phrase blends autonomy with control and security assurance. In practice, it often supports domestic substitution in software, semiconductors, and infrastructure.

Dual circulation (双循环). Xi's framework for anchoring growth in the domestic economy while still using global markets, capital, and technology when they serve Chinese objectives. Internal circulation is primary, but external circulation remains instrumental. The term signals a selective integration strategy, not autarky.

New infrastructure (新基建). The Xi-era label for digital and advanced infrastructure, such as 5G, data centers, industrial internet, AI-related computing, and EV charging networks. Unlike traditional infrastructure, it is meant to support technological upgrading and future productivity. The term became a key way to align macro stimulus with industrial strategy.

Modernized industrial system (现代化产业体系). A broad Party-state goal for upgrading the industrial base so that it is technologically advanced, secure, resilient, and better integrated across sectors. The phrase is deliberately expansive. It signals that industrial structure, supply chains, and innovation capabilities are now treated as a core pillar of national power.

Real economy and foundation of the real economy (实体经济, 实体经济根基). Real economy refers to the production of goods and nonfinancial services, but in official Chinese usage, it carries a strong manufacturing emphasis. The foundation of the real economy underscores Beijing's view that national strength rests on tangible industrial capacity rather than finance or property. In this report, the phrase captures Xi's turn back toward manufacturing as a strategic base.

New quality productive forces (新质生产力). A recent slogan for productivity growth driven by advanced technology, new forms of industrial organization, high-end manufacturing, and digital transformation. It is

meant to signal a new growth model beyond property, infrastructure, and low-cost scale. In practice, it is an umbrella term that legitimizes support for AI, semiconductors, clean energy, advanced equipment, and similar sectors.

New industry tracks (新产业新赛道). A planning term for emerging sectors or technology domains that the state wants to cultivate before market leadership is settled. The racing metaphor matters. It frames industrial competition as a contest for first-mover advantage in future industries.

Extraordinary measures (超常规措施). Official language for unusually forceful or flexible policy intervention when ordinary tools are judged insufficient. The phrase signals political urgency and a willingness to concentrate resources. In the 15th FYP context, it points to intensified support for key core technologies.

Key core technologies (关键核心技术). A high-priority label for technologies that are foundational, strategically sensitive, or vulnerable to foreign chokepoints. The term usually implies concentrated state attention, not just technical importance. It is one of the clearest verbal signals that a field is treated as strategic.

Comprehensive national security (总体国家安全观). Xi's overarching security concept, which extends security far beyond the military realm to include technology, finance, industry, data, food, energy, and ideology. Under this framework, techno-industrial policy becomes a security instrument. The term helps explain why supply chain resilience and technological autonomy are treated as strategic imperatives.

Comprehensive national power (综合国力). A broad strategic concept that aggregates economic, technological, military, institutional, and political strength. The term matters because it helps explain why manufacturing capability is treated as a strategic asset rather than merely an economic variable.

Common prosperity (共同富裕). A Party slogan for broadly shared affluence rather than egalitarian income leveling. It combines growth, redistribution, regional balancing, and social stability under continued Party leadership. In this report, it matters because Xi links industrial upgrading to better jobs, stronger manufacturing, and a more socially defensible development model.

Systems competition (制度竞争). The claim that geopolitical rivalry is not only about power or markets but also about the comparative performance of political and economic systems. In official usage, technological and industrial success helps validate China's governance model. The term turns industrial policy into an argument about regime efficacy.

Socialism with Chinese characteristics (中国特色社会主义). The core ideological formula for China's post-Mao political economy. It legitimizes market activity, private firms, and foreign engagement under continued Party leadership. When it appears in industrial policy discourse, it signals that state direction and market instruments are viewed as compatible rather than contradictory.

Governance, Implementation, and Market Order

Top-level design (顶层设计). A mode of governance in which Party leadership sets the architecture, priorities, and authority relationships for a large policy system and lower levels are expected to implement within that frame. The phrase signals centralized design, not detailed central execution of everything. In this report, it is central to Xi's effort to structure techno-industrial policy from the top down.

Tiao and kuai (条, 块). A classic Chinese bureaucratic distinction between vertical functional authority, or tiao, and territorial authority, or kuai. The first runs through ministries and sectoral lines, the second through local governments. The balance between them helps explain why implementation in China can be both highly centralized in strategy and highly decentralized in practice.

Unified national market (全国统一大市场). Beijing's program to reduce local protectionism, standardize rules, harmonize enforcement, and make China's domestic market function more as a single national space. The term sounds liberalizing, and it partly is. But in this report, it also matters as a tool for central coordination, industrial rationalization, and discipline of local subsidy races.

National chessboard (全国一盘棋). A common political metaphor for treating the country as one integrated strategic space in which local actors should align with national priorities. The image emphasizes unified coordination rather than local autonomy. In industrial policy, it supports pooled demand, standardized rules, and centrally directed adjustment.

Involution and anti-involution (内卷, 反内卷式竞争). Involution refers to intense competition that multiplies effort, capacity, or investment without producing equivalent gains in innovation, quality, or profits. In current Chinese industrial debate, it usually points to subsidy races, price wars, imitation, and redundant expansion. Anti-involution is Beijing's campaign language for curbing those dynamics through stronger discipline, competition policy, and exit pressure.

Fair Competition Review System (公平竞争审查制度). A regulatory mechanism that requires governments to screen policy measures for anticompetitive bias, especially local protectionism or discriminatory support. In principle, it is a market-ordering reform. In practice, it also gives Beijing another lever to discipline local industrial policy behavior.

Two-high (两高). Shorthand for projects that are both high in energy consumption and high in emissions. The label is widely used in China's climate and industrial governance. It is important because Beijing now uses energy and environmental discipline to reshape industrial composition, not just to reduce pollution.

Chain chief system (链长制). A local or regional governance mechanism that assigns officials responsibility for developing, stabilizing, or upgrading a specific industrial chain. It turns supply chain management into an administrative task. The system often links local governments, lead firms, financing, and infrastructure around targeted sectors.

Two-way entry, cross appointment (双向进入、交叉任职). A Party governance formula in which Party secretaries and company executives or board leaders hold overlapping roles. The purpose is to tighten Party influence over enterprise decisionmaking without abandoning corporate forms. It is especially important in Xi-era SOE governance.

Finance, Firms, and Policy Instruments

Big Fund (大基金, 国家集成电路产业投资基金). The flagship state-backed semiconductor investment fund. More than a single fund, it symbolizes Beijing's effort to build whole-of-sector financing capacity for chips and related equipment. In this report, it illustrates both the reach of the investor state and the governance problems that follow from it.

Special management shares or golden shares (特别管理股). Small state equity stakes that carry outsized governance rights, such as board seats, veto power, or control over key decisions. They are especially important in digital platforms and data-intensive firms. The point is not financial return but political oversight and embedded control.

From managing enterprises to managing capital (管企业, 管资本). A shift in state-asset governance under which SASAC and related bodies emphasize control over capital allocation, shareholding structures, and portfolio strategy rather than day-to-day enterprise administration. The phrase captures the investor-state logic. It seeks to modernize state control, not reduce it.

Information technology application innovation (信创). A policy push to replace foreign IT systems with domestic alternatives in government and other sensitive sectors. The awkward English translation hides how specific the program is. In practice, it has become a major vehicle for software, server, database, and operating-system substitution under the banner of security.

Secure and controllable (安全可控). A procurement and regulatory standard that prioritizes technologies judged safe from external disruption and sufficiently subject to domestic oversight. It is related to, but

not identical with, independent and controllable. The first stresses security and governance assurance, the second broader autonomy.

First set and first batch (首台套, 首批次). Policy terms for the first commercial deployment of domestically developed major equipment or the first production batch of an innovative product. These schemes lower the market-entry barrier for new domestic technologies. In practice, they help shift commercialization risk from producers to public buyers, insurers, or state-linked purchasers.

Two New programs (两新). Shorthand for the trade-in and equipment-renewal programs used to support consumer demand and industrial upgrading at the same time. The phrase matters because it shows how macroeconomic support and industrial policy have fused. These are targeted demand-side supports for favored sectors, not generic stimulus.

Little Giants and SRDI (专精特新小巨人, 专精特新). SRDI stands for specialized, refined, distinctive, and innovative and describes firms that are highly focused, technically capable, and strong in niche segments. Little Giants are the most prominent firms within that policy ladder. The program channels credit, recognition, and policy support to smaller firms expected to fill supply chain gaps or become national industrial specialists.

Single Champions (制造业单项冠军, 单项冠军). Manufacturers recognized for global or national leadership in a narrow product category or production niche. The term signals extreme specialization rather than conglomerate scale. In Beijing's view, these firms strengthen industrial depth, component self-sufficiency, and export competitiveness.

31 Measures (31条). The shorthand used for the 2023 policy package supporting the private economy. In Chinese policy discourse, numbered measure packages often become labels in their own right. In this report, the term matters because it captures Beijing's efforts to reassure private firms even as oversight and Party control deepen.

Challenge-based leadership (揭榜挂帅). A mission-oriented mechanism in which teams bid to solve a defined technical problem and the winner is tasked with delivering results. The phrase evokes publicly listing the challenge and appointing a lead commander. It reflects a preference for mobilizing targeted breakthroughs rather than simply distributing research funds.

Four only (破四唯). Literally breaking the four onlys, meaning overreliance on papers, titles, diplomas, and awards in talent and research evaluation. The slogan is an internal reform critique. It signals dissatisfaction with credentialism and a desire to judge scientists and engineers more by contribution and mission performance.

World-important talent center and strategic talent forces (世界重要人才中心, 战略人才力量). These terms frame talent as a national strategic resource, not just a labor-market issue. The first sets an ambition to make China a globally important hub for high-level talent. The second emphasizes building politically reliable, mission-oriented pools of scientists, engineers, and managers for priority sectors.

Hukou (户口). China's household registration system. In this report, it matters because local governments use hukou-related benefits (such as access to schooling, housing, and public services) to attract high-level talent. The term shows how industrial policy in China often depends on both administrative and financial incentives.

Fault-tolerance mechanism (容错机制). A governance term for protecting officials or fund managers from punishment when policy-aligned risk-taking fails despite following procedure. The goal is to reduce excessive caution in state-guided investment. It reflects Beijing's recognition that anticorruption and audit pressure can suppress experimentation.

Hard technology (硬科技). Chinese policy and venture-capital shorthand for deep-tech sectors that require overcoming substantial engineering or scientific barriers, such as semiconductors, advanced materi-

als, industrial equipment, or frontier manufacturing. The label distinguishes these sectors from consumer internet business models. In practice, it justifies longer horizons and more patient capital.

Standards Connectivity (标准联通). The BRI term for aligning, translating, recognizing, and codeveloping standards with partner countries. The phrase sounds technical, but its strategic function is commercial and geopolitical. It helps lock Chinese equipment, testing systems, and technical rules into overseas markets.

Catfish effect (鲶鱼效应). A business metaphor for introducing a strong outsider to force incumbents to improve. In this report, it is used to explain how foreign entrants, such as Tesla, can be welcomed when they accelerate domestic upgrading. The phrase illustrates how selective opening is sometimes used instrumentally to sharpen local competition.

New Nine Rules (新国九条). A 2024 capital-market reform package that tightened listing standards, delisting, and investor protection while reaffirming support for technology financing. The nickname follows a long Chinese habit of referring to important policy packages by the number of provisions they contain. In this report, it marks the more disciplined use of capital markets for industrial policy.

Additional Policy Phrases Used in Table 2.1

Science and technology power (科技强国). The aspiration to make China a leading scientific and technological power. The phrase links innovation directly to national status.

High-quality development (高质量发展). Xi-era language for growth that is more advanced, efficient, innovative, and environmentally sustainable than the old high-speed model. It is one of the regime's main justifications for industrial upgrading.

Integration of industry, university, and research (产学研融合). A standard Chinese innovation-policy phrase for closer links among firms, universities, and research institutes. It refers to commercialization and system integration, not just academic collaboration.

Transformation of scientific and technological achievements (科技成果转化). The policy goal of turning research outputs into usable products, processes, and firms. The phrase is central to China's effort to convert R&D inputs into industrial capacity.

Great rejuvenation of the Chinese nation (实现中华民族伟大复兴). The master political narrative under Xi. In industrial policy discourse, technological and manufacturing strength are presented as core components of national rejuvenation.

Great-power heavy equipment (大国重器). A celebratory term for large, advanced, strategically important hardware, such as major transport systems, energy equipment, aerospace platforms, or heavy industrial machinery. The phrase mixes industrial capability, prestige, and national strength.

World-class (世界一流). A common policy benchmark used for firms, universities, laboratories, and industrial clusters that are supposed to compete at the highest global standard. The term often functions as a performance target as much as a description.

Abbreviations

AE	advanced economy
AI	artificial intelligence
BRI	Belt and Road Initiative
BSE	Beijing Stock Exchange
CCP	Chinese Communist Party
CDB	China Development Bank
CNKI	China National Knowledge Infrastructure
COVID-19	coronavirus disease 2019
CSIS	Center for Strategic and International Studies
DSR	Digital Silk Road
EMDE	emerging and developing economy
EV	electric vehicle
Exim	Export-Import Bank of China
FDI	foreign direct investment
FYP	Five-Year Plan
GDP	gross domestic product
ICT	information and communications technology
IEC	International Electrotechnical Commission
IMF	International Monetary Fund
IPO	initial public offering
ISO	International Organization for Standardization
IT	information technology
LGFV	local government financing vehicle
MIIT	Ministry of Industry and Information Technology
MLP	Medium- and Long-Term Plan for Science and Technology Development
MOF	Ministry of Finance of the People's Republic of China
NDRC	National Development and Reform Commission
OECD	Organisation for Economic Co-operation and Development
PBOC	People's Bank of China
R&D	research and development
S&T	science and technology
SASAC	State-owned Assets Supervision and Administration Commission
SEI	strategic emerging industry
SMEs	small and medium-sized enterprises
SOE	state-owned enterprise
SRDI	specialized, refined, distinctive, and innovative
STAR Market	Shanghai Science and Technology Innovation Board
TFP	total factor productivity

UNM	unified national market
VAT	value-added tax
WFT	Wind Financial Terminal
WTO	World Trade Organization

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